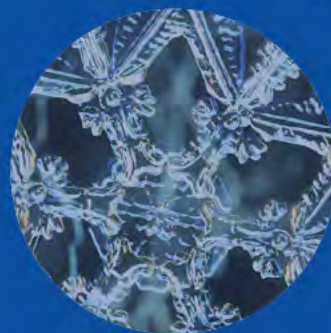
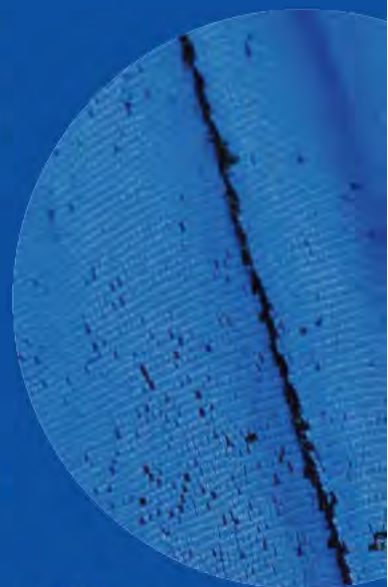
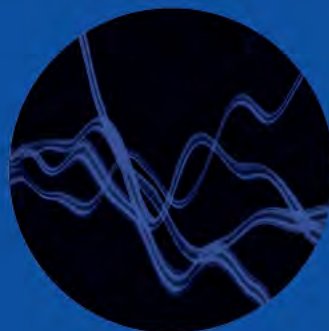


Review

ERICSSON
TECHNOLOGY



INDUSTRIAL REMOTE
OPERATION
5G RISES TO
THE CHALLENGE

HARALD LUDANEK
ON ICT AND INTELLIGENT
TRANSPORTATION SYSTEMS

**MICROWAVE
BACKHAUL**
GETS A BOOST
WITH MULTIBAND







08 CRYPTOGRAPHY IN AN ALL ENCRYPTED WORLD

Cyber attacks are on the increase, global fears over personal security and privacy are rising, and quantum computing might soon be reality. These concerns have created a number of shifts in how encryption technologies are being developed and applied. Today, it is no longer sufficient to encrypt data as it passes through the access part of the network; information needs to be protected from source to destination.

20 MICROWAVE BACKHAUL GETS A BOOST WITH MULTIBAND

Is there a spectrum shortage? The answer to the question is both yes and no; in some locations spectrum is severely congested, while in other places it is highly underutilized. New methods that will maximize spectrum efficiency, and new technologies that can exploit unused spectrum are needed. Multiband booster is one such method, fundamentally shifting the way spectrum can be used, with a promise to deliver a massive improvement in the performance levels of microwave backhaul.

30 LUDANEK ON ICT AND INTELLIGENT TRANSPORTATION SYSTEMS

Over the past 50 years, the automotive industry has undergone what could be described as a technology revolution. Fuel efficiency, environmentally sound vehicle powertrain concepts, increased electronics, driver assistance, and safety features like ABS and airbags are just a few of the improvements that have taken place, which have led to sustainable, safer, and more comfortable driving.

40 FLEXIBILITY IN 5G TRANSPORT NETWORKS: THE KEY TO MEETING THE DEMAND FOR CONNECTIVITY

As applications like self-driving vehicles and remotely operated machinery evolve, become more innovative, and more widespread,

the level of performance that 5G networks need to deliver will inevitably rise. Keeping pace with ever-increasing demand calls for greater flexibility in all parts of the network.

54 INDUSTRIAL REMOTE OPERATION: 5G RISES TO THE CHALLENGE

Ericsson and ABB are collaborating to determine how to make the most of 5G and cellular technologies in an industrial setting. This article presents some of the use cases being assessed, highlights the challenges posed by remote operations, and describes how 5G technology can be applied to overcome them.

(This article was written in collaboration with ABB)

68 IDENTIFYING AND ADDRESSING THE VULNERABILITIES AND SECURITY ISSUES OF SDN

The promises of agility, simplified control, and real-time programmability offered by software-defined networking (SDN) are attractive incentives for operators to keep network evolution apace with advances in virtualization technologies. But do these capabilities undermine security? To answer this question, we have investigated the potential vulnerabilities of SDN.

80 A VISION OF THE 5G CORE: FLEXIBILITY FOR NEW BUSINESS OPPORTUNITIES

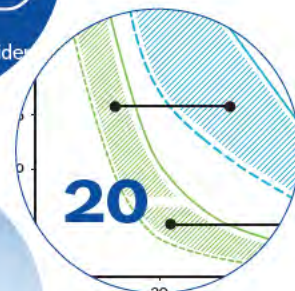
Next generation 5G networks will cater for a wide range of new business opportunities, some of which have yet to be conceptualized. Being able to provide customized connectivity will benefit many industries around the world. But how will future networks provide people and enterprises with the right platform, with just the right level of connectivity?



40



Cloud service provider



20



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ERICSSON TECHNOLOGY REVIEW

Bringing you insights into some of the key emerging innovations that are shaping the future of ICT.

Our aim is to encourage an open discussion on the potential, practicalities, and benefits of a wide range of technical developments, and help provide an insight into what the future has to offer.

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WHY FLEXIBILITY COUNTS...

■ **EVERY MORNING**, I get out of bed and go to work because I believe technology makes a difference. I believe that in the midst of global growth, numerous humanitarian crises, the increasing need for better resource management, and an evolving threat landscape, a new world is emerging. And I believe technology is playing a key role in making that world a better, safer, and healthier place for more people to enjoy. It feels good to be part of that.

Fundamentally, I believe the breakdown of traditional industry boundaries and increased cross-industry collaboration have enabled us to maximize the benefits of technology. Today, Ericsson works with partners in many different industries that all rely on connectivity embedded into their solutions, services, and products. Our early collaborations, which were with utilities and the automotive industry, have led to innovations like the Connected Vehicle Cloud and Smart Metering as a Service.

I am delighted that Harald Ludanek, Head of R&D at Scania (a leading manufacturer of heavy trucks, buses, coaches, and industrial and marine engines) agreed to contribute to this issue. His article on the significance of ICT—how digitalization and mobility will impact the automotive industry and bring about the intelligent transportation system (ITS)—illustrates the importance of new business relationships, ensuring that different sectors create innovative solutions together, and maximize the value they bring to people and society.

Technology is making it easier for people to protect their homes, families, and belongings. The standardization of anti-theft systems in automobiles, for example, has led to a decline in car theft in most parts of the world. However, while technology offers improved security, somehow criminal countermeasures manage to keep up. In an article about end-to-end cryptography, a number of Ericsson experts highlight how car theft is no longer carried out with a slim jim and a screwdriver, but rather with highly sophisticated decryption algorithms, smartphones, and illegal access to software keys.

The protection of data — and the people who own it — as it travels across the network has always been a ■■■

■ cornerstone of the telecoms industry. But in today's world, no single organization can maintain end-to-end control over information as it is carried from source to destination, and so upholding the right to privacy is becoming an increasingly complex issue. And with quantum computing posing a threat to our current security systems, our experts point out that this will render certain existing methods of protection useless. Not only do protocols need a shake up, so does software — so it can work in lightweight mode for constrained or hardware-limited devices.

The idea that technology can manage an underground mine efficiently, operate construction machinery from a distance, or carry out a complex surgical procedure on a remote basis, is not far from magical. Imagine a world in which the hazardous work environment is a thing of the past, where manufacturing operations are run smoothly using remotely operated machines and robots, where everyone has access to vital medical expertise... This is the stuff of my boyhood science fiction comics. But today, these are the technical innovation challenges my colleagues intend to solve — and in some cases, they already have.

The article on 5G remote control, which was cowritten with experts from ABB, is yet another example of how collaboration has become embedded in our ways of working, and how different industries can help each other to create more innovative solutions.

If you were to ask me to pick a few words to summarize this issue of Ericsson Technology Review, I would choose security, new business opportunity, flexibility, SDN, virtualization, and 5G. But, it is flexibility that clearly stands out for me. If networks are going to provide the kind of connectivity that industry needs, flexibility is not only required in the technical solution, but at all other levels too — even in business models and internal processes.

Flexibility will be achieved in the network through greater abstraction, programmability, and a core built on the concept of network slicing — which is where 5G comes in. As the article on the 5G core shows, a flexible

BY 2021, OVER 90% OF THE WORLD'S POPULATION WILL BE COVERED BY MOBILE BROADBAND NETWORKS*

network architecture is needed by service providers and industries that depend on connectivity to develop new solutions. It will enable them to fail fast, and to adapt their networks as quickly as business models change. In his article on the multiband booster for microwave backhaul, Jonas Edstam points out that in a 5G world, capacity needs will no longer represent the main determining factor for network architecture; instead, total cost of ownership will take over, with a more holistic approach to networking.

As always, I hope you find our stories relevant and inspiring. All of our content is available at www.ericsson.com/ericsson-technology-review, through the Ericsson Technology Insights app, and on SlideShare.



ULF EWALDSSON

SENIOR VICE PRESIDENT,
GROUP CTO, AND HEAD OF GROUP FUNCTION
TECHNOLOGY

*Ericsson Mobility Report, November 2015

Cryptography

IN AN ALL **ENCRYPTED** WORLD

Ensuring that communication is secure, including the ability to encrypt sensitive traffic, has always been a fundamental pillar of the telecom industry. Users expect their right to privacy to be respected, and operators expect to be able to protect themselves and their customers from various kinds of attacks. But the world is changing. Encryption technologies are advancing, regulations are changing, criminals are becoming highly tech savvy, and security awareness has become a popular conversation topic. So, in light of new threats and security demands, security protocols need a shake-up.

CHRISTINE JOST
JOHN MATTSSON
MATS NÄSLUND
BEN SMEETS

TRADITIONALLY, ENCRYPTION has been applied to data carried over the access network — other parts of the network being trusted inherently. But the shift to cloud networking, the increased awareness of threats, exposure of the weaknesses of traditional security algorithms, and the rise in the value of owning data, have all contributed to the need to protect data in all parts of the network, and tighten encryption methods against unwanted intrusion.

■ In the post-Snowden era, revelations relating to the apparently indiscriminate way pervasive surveillance is carried out have heightened public awareness of privacy issues. Security and privacy have since moved up on the list of top priorities for standardization groups in many industries. Strong reactions to the sabotage of an encryption standard have led to mistrust and eroded confidence in some standards that are widely used to protect data. Our collective dependence on networks has made protecting the data they carry a topic of concern for governments, regulators, and security companies,

but heightened public and media awareness is signaling a move to a more conservative approach.

As the sensitivity of data is not an easily defined concept, many standardization groups, such as the IETF, have chosen to adopt the same approach as modern mobile networks; in other words, encrypt everything — not just data as it is carried over the access network, but over the entire path, end-to-end.

Encryption-enforcing protocols such as HTTP/2, WEBRTC, and TLS 1.3 are essential for OTT service providers. They are also required when operators introduce IMS, VoLTE, RCS, CDN and cloud services on top of the core mobile network.

The increased use of encryption is good for enterprise security and privacy, but comes at the expense of more complicated network management, more complex content delivery optimization, and hampered ability to offer value-added services. Heuristic mechanisms, like those based on the frequency and size of packets, as well as IP-based classification, will help to overcome these difficulties and continue to work well in many cases, even where traffic classification is required.

The global rise in awareness and impending stricter regulations surrounding individual security and privacy requirements have driven the need for communication standards that enable levels of security. Industry use of encryption, however, is being driven by a desire to control delivery end-to-end. For example, enterprises need to be able to avoid potential problems caused by network intermediaries, such as ad injectors or application

layer firewalls, ensuring that the integrity and exclusive ownership of valuable analytics data continue to be protected.

Communication security in cellular networks is changing. The algorithms developed by 3GPP and GSM for confidentiality, integrity, authentication, and key derivation have evolved dramatically since they were first introduced. The original algorithms deployed in 2G/GSM networks were kept secret — security by obscurity — and designed to meet the import/export restrictions related to encryption of the time (early 1990s). These algorithms were subsequently leaked and found to have weaknesses. The encryption algorithms developed for 3G and LTE have been made available for public analysis. They use well-known and standardized cryptographic algorithms such as AES, SNOW, and SHA-3, and to date, no weaknesses have been found. Communication security has not only evolved in terms of how to encrypt data but also what to protect: traditionally, only the access part of the network was encrypted. In today's networks, protection has been extended to cover backhaul, core node communication links using IPsec or TLS as well as services using SRTP, TLS, DTLS, or through object security provided by, for example, XML encryption.

Complementing protection on trusted interfaces and nodes provides additional assurance against unexpected compromises, secures operational ownership, and enables end-to-end security — making it easier to create the right services for

Terms and abbreviations

ABE—Attribute-Based Encryption | **AEAD**—Authenticated Encryption with Associated Data | **AES**—Advanced Encryption Algorithm | **CDN**—content delivery network | **IRTF CFRG**—IRTF Crypto Forum Research Group | **DTLS**—Datagram TLS | **ECC**—Elliptic Curve Cryptography | **ECDSA**—Elliptic Curve Digital Signature | **GCM**—Galois Counter Mode | **IOT**—Internet of Things | **IPsec**—Internet Protocol Security | **IRTF**—Internet Research Task Force | **OTT**—over-the-top | **PQC**—post-quantum cryptography | **QUIC**—Google's Quick UDP Internet Connections | **RCS**—Rich Communication Services | **RSA**—Rivest-Shamir-Adelman cryptosystem | **SHA**—Secure Hash Algorithm | **SNOW**—synchronous stream cipher | **SRTP**—Secure Real-time Transport Protocol | **TLS**—Transport Layer Security

security-aware customers like the IT department of an organization.

From an ethical standpoint, strong user protection is probably the best guide to how to use security mechanisms in standardization as well as in products. At the same time, law enforcement authorities need to be able to intercept the communication of an individual or of an organization — in other words, networks need to support lawful intercept (LI) authorized by a court order. However, as the application of LI may intrude upon private communication, a trade-off between the overall safety of society and user/enterprise privacy is necessary. In many cases, it is sufficient to supply law enforcement with signaling and/or network management data; access to the actual content of a communication tends to be less frequent. However, in the light of the increasing threat of attack, the scope and concept of LI is changing, and some countries like France and the UK are already amending their regulations.

With the right technical solutions and standards in place, the need for next generation networks to work in an all-encrypted manner is not in conflict with providing value to all stakeholders. However, in a new world where encryption is applied in access networks, as well as in backhaul, core, and for services, new demands are placed on cryptographic primitives and how they are used. Ericsson is therefore actively pushing standardization, and the development of products and services with this goal in mind.

Developments and challenges

As algorithm design and technology develop, giving rise to powerful computers and large memory capacity, the need to strengthen current cryptography methods against brute-force key-recovery attacks has become a widely accepted fact.

At the same time, new capabilities resulting from advances in computing can be applied to increase the strength of encryption algorithms. Aside from the practical issues related to key management, strengthening encryption can be quite simply achieved by using longer keys. However, the heightened security environment of 2015 has drastically altered expectations from individuals

and society as a whole. Demand for security and privacy capabilities has soared, and so the requirements placed on cryptographic techniques have risen accordingly. This situation has put existing algorithms into question, leading to efforts to standardize new algorithms and protocols.

Security issues are not the only factor shaping the design of new security protocols and cryptographic algorithms. Performance characteristics like latency and energy efficiency, as well as new business opportunities are significant factors that need to be included in the design. High-performance algorithms need to be developed, and challenges such as providing security in the virtualized world need to be overcome. But how will developments like these affect the ICT industry, and what business opportunities do they bring?

High-performance algorithms and protocols

Some legacy algorithms no longer meet the increased security and performance demands in today's technological environment. In some cases, they are perceived as too slow and consume too much energy. The ability to ensure the security of information is fundamental in an all-encrypted world. Yet in this environment, the performance and efficiency of cryptographic algorithms has become an additional essential, so that systems can deliver the expected service performance with minimum impact on the energy budget.

Keyed cryptographic algorithms come in two varieties: symmetric, and asymmetric (public key), and provide encryption and integrity protection. In a symmetric algorithm, the sender and the receiver share an identical secret key. Symmetric algorithms, such as AES, are relatively fast and are as such often used to protect traffic or stored data. To reveal the key, it would take an attacker 2^n evaluations of the decryption algorithm, where n is the key length, which for AES-128 is 2^{128} evaluations.

In legacy symmetric algorithms, the processes of encryption and integrity protection are separated. By instead combining them, newer AEAD algorithms achieve huge performance gains over their legacy counterparts. For example, when AES is used in Galois Counter Mode (AES-GCM), it has

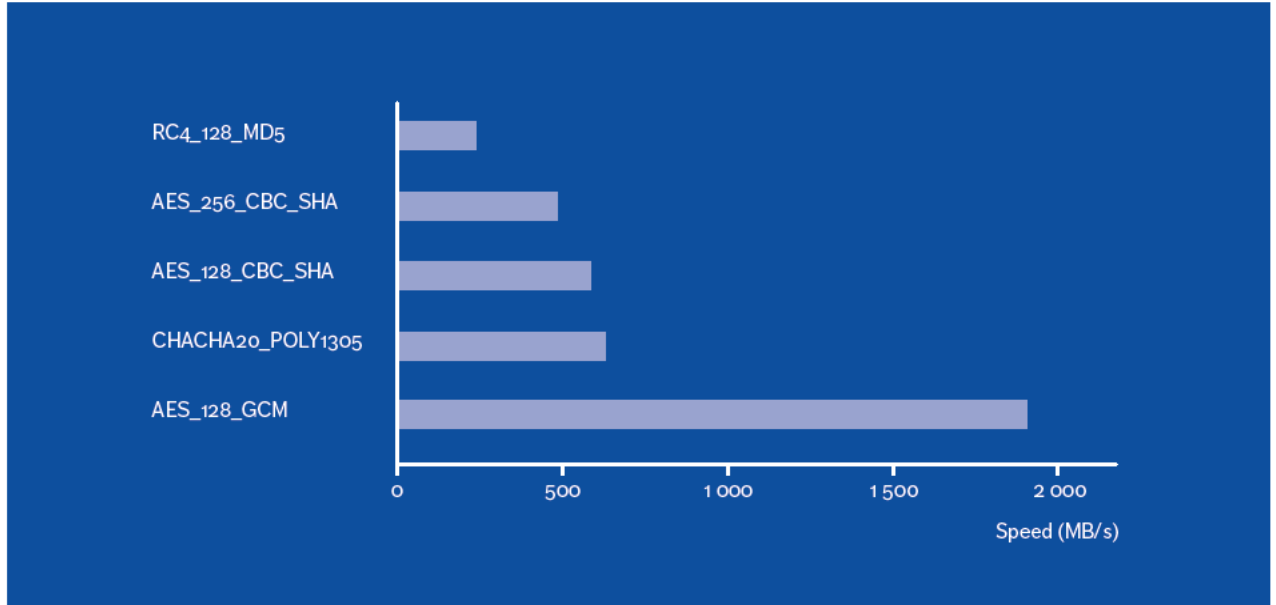


Figure 1
Data rate transfer
of various ciphers

outstanding performance on modern processors, and is today's solution of choice for many high-end software applications. However, alternative solutions are needed for constrained devices or devices without hardware support for AES. AEAD algorithms, such as AES-CCM or CHACHA20-POLY1305, might be preferable in such cases. To get a feeling for the gains that can be made, TLS 1.2 with AES-GCM is about three times faster than AES-CBC with SHA-1, and can be up to 100 times faster than TLS with 3DES. *Figure 1* shows the performance gains that can be achieved with various ciphers in `OPENSSL` running on a 2GHz Intel Core i7. In addition to the speed gains that AEAD algorithms can achieve, some of the security weaknesses found in older versions of TLS have also been resolved.

In asymmetric algorithms, data encrypted with the public key can only be decrypted by the private key, and signatures created with the private key can be verified with the public key. As its name implies, the public key is not secret and is freely distributed. Typically, public-key algorithms like RSA and DH are

used for authentication and key exchange during session setup, and not for the protection of data traffic; these algorithms are far less performant for bulk encryption compared with symmetric cryptographic algorithms.

Similar to the way AEAD algorithms have led to improved security and performance of symmetric cryptography, Elliptic Curve Cryptography (ECC) is enabling smaller key sizes and better performance for public-key cryptography. The key sizes used in public-key algorithms need to be longer than those used in symmetric algorithms of comparable strength, and are chosen so that recovery takes roughly 2^{128} operations. Such key sizes are said to provide 128-bit security. To provide security at the 128-bit level, the ECC signature algorithm ECDSA (with the NIST P-256 curve) uses significantly smaller key sizes than RSA (256 bits compared with 3072 bits) and delivers significantly better performance in use cases where both signing and verification are needed. The new ECC curves [1] and signature algorithm ED25519 [2] standardized

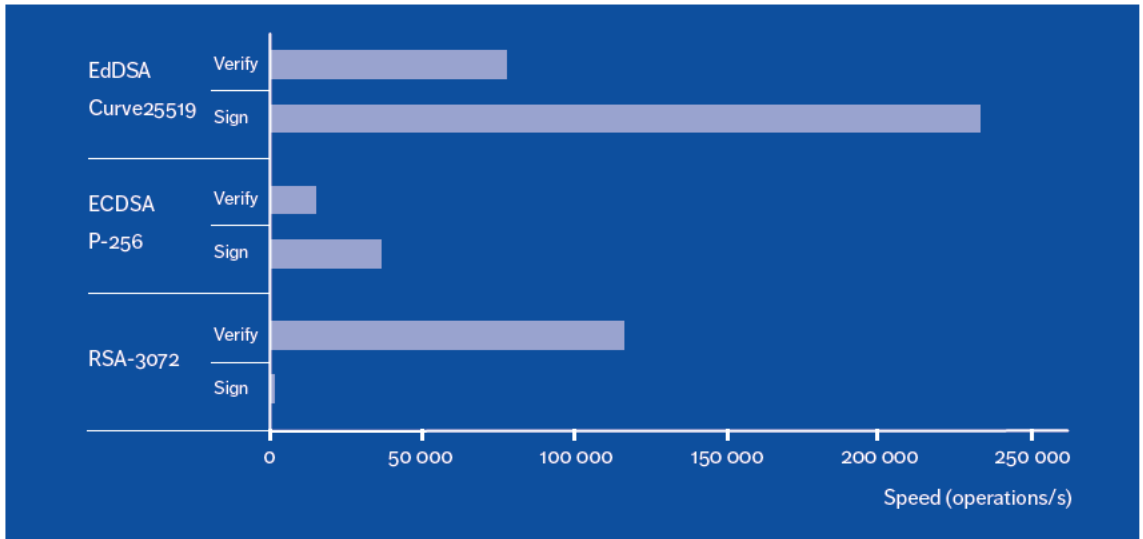


Figure 2:
Signing and verification speeds of 59 byte messages with 128-bit security algorithms on Intel Xeon E3-1275 [3]

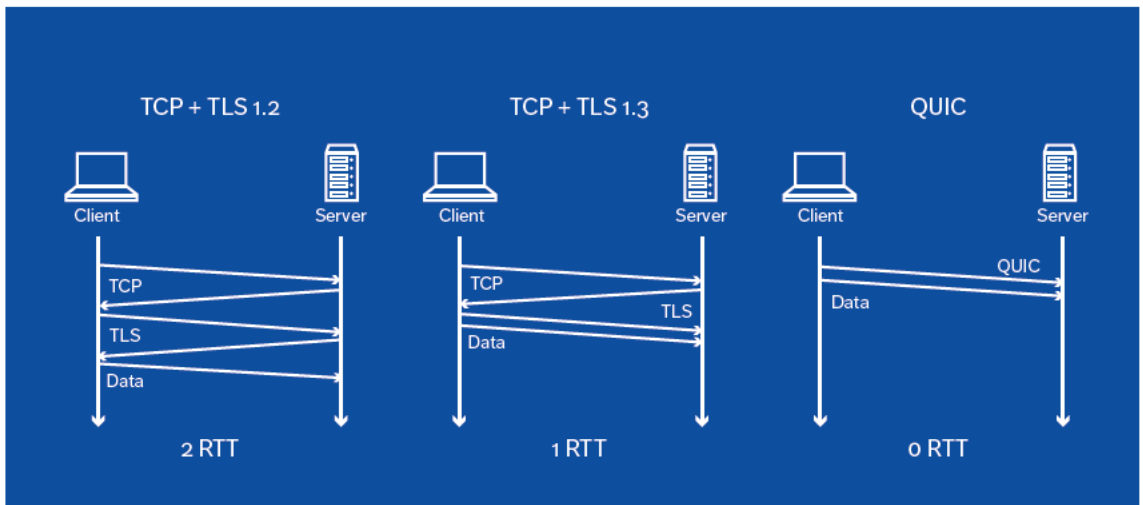


Figure 3:
Repeated connection establishment using TLS 1.2, TLS 1.3, or QUIC

by the IETF CFRG will further improve the performance of ECC, so that it will be able to offer over 100 times better signing performance than RSA. [Figure 2](#) shows the performance comparison of ECC to RSA. Not only are new standards such as CURVE25519 and EDDSA much faster than their predecessors, they are also more secure, as their designs take into account two decades worth of security improvement suggestions developed by the scientific community.

New protocols such as TLS 1.3 and the soon-to-be-standardized QUIC significantly reduce connection setup latency by lowering the number of messages needed to complete security association on setup. These protocols disable any options that weaken security and forward secrecy is on by default. Forward secrecy protects a communication session against future compromise of its long-term key. Old versions of TLS, except TCP, required two round trips to set up a connection to a new server, and one round trip for a repeat connection. Newer versions such as TLS 1.3 only use one round trip message exchange to set up a connection to a new server, and no additional trips for subsequent secure connection establishments. QUIC takes this improvement one step further, requiring just a one-directional (client to server) message link to establish server connections. [Figure 3](#) explains the latency reductions obtained by the improved connection establishment of TLS 1.3 and QUIC.

The ICT industry is in the process of abandoning the use of several legacy algorithms and protocols including 3DES, RC4, CBC-mode, RSA, SHA-1, and TLS 1.1 opting for newer, more secure, and faster algorithms such as AES-GCM, ECC, SHA-2, and TLS 1.2, and later versions.

This shift is embraced in Ericsson's strategy on the use of next generation cryptography and in the product roadmaps. In addition, Ericsson has recently initiated an upgrade of the 3GPP security profiles for certificates and security protocols such as TLS, IPSEC, and SRTP [4]. Ideally, all security should be implemented using efficient and well-tested algorithms that offer a cryptographic strength that is equivalent of at least 128-bit security for AES — even the world's fastest supercomputer, breaking

this level of security by brute-force exhaustive search would be expected to take longer than the time that has elapsed since the Big Bang.

IoT and the Networked Society

The two prominent messaging patterns used in IoT device communication are store-and-forward and publish-subscribe. IoT device communication occurs in a hop-by-hop fashion and relies on middleboxes, which limits the possibility for end-to-end security. Traditional transport-layer security protocols, such as DTLS, have difficulty in providing end-to-end data protection for this IoT-type traffic — DTLS, for example, only offers hop-by-hop security. To overcome this issue, fully trusted intermediaries are necessary, which makes it harder to offer IoT communication services to enterprises and governments that are highly security and privacy sensitive.

The debate regarding pervasive monitoring has illustrated the need to protect data even from trusted intermediary nodes — as they can be compromised. To respond to this need, the IETF (supported by Ericsson) is working on object security for the IoT [5] — as illustrated in [Figure 4](#). The aim of object security is to provide end-to-end protection of sensitive data, while at the same time enabling services to be outsourced. For example, data collection from a large IoT sensor deployment is a typical service that could be outsourced to a third party.

The security properties of cyber-physical systems (CPSs), such as a smart power grid, are quite different to those of a typical IoT deployment, which tend to contain a mass of sensors. The ability to control a CPS in a secure manner is essential in a world where billions of connected and networked things interact with the physical world. The purpose of a remote-controlled CPS, like a drone or a group of robots, can often be mission-critical. These systems tend to be open or closed-loop controlled, and any denial-of-service attacks such as the blocking, delaying, or relaying of messages can have serious consequences. For example, by relaying messages out-of-band, using say walkie-talkies, attackers can unlock and drive away with

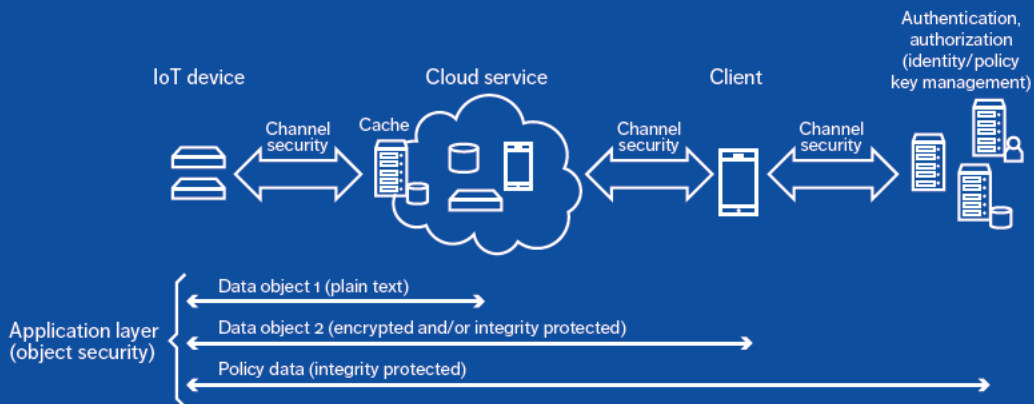


Figure 4:
Object security in the IoT

exclusive vehicles using automatic car keys based on proximity (an attack that has been executed against real assets).

Cloud data security

In the post-Snowden era, the significance of data security and privacy, as key selection criteria for cloud-infrastructure providers, has risen considerably [6]. To make it easier for organizations to outsource their communication solutions, Ericsson's approach is to push standardization, so that end-to-end protection of content can be combined with hop-by-hop protection of less sensitive metadata [7]. Many cloud-storage providers have adopted client-side encryption to prevent unauthorized access or modification of data, which solves the issues surrounding secure storage and forwarding for cloud data.

Data encryption has other benefits; in many jurisdictions users need to be informed of data breaches unless their information was encrypted. However, encryption does not necessarily mean better compliance with privacy regulations.

Homomorphic encryption is one of the key breakthrough technologies resulting from advances in cryptographic research. In contrast to AES, for example, this approach allows operations to be performed directly on encrypted data without needing to access data in its decrypted form. Unfortunately, fully homomorphic encryption, which includes methods that allow arbitrary computations on encrypted data, have yet to overcome some performance issues. However, a number of specialized methods like partially homomorphic encryption, deterministic encryption, order-preserving encryption, and searchable encryption allow a specific set of computations to be performed on encrypted data, with a sufficient level of performance so that they can be applied to real-life scenarios. By combining these methods, it is possible to cover many types of computations that arise in practice. For example, different proofs of concept have shown that by combining encryption methods, typical SQL operations such as **SUM**, **GROUP BY**, and **JOIN** can be carried out on encrypted databases [8]. Many computations,

best outsourced to the cloud, use a restricted set of operations that can be dealt with using these specialized methods with good performance. For example, sums, averages, counts, and threshold checks can be implemented. However, further research is needed to make these methods applicable to real-world use cases. For example, data encryption performance is crucial for use cases with high data throughput. Ericsson's research [9] into the encryption performance of the most popular partially homomorphic cryptosystem (the Paillier system) has shown a performance increase of orders of magnitude, which makes Paillier suitable for high-throughput scenarios.

Specialized methods, like homomorphic encryption, used for carrying out computations on encrypted data, could also be used for preserving confidentiality in cloud computation and analytics-as-a-service. With these methods, clients with large datasets to be analyzed — such as network operators, health care providers, and process/engineering industry players — would be able to outsource both storage and analysis of the data to the cloud service provider. Once outside the client's network, data is encrypted, thereby preserving confidentiality, and allowing the cloud provider to perform analytics directly on the encrypted data. As illustrated in [Figure 5](#), such an approach enables cloud computation for analysis of confidential data.

Identity and attribute-based encryption

Strong cryptography alone does not work without proper key management. Specifically, management covers how keys are generated and distributed, and how authorization to use them is granted.

Protecting data exchange between n endpoints using symmetric key cryptography requires the secure generation and distribution of roughly n^2 pair-wise symmetric keys. With the breakthrough invention of public key cryptography in the works of Diffie, Hellman, Rivest, Shamir, and Adleman in the mid-1970s, the use of asymmetric key pairs reduced the quadratic complexity, requiring only n key pairs. However, this reduction in the number of keys is offset by the need to often ensure that the public portion of the key pair can be firmly associated

with the owner of its private (secret) portion. For a long time, a Public Key Infrastructure (PKI) was the main way to address this issue. But PKIs require management and additional trust relations for the endpoints and are not an optimal solution.

Identity-Based Encryption (IBE) allows an endpoint to derive the public key of another endpoint from a given identity. For example, by using an e-mail address (name.surname@company.com) as a public key, anyone can send encrypted data to the owner of the e-mail address. The ability to decrypt the content lies with the entity in possession of the corresponding secret/private key — the owner of the e-mail address — as long as the name space is properly managed.

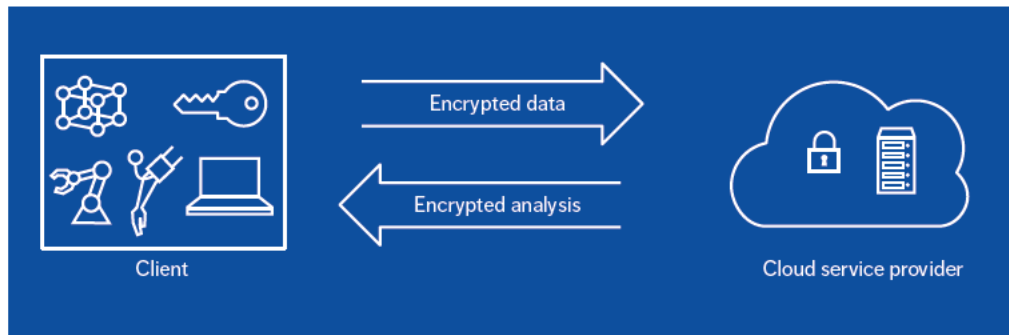
Attribute-Based Encryption (ABE) takes this idea further by encoding attributes, for example, roles or access policies, into a user's secret/private keys. IBE and ABE allow endpoints without network connections to set up secure and authenticated device-to-device communication channels. As such, it is a good match for public safety applications and used in the 3GPP standard for proximity-based services for LTE.

Post-quantum cryptography

Although the construction of quantum computers is still in its infancy, there is a growing concern that in a not too distant future, someone might succeed in building much larger quantum computers than the current experimental constructions. This eventuality may have dramatic consequences for cryptographic algorithms and their ability to maintain the security of information. Attack algorithms have already been invented and are ready for a quantum computer to execute on.

For symmetric key cryptography, Grover's algorithm is able to invert a function using only $\sqrt{N}$ evaluations of the function, where N is the number of possible inputs. For a symmetric 128-bit key algorithm, such as AES-128, Grover's algorithm enables an attacker to find a secret key 200 quintillion times faster, using roughly 2^{64} evaluations instead of 2^{128} — the complexity of an exhaustive search. Quantum computing therefore weakens the effective security of symmetric key cryptography by

Figure 5:
Cloud-based analytics on
encrypted data



half. Symmetric key algorithms that use 256-bit keys such as AES-256 are, however, secure even against quantum computers.

The situation for public-key algorithms is worse; for example, Shor's algorithm for integer factorization directly impacts the security of RSA. This algorithm is also effective in dealing with all other standardized public-key crypto systems used today. With Shor's algorithm, today's public-key algorithms lose almost all security and would no longer be secure in the presence of quantum computing. Figure 6 shows the effect of quantum computing on today's algorithms.

Although current research is far from the point where quantum computing can address the size of numbers used today in crypto schemes, the ability to perform quantum computing is increasing. The largest number factored by a quantum computer used to be the integer 21 (3×7), but in 2014, a quantum computer factored 56,153 (233×241). The term post-quantum cryptography (PQC) is used to describe algorithms that remain strong, despite the fledgling capabilities of quantum computing. In 2014, ETSI organized a workshop on quantum-safe cryptography, and in 2015 the US National Security Agency (NSA) said [10] it would initiate a transition to quantum-resistant algorithms. The potential impact of quantum computing has reached the level of industry awareness.

So, where does research stand today with respect to PQC? Understandingly, most effort is being

focused on finding alternatives for the potentially broken public-key algorithms — particularly those that produce digital signatures. In their efforts, researchers follow different tracks such as the use of coding theory, lattices, hash functions, multivariate equations, and supersingular elliptic curves. For example, some schemes go back to ideas set forth by Merkle and use hash functions in Merkle trees as a component. As quantum computing becomes a reality, such schemes would reduce the effective key size by 33 percent, still enabling them to remain practically secure. The challenge for new schemes is to find solutions that have the same properties, such as non-repudiation, that digital signatures have today or provide data integrity with public verification. From this perspective, the blockchain construction used in Bitcoin is interesting. Although Bitcoin itself is not quantum immune, there is an interesting ingredient in its construction: when the chain has grown long enough, the integrity of hash value does not rely on verification against a digital signature but by having it endorsed by many users. By creating a public ledger, any tampering of a hash value is revealed by comparing it with the public value. The idea of a public ledger is significant in the KSI solution [11] for data integrity available in Ericsson's cloud portfolio. Yet the search for PQC schemes that can provide digital signatures with non-repudiation continues.

Today's systems that use or introduce symmetric schemes, should be designed with sufficient margin

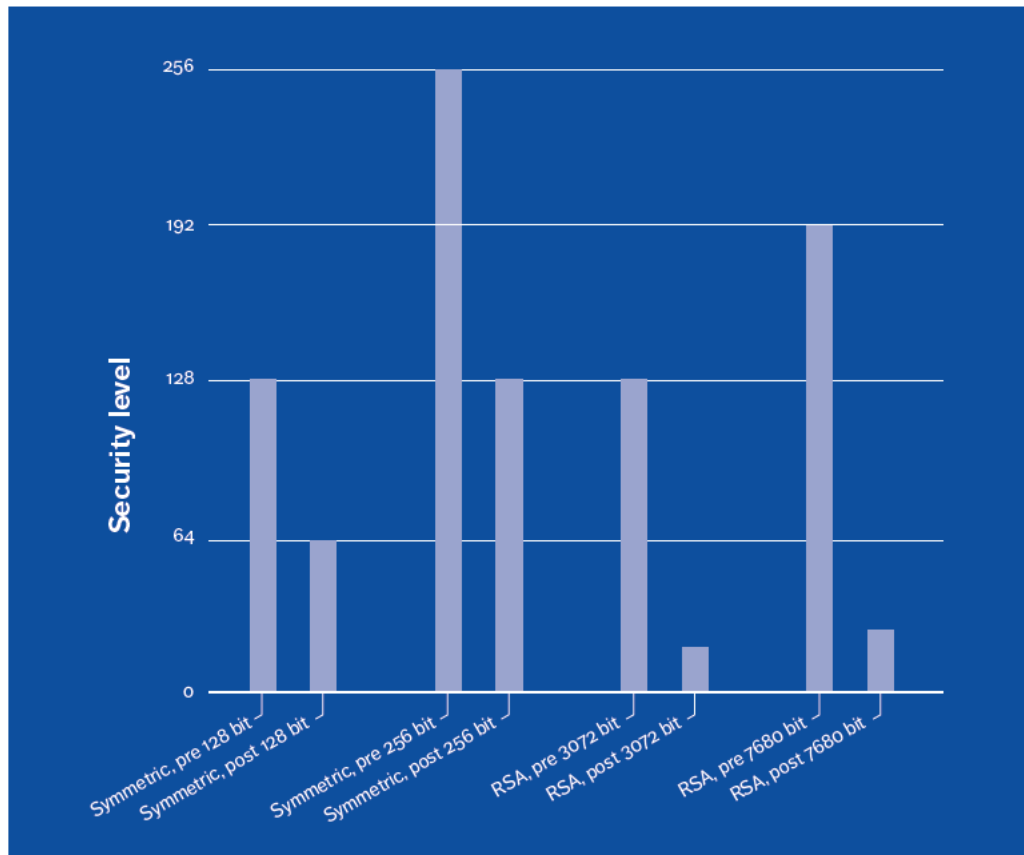


Figure 6:
Relative complexities for breaking cryptographic algorithms
before quantum computers and post-quantum computers

in key size, so they can cope with the potential capability of quantum computers. However, just as advances have been made in the fields of computer engineering and algorithm design over the past half-century, developers may well bring us new cryptographic schemes that will change the security landscape dramatically.

Summary

Concerns about security and privacy now rank among the ICT industry's top priorities. For Ericsson, overcoming these concerns is a non-negotiable element of the Networked Society. The world is heading in the direction of comprehensive protection of data (in transit and at rest), where encryption techniques are not just reserved for access networks, but are applied across the entire communication system. This, together with new, more complex communication services places new

demands on cryptography technology.

New cryptographic algorithms such as AEAD and ECC overcome the performance and bandwidth limits of their predecessors, in several cases offering improvements of several orders of magnitude. On the protocol side, TLS 1.3 and QUIC significantly reduce latency, as they require fewer round trips to set up secure communications.

Homomorphic encryption may create new business opportunities for cloud-storage providers. Should quantum computers become a reality, the future challenge will be to replace many established algorithms and cryptosystems. Ericsson has a deep understanding of applied cryptography, its implications, and the opportunities it presents for the ICT industry. We actively use this knowledge to develop better security solutions in standardization, services, and products, well in advance of their need in the world. ●

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most aspects of network and information security, making contributions to various standards (3GPP/ETSI, IETF, ISO, CSA). He has taken part in external research collaborations such as EU FP7 ECRYPT (Network of Excellence in Cryptography). He is also a very active inventor, and was a recipient of Ericsson's Inventor of the Year Award in 2009. Recently, he was appointed adjunct professor at KTH in the area Network and System Security.

Ben Smeets

◆ is a senior expert in Trusted Computing at Ericsson Research in Lund, Sweden. He is also a professor at Lund University, from where he holds a Ph.D.



in information theory. In 1998, he joined Ericsson Mobile Communications, where he worked on security solutions for mobile phone platforms. His work greatly influenced the security solutions developed for the Ericsson mobile platforms. He also made major contributions to Bluetooth security and platform security-related patents. In 2005, he received the Ericsson Inventor of the Year Award and is currently working on trusted computing technologies and the use of virtualization.

The authors greatly acknowledge the support and inspiration of their colleagues Christoph Schuba, Dario Casella, and Alexander Pantus

Microwave backhaul

GETS A BOOST WITH
MULTIBAND

JONAS EDSTAM

Is there a spectrum shortage? The answer to the question is both yes and no; in some locations spectrum is severely congested, while in other places it is highly underutilized. As the performance level demands on services like mobile broadband continue to rise, networks are going to need some innovative tools. New methods that will maximize spectrum efficiency, and new technologies that can exploit unused spectrum are going to be needed. Multiband booster is one such method. This concept fundamentally shifts the way spectrum can be used, with a promise to deliver a massive improvement in the performance levels of microwave backhaul, while at the same time accelerating the much needed shift toward the use of higher frequency bands.

TECHNOLOGY EVOLUTION, increased mobility, and massive digitalization continue to place ever more demanding performance requirements on networks — a trend that shows no signs of leveling off. As the dominant backhaul media in today's networks, microwave plays a significant role in providing good mobile network performance. However, the constant pressure to increase performance levels translates into a need for more spectrum, and more efficient use of it — not just when it comes to radio access, but for microwave backhaul as well.

■ As a finite natural resource, radio spectrum is governed by national and international regulations to ensure that social and economic benefits are maximized. Spectrum is divided into frequency bands that are allocated to different types of radio services, such as communication, broadcasting, radar, as well as scientific use. Allocation is based on propagation characteristics, which vary with frequency. Lower frequencies, for example, enable radio signals to be transmitted over longer distances, and can penetrate building facades. Higher frequencies, on the other hand, are more limited in terms of reach and coverage, but they can generally provide wider frequency bands, and as such have high data-carrying capacities. Driven by growing communication needs, ever higher frequencies have been taken into use over the past few decades. Historically, microwave backhaul has used much higher frequencies (from about 6GHz to 86GHz) than mobile radio access, which today uses spectrum ranging from about 400MHz to 4GHz. For 5G radio access, research is currently underway on the use of much higher frequencies (above 24GHz). The

findings of this work will be presented at the next ITU World Radiocommunication Conference, due to be held in 2019 (WRC-19) [1].

By 2020, 65 percent of all cell sites (excluding those in Northeast Asia) will be connected to the rest of the network using microwave backhaul technology [2]. Between now and then, the performance of microwave backhaul will continue to improve, supporting growing capacity needs through technology evolution and more efficient use of spectrum. The decision-making process used to establish what media can best provide backhaul to a given site will also change; it will no longer be determined by capacity needs, but rather which solution — fiber or microwave backhaul — provides the lowest total cost of ownership (TCO).

Multiband solutions, which enable enhanced data rates by combining resources in multiple frequency bands, already constitute an essential part of modern radio access systems. Their significance will, however, increase in the coming years, as they enable efficient use of diverse spectrum assets, and as such will support the evolution of LTE and 5G technologies.

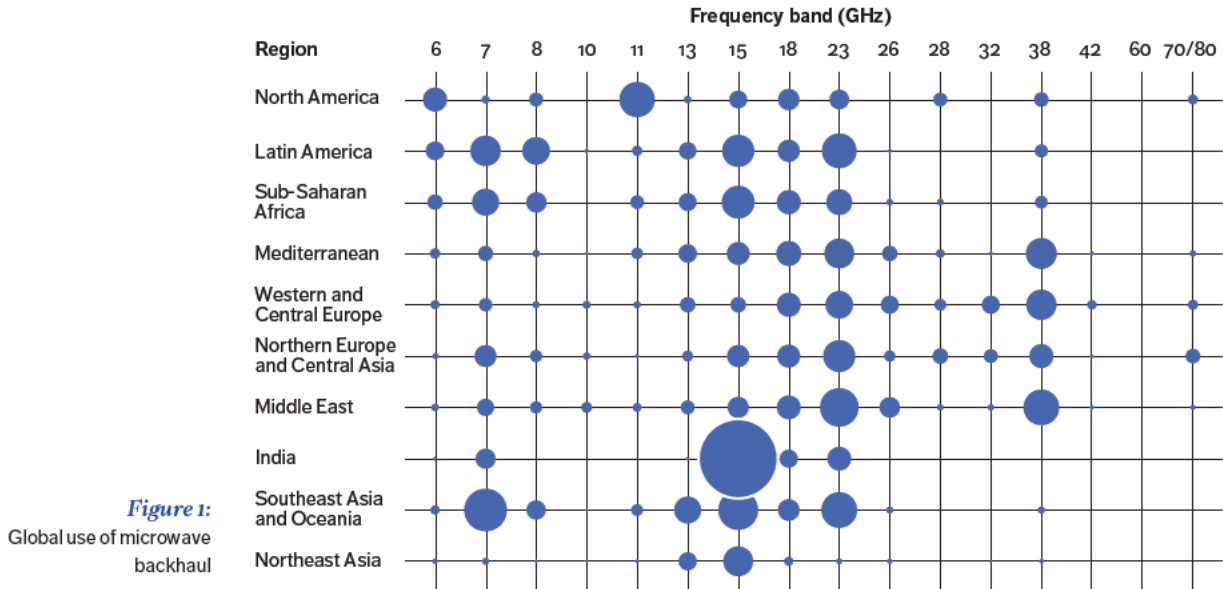
The question today, however, is how to exploit the multiband concept for backhaul. And how can a holistic view enable more efficient use of diverse backhaul spectrum assets.

Use of spectrum for backhaul

Spectrum in different frequency ranges is used by backhaul solutions to support communication in many types of locations, from sparsely populated rural areas to ultra-dense urban environments. Globally, about 4 million microwave backhaul hops are in operation today. [Figure 1](#) illustrates the extent of microwave backhaul usage by region and band — the size of each circle is relative to the

Terms and abbreviations

PDH – Plesiochronous Digital Hierarchy | **QAM** – quadrature amplitude modulation | **SDH** – Synchronous Digital Hierarchy



Source: Ericsson 2015

number of microwave hops in operation. Which frequency band is used varies greatly from one place to the next, because the most appropriate band is chosen depending on regional climate and national spectrum regulations [3]. Other factors like inter-site distance, target performance requirements, and fiber penetration are also taken into consideration when selecting the backhaul frequency band that best fits a given location.

As capacity needs have grown, the use of spectrum has shifted. Higher, previously less utilized frequencies have grown in popularity. About a decade ago, new 26GHz, 28GHz, and 32GHz bands were introduced, and since then, the use of these bands to support LTE backhaul has become popular in parts of Europe, Central Asia, the Mediterranean, and the Middle East. The older 38GHz band is quite popular in these regions, and its attractiveness is currently growing in the rest of the world. The newer 70/80GHz band is today gaining popularity [2, 4], as

it offers wide spectrum and channels alike, enabling capacities in the 10Gbps range over a few kilometers.

Looking to the future, industry has an interest in the use of frequencies above 100GHz, as they will enable capacities in the 40Gbps range over hop distances of about a kilometer [2].

Technologies are being investigated [5], and regulatory studies are examining channel arrangements and deployment scenarios in the 92-114.5GHz, and 130-174.7GHz frequency ranges, commonly referred to as the w- and d-band for microwave backhaul [6].

Unfortunately, the use of spectrum is unbalanced: hotspots occur in bands that are heavily used, while there are large geographical areas with untapped spectrum in all frequency bands.

Microwave backhaul technology

Unlike the various generations of radio access technology (2G, 3G, and 4G), there is no formal

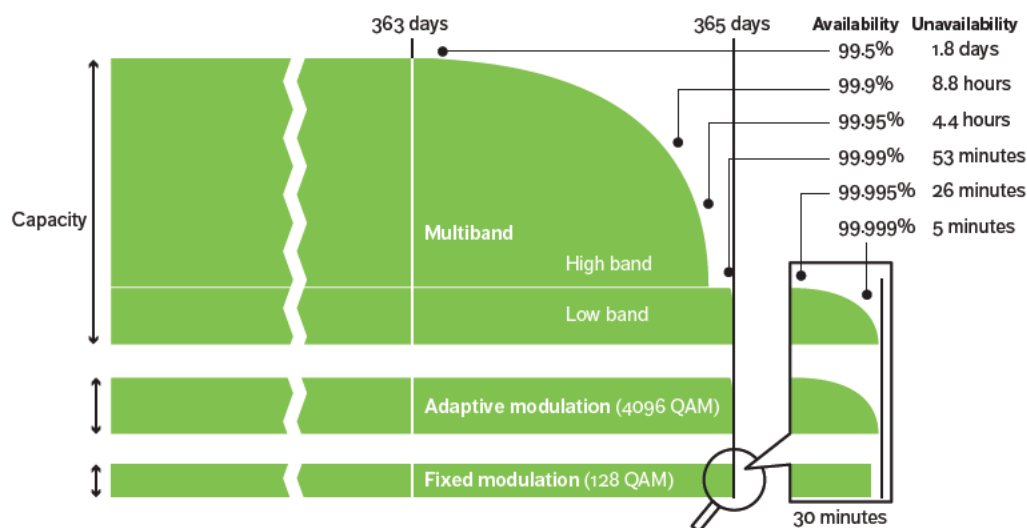


Figure 2:
Evolution of microwave
backhaul technology

classification for microwave backhaul technology evolution. Nevertheless, its performance has improved tremendously over the past few decades with the introduction of innovative technologies and enhanced features [2, 7, 8].

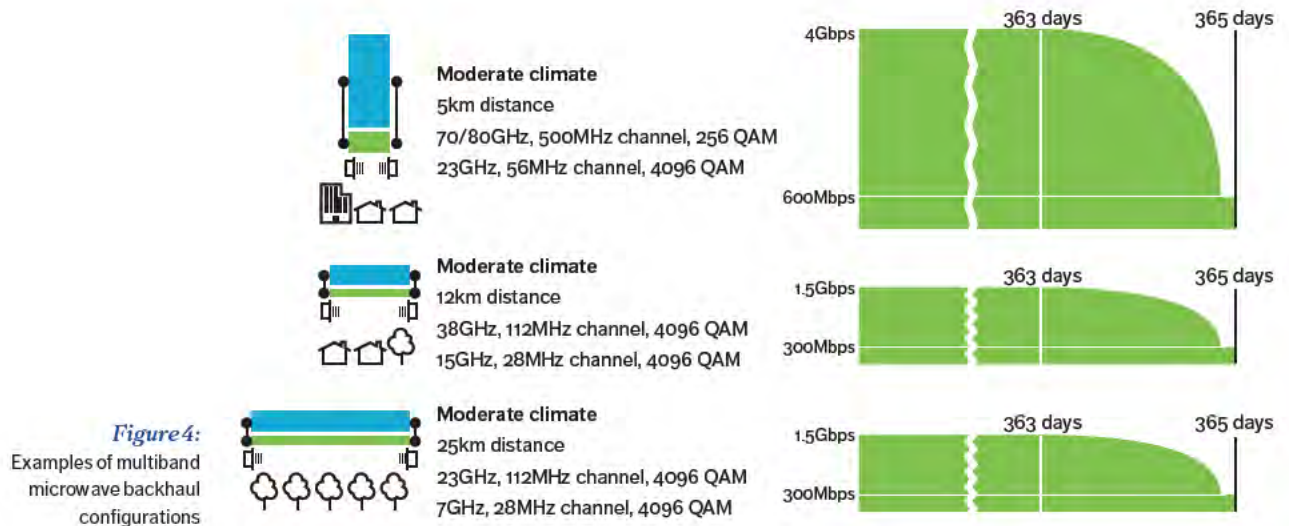
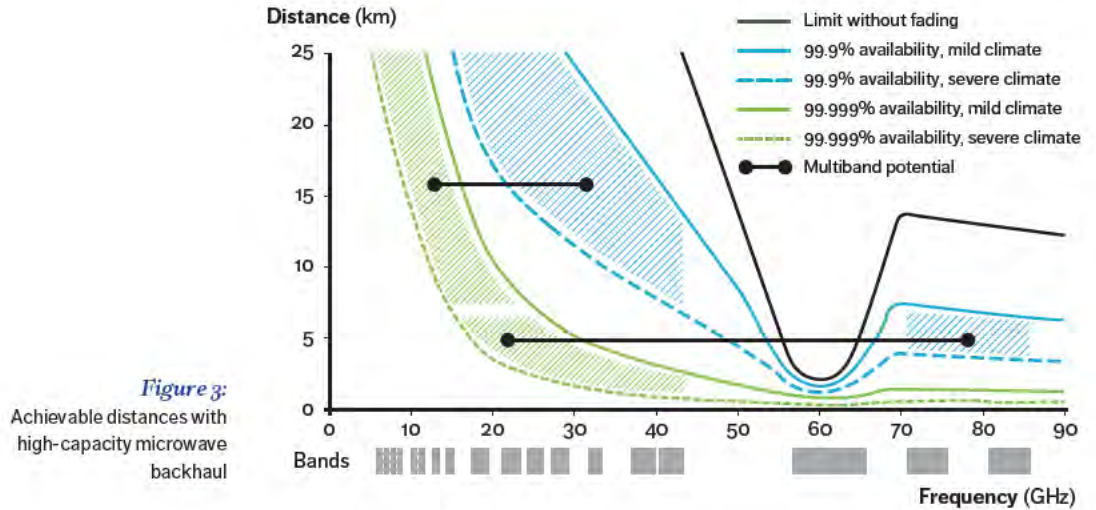
One issue that in some way characterizes microwave backhaul is the impact on signal strength of adverse propagation effects, such as those caused by rain. Planning and dimensioning of microwave links need to be carried out using recommended propagation prediction methods and long-term statistics, to ensure that targeted service availability (the ratio of actual service provided to the targeted service level, measured over 365 days, and expressed as a percentage) can be secured [9].

Originally, microwave supported PDH and SDH transport using fixed modulation designed for a service availability of up to 99.999 percent (five-nines availability), which allows for five minutes of total outage in a year.

Since then, adaptive modulation has been introduced for packet transport: a technique that is now well established, and supports extreme order modulation with up to 4096 QAM. Adaptive modulation maximizes the bit-error-free throughput under all propagation conditions. It can be configured to provide guaranteed capacity for high-availability services, and still provide more than double the capacity with somewhat lower availability, as illustrated in [Figure 2](#).

Multiband booster for backhaul

Radio-link bonding is a well-established method for microwave backhaul, enabling multiple radio carriers to be aggregated into a single virtual one [7] — somewhat similar to carrier aggregation in radio access. Bonding not only enhances peak capacity, it also increases effective throughput by using statistical multiplexing. Since its introduction, the technology has evolved continuously, supporting



ever-higher capacities and more flexible carrier combinations. So far, focus has been on bonding carriers within the same frequency band. The beauty of the multiband booster concept lies in the fact that it uses radio-link bonding to aggregate carriers in different frequency bands, enabling the full spectrum potential to be unleashed.

Wider channels are easier to obtain at higher frequencies, but as rain attenuation increases with frequency, availability drops for a given distance. Multiband booster overcomes this issue by bonding a wide high-frequency channel with a narrow low-frequency channel, as illustrated in [Figure 2](#). The resulting combination provides the best of both channels, giving higher capacities over much longer distances — drastically changing the way spectrum can be used for backhaul. Multiband booster brings about a huge increase in performance, and introduces a high degree of flexibility into the design of the backhaul solution. Ultimately, it enables the performance and availability requirements for different services to be met.

How different microwave backhaul frequency ranges can be used is to a large extent determined by propagation properties [9]. As rain attenuation and free-space losses increase with frequency, the achievable hop distance at higher frequencies is limited. The maximum distances for high-capacity microwave are shown in [Figure 3](#) for different climates and levels of availability. The mild climate has a rain zone of about 30mm per hour (rain rate exceeded for 0.01 percent of the year), and is typical for large parts of Europe. The severe climate is for a rain zone of about 90mm per hour, which is typical for India. The availability targets in this example are set for half the maximum link capacity, which corresponds to 64 out of 4096 QAM in the 6-42GHz range, and to 16 out of 256 QAM in the 60GHz and 70/80GHz bands. The full link capacity has lower availability, but is maintained for most of the year. For applications that require lower capacities, longer distances can be achieved using lower modulation levels. Figure 3 also shows the limit for maximum modulation without fading, still including free-space loss and atmospheric attenuation. The

oxygen absorption peak, which occurs at around 60GHz, severely limits hop distance — this phenomenon is clearly illustrated by the dip in the curve.

The width of a frequency band generally scales with frequency; the higher the frequency, the more bandwidth it offers. Backhaul frequency bands can be roughly categorized into three frequency ranges:

- » 6-15GHz bands with an average of 750MHz per band
- » 18-42GHz bands with an average of 2.2GHz per band
- » 70/80GHz band that is 10GHz wide.

For a given hop distance, the typical multiband combinations that would boost capacity are illustrated in Figure 3. They include: 18-42GHz bands bonded with the very wide 70/80GHz band for hop distances of up to about 5km; or the narrow 6-15GHz bands bonded with the wider 18-42GHz bands for longer hop distances. The multiband solution is, however, highly flexible, and any locally available frequency combinations that meet the targeted performance can be used.

Boosting backhaul performance

The multiband booster is an excellent tool for upgrading the capacity of microwave backhaul networks up to tenfold. [Figure 4](#) shows three different multiband examples with typical hop distances found in different parts of the network — in suburban areas hops tend to be a few kilometers long, and tens of kilometers in remote rural regions. Examples are given for a moderate climate, with a rain zone of about 60mm per hour (which is typical for places like Mexico). Bearing in mind that these configurations are just examples, the multiband booster provides a highly flexible way to bond different carrier and frequency band combinations. Combining different frequency bands makes it possible to get more out of available spectrum,

AS RAIN ATTENUATION AND FREE-SPACE LOSSES INCREASE WITH FREQUENCY, THE ACHIEVABLE HOP DISTANCE AT HIGHER FREQUENCIES IS LIMITED.

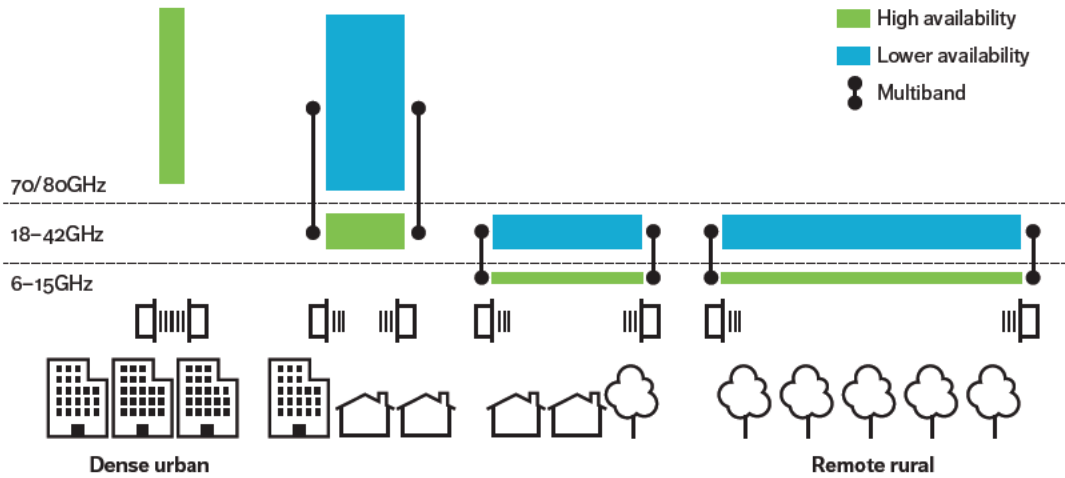


Figure 5:
Efficient use of
microwave backhaul
spectrum

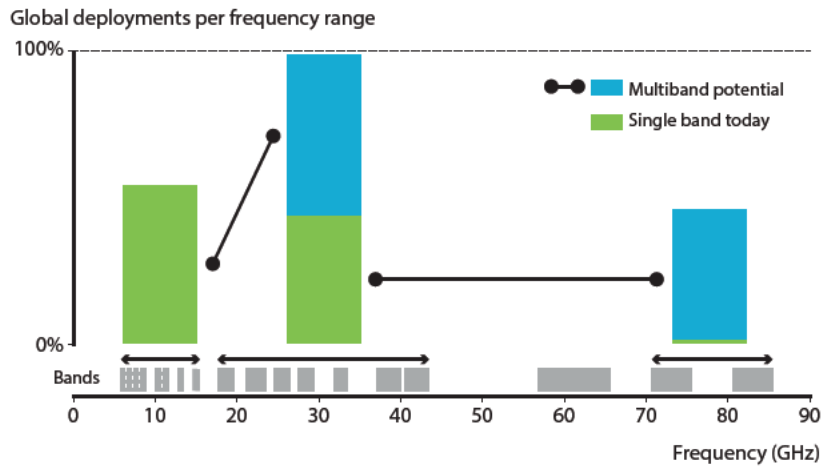


Figure 6:
Increased use of high
frequencies with multiband
microwave backhaul

and so help networks meet the performance and availability requirements of future services.

Unleashing spectrum potential

It is clear that if networks are to meet future performance requirements, efficient use of spectrum is essential. There are, however, many different aspects to spectrum efficiency, and the level that can be achieved directly depends on the local deployment density and topology of microwave hops.

Microwave backhaul performance is, for the most part, determined by the propagation properties of different frequencies. At higher frequencies, rain attenuation and free-space losses are greater, while antenna size drops for the same antenna gain. Best practice dictates using the highest frequency band possible that can still meet the availability and performance objectives for a given link distance. This approach preserves the lower frequency bands for use by greater link distances. In many countries, regulatory incentives promote the use of higher bands through lower spectrum-licensing costs, and by imposing policies that dictate minimum hop-length distances.

Increasing the spectral efficiency of a link can be achieved by using higher-order modulation, but this comes at the cost of increased sensitivity to interference — which may in turn limit the use of more extreme order modulation in local hotspots. Consequently, the significance of interference mitigation technologies, like super high performance antennas (Class 4 [10]), is growing.

Today, the use of higher frequency bands is limited to shorter hop lengths, which tend to be most common in urban environments. As a result, higher-frequency spectrum is seldom used outside these areas. Clearly such biased usage is inefficient, and a significant amount of valuable spectrum remains untapped.

As *Figure 5* illustrates, the multiband booster enables higher backhaul frequency bands to be used over longer distances and much wider areas. The concept can be applied to advantage in all geographical areas, although different frequency bands are appropriate depending on the desired

hop distance. Wider channels should also be much easier to obtain in these less congested areas, further increasing the benefit of multiband solutions.

Regulatory authorities can apply different licensing models to encourage efficient use of spectrum, weighing in factors like frequency bands, geographic region, and local microwave hop density. Introducing and allowing wider channels in less deployed areas would further encourage the use of multiband solutions.

Future backhaul spectrum use

In most geographical areas, hop distances are generally becoming shorter due to the densification of the macro cell network and introduction of small cells. Likewise, the distance to a fiber point-of-presence is dropping as fiber penetration increases.

As hop distances fall, the use of higher frequency bands rises. For example, use of the 70/80GHz band is growing significantly, and if the growth curve continues, will account for 20 percent of new deployments by 2020 [2]. Today, bands in the 26-42GHz range are predominately used in Europe, the Mediterranean, Central Asia and Middle East (see *Figure 1*), but use in other regions is beginning to show signs of growth.

Figure 6 shows the relative amount of single-band microwave hops in global operation today in the 6-15GHz, 18-42GHz, and 70/80GHz frequency ranges (see also *Figure 1*). The multiband booster is a highly attractive solution to enhance performance for microwave backhaul. Upgrading existing single-band microwave links to multiband solutions will result in the accelerated use of higher frequency bands, as illustrated in *Figure 6*.

Denser networks, increasing performance needs, and new efficient technologies, such as multiband booster, will all lead to a dramatic increase in the use of the 70/80GHz band, as well as a large increase in the use of bands in the 18-42GHz range.

Summary and conclusions

The performance of microwave backhaul has evolved continuously with new and enhanced technologies and features that make ever better

use of available spectrum [2, 7, 8]. Today, microwave backhaul can provide fiber-like multi-gigabit capacity — even in locations where there is no direct line-of-sight [11].

Multiband solutions are essential for mobile systems, as they enable diverse spectrum assets to be used efficiently. The importance of these types of solutions for mobile communication will rise as LTE evolves and 5G becomes a reality. A number of years ago, we documented the benefits of adapting multiband for microwave backhaul in a previous article [7]. It's now time to fully exploit the concept.

Multiband booster provides a massive increase in the performance of microwave backhaul, and is an excellent tool that can increase network capacity up to tenfold. It supports flexible bonding of different carriers and frequency band combinations, enabling networks to meet the performance and availability requirements for future services. Multiband booster represents a paradigm shift toward much more efficient use of diverse backhaul spectrum assets, unleashing the use of higher frequencies over much wider geographical areas.

The technology evolution for spectrum — how it is used and how it is allocated — is moving fast, with many new innovations becoming available for both radio access and microwave backhaul. Regulatory authorities are carefully considering the current and future use of frequency bands, not only for mobile systems but also for microwave backhaul.

As networks become denser, and performance needs grow, new efficient technologies, like the multiband booster, will dramatically increase the use of the 70/80GHz band, as well as the bands in the 18-42GHz range. To support evolving technology, and ensure good backhaul performance, regulatory incentives that promote efficient and holistic use of backhaul spectrum are key. ●

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THE AUTHOR

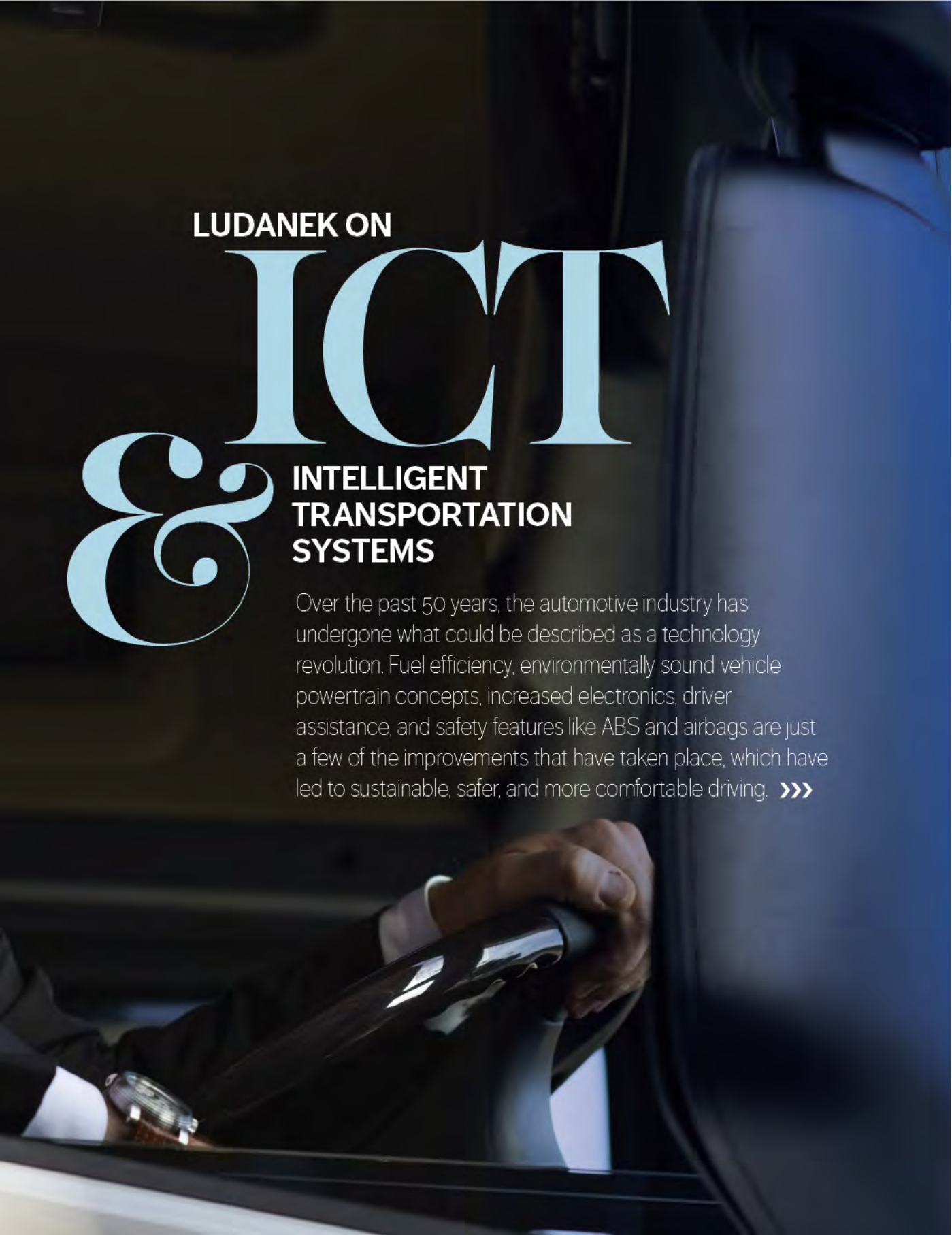
**Jonas Edstam**

◆ joined Ericsson in 1995, and is responsible for technology strategies and industry-wide collaborations at Product Area Microwave Networks, Business Unit Radio. He is an expert in microwave backhaul networks, having more than 20 years of experience in this area. Throughout his career, he has fulfilled various roles, working on a wide range of topics including detailed microwave technology and system design. His current focus is on the strategic evolution of mobile networks and wireless backhaul to 5G. He holds a Ph.D. in physics from Chalmers University of Technology, Gothenburg, Sweden.

A portrait of Harald Ludanek, a middle-aged man with short grey hair and glasses, wearing a dark suit, white shirt, and a dark red tie. He is smiling and leaning against a dark blue vertical structure, possibly part of a vehicle. The background is dark and out of focus.

“BY CONNECTING VEHICLES
AND COMBINING THE VALUABLE
DATA THEY TRANSMIT WITH
INFORMATION ABOUT THEIR
ENVIRONMENT, WE CAN CREATE
A PLATFORM THAT CAN HELP
IMPROVE TRAFFIC FLOW AND
INCREASE SAFETY”

—Harald Ludanek



LUDANEK ON

ICT

& INTELLIGENT TRANSPORTATION SYSTEMS

Over the past 50 years, the automotive industry has undergone what could be described as a technology revolution. Fuel efficiency, environmentally sound vehicle powertrain concepts, increased electronics, driver assistance, and safety features like ABS and airbags are just a few of the improvements that have taken place, which have led to sustainable, safer, and more comfortable driving. >>>

HARALD LUDANEK
Executive Vice President
and Head of Research
and Development, Scania
CV AB

TODAY, WE ARE in the era of connectivity. Vehicles are no longer isolated entities moving from one place to another, but are an intricate part of a greater transportation system. In the future, we can look forward to increased levels of comfort in vehicles, greater degrees of driver assistance, and more advanced safety features. To achieve this, we need to partner up and develop solutions together with a holistic and end-to-end approach. We need to learn from each other and share advancements in technology. Thankfully, today's industries are ripe for the collaboration that is needed to build integrated solutions. How Scania and Ericsson work today highlights just how greater we are together.

■ *How do you see the automotive industry evolving in the context of digitalization and mobility?*

Throughout its history, both the automotive industry and ICT have relied heavily on technology, standardization, continuous improvement, and not least R&D. New technologies are shaped by external influences and regulations, but the direction development takes is primarily determined by customer demand. The customers in my industry include a wide range of enterprises and individuals — from professional truckers and bus drivers, to regular citizens who need a vehicle to get around. The enterprise sector — including logistics, shipping, and tourism, for example — has a significant influence on the technological innovations we prioritize. Once again, clear similarities arise between my industry and ICT.

The technological advances that have taken place in the automotive industry, along with the developments that have come about in a number of tangential sectors like materials and electronics, and governmental regulations that have come into force, have shaped several waves of innovation (illustrated in [Figure 1](#)) over the past 65 years. The result of all of these developments is a safer, more efficient, and more comfortable driving experience.

The 1970s oil crisis has had a long-lasting

impact on the automotive industry all over the world, putting fuel-efficiency firmly at the top of our list of technological development priorities. The crisis led to a dramatic shift in R&D, as fuel-saving technologies, and more efficient engines became top priorities. The powertrain, for example, was improved with innovations like gasoline direct injection and start-stop systems, which, along with new lightweight vehicle materials, led to improved fuel consumption and fewer efficiency losses. These technologies are pretty much standard components in the vehicles being built today.

The 1990s were marked by the birth of mechatronics. The introduction of sensor technologies and affordable electronic control units (ECUs) led to the replacement of control and mechanical systems with electrical and electronically steered actuators.

The boom in the consumer electronics market began at the turn of the 21st century. User demand for new functionalities like navigational support systems, air bags, and driver assistance had to be met, and so the era of automotive electronics began.

Looking ahead, [Figure 2](#) illustrates some of the developments that drivers can look forward to. While today, development focus is on end-to-end resource management (during manufacturing, operation, as well as the end-of-life phase of a vehicle), in the future, we can look forward to much greater levels of driver assistance. The way I see it, manufacturing and production processes have undergone four revolutions, becoming more efficient with each one. In the beginning of mass production, engines were powered by steam, then electricity took over. Later on, computing power took control, and now the Internet of Things (IoT) has ushered in a whole new era of possibilities.

The fourth industrial revolution of production — which we refer to as Industry 4.0 — is not actually limited to the IoT, but encompasses other aspects like cybersecurity, big data analytics, and integration across traditional organizational boundaries. But, as more things become connected, the significance of each aspect rises. When people, for example, share their location data, a lot of information is generated.



Figure 1
Waves of innovation

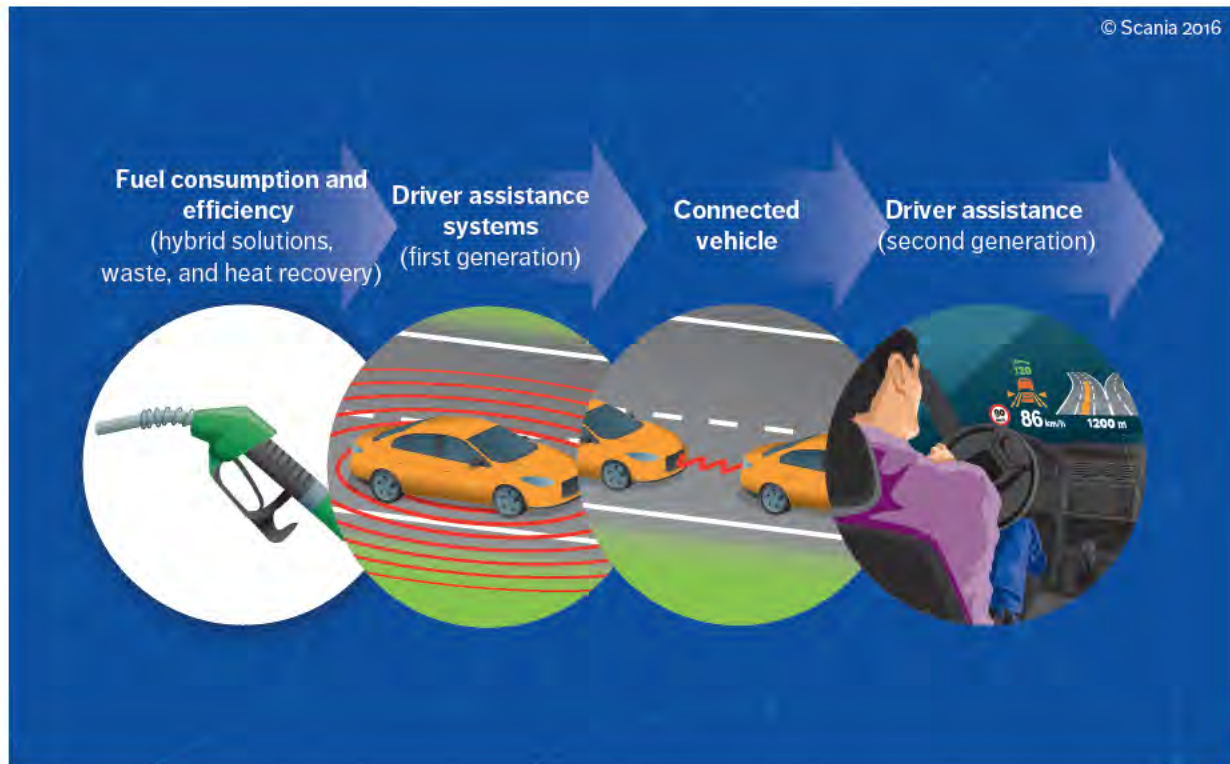


Figure 2
Evolution of technology

But to make any use of it, big data analytics must come in to play.

Today, we live in a world based on connectivity and digitalization. Individuals and enterprises alike are taking advantage of the capability to connect almost anything to a network, the possibility to make data available through the cloud, and the ability to mash massive amounts of data together to create an enriched understanding of everything, everyone, and every inch of space on the globe. The opportunities opened up by mobility and digitalization have enabled the automotive industry to create new functionalities and capabilities,

boosting efficiency and safety, while offering a higher level of comfort.

By connecting vehicles and combining the valuable data they transmit with information about their environment, we can create a platform that can help improve traffic flow and increase safety. In this new business model, the car manufacturer turns into a provider of mobility, and the truck manufacturer shifts into the transport management domain.

But throughout the whole process of transformation, digitalization, development, and connectivity, the automotive industry has remained true to its basic principle: to come up with ever

more efficient, and environmentally sound vehicle powertrain concepts.

What is your view on the intelligent transportation system (ITS), and what kind of technology evolution is required for it to be a business success?

Logistics, both in Europe and in the US, are well-developed and optimized systems. Europe spends about 8.2 percent of its GDP on the transportation of persons and goods: below US spending of about 9.4 percent. China, however, spends 18 percent of its GDP on logistics. That China's relative spend is around double that of Europe/US is an indication of the degree to which European/American transportation systems (a fundamental parameter for a successful economy) have been optimized. Yet despite these seemingly positive figures, 24 percent of trucks run empty, and transportation utilization capacity is just 54 percent – highlighting a common issue shared by cellular and transportation networks.

In theory, utilization of transportation systems could be raised to about 85 percent. Achieving such a level, however, would require improved flow control and a connected system that incorporates order, supply, as well as all the transportation partners. In short, what we need is an ITS that can connect the various stakeholders to each other.

These stakeholders include suppliers, infrastructure owners, society, and logistics providers. An additional challenge for the ITS is the global trend toward urbanization. Transportation of goods and persons across bustling city centers is a key element of modern urban logistics. However, implementing an ITS to cope with our complex city structures requires state-of-the-art connectivity, as well as new business and governance models that give due weight to the needs and wishes of all stakeholders.

The Integrated Transport and Research Lab (ITRL) at KTH Royal Institute of Technology was established to address this very issue. Here, under one collaborative umbrella, Scania, Ericsson, and KTH have begun to develop innovative and holistic technical solutions to address global environmental

transport challenges, by taking a long-term and multidisciplinary approach to the matter (as illustrated in [Figure 3](#)). As partners, we are working together to develop seamless transportation services for use within modern infrastructures, novel vehicle concepts, as well as new business models and policies — all of which need to be tuned and optimized.

What are the key use cases and connectivity requirements for ITS/ICT?

Fundamentally, the future ITS needs to be able to deliver economical and ecological benefits to everyone and everything it encompasses. This includes commuters and drivers, enterprises (like shipping companies and couriers), and the organizations that control them (like transportation operators). Scania's aims and commitment lie in the development and delivery of customized solutions for sustainable transportation. In this context, our aim is not only to satisfy the needs of our direct customers (such as trucking companies), but also those of the people and enterprises that use our solutions daily as they commute to work, travel around, or ship goods from one place to another.

To develop the future ITS, we need to identify the opportunities for improvement from a holistic point of view, so the overall solution can be integrated in the logistics chain end-to-end. The ICT industry is a fundamental enabler in this chain, as it provides the vital ingredient of connectivity, allowing the various transportation stakeholders to connect. A key element for the future system is guaranteed and controlled data security, with defined access and handling responsibility. Together with the users of transportation, the technology provider for the connected infrastructure needs to develop methods and techniques that will provide the right level of security and the right tools for access and responsibility.

 FUNDAMENTALLY, THE FUTURE ITS NEEDS TO BE ABLE TO DELIVER ECONOMICAL AND ECOLOGICAL BENEFITS TO EVERYONE AND EVERYTHING IT ENCOMPASSES 

TO GAIN THIS GLOBAL, LONG-TERM PERSPECTIVE, SPECIALISTS WITH DIFFERENT KINDS OF EXPERTISE WORKING IN A VARIETY OF INDUSTRIES NEED TO BE ABLE TO COME TOGETHER AND COLLABORATE ON POSSIBLE USE CASES.

The key ICT technologies are mobility, broadband, and cloud. Will they all be adopted by the ITS?

Yes, these key technologies will be adopted by ITSs with permanent availability and a high level of security. The connectivity requirements for transportation are vastly different from other applications such as providing connectivity to

consumers, say, or remote operation of machinery in an underground mine or on a construction site. The demands of ITS in terms of availability and security, for example, are high. And while the latency of the link for communication with response services needs to be low, it needs to be even lower for haptic systems, where the controller needs instant feedback — such as is the case for telesurgery.

What are the greatest opportunities and challenges involved, and what specific kind of security technology is needed?

Cars, trucks, buses, trains, and even people will deliver high volumes of data — including information on location, a given traffic situation, speed, and weather — to cloud computing centers over a broadband connection. All this data needs to be mashed with information delivered by other stakeholders in the transportation system to create a holistic view of the flow of people and all modes of transportation in a given geographical area. Fast, intelligent analytics are needed to assess the aggregated data, and offer an overall view before a real-time transportation flow control can be carried out. In the future, transportation systems — both within and outside urban conglomerations — will become highly dependent on analytics, so the failure or incorrect results of data mining will risk collapsing the entire value chain.

In my opinion, a two-step approach should be taken to providing a solution. First, players like

Scania should develop solutions with partners in ICT. Second, a cloud architecture and a data infrastructure are needed to test use cases for a wide variety of applications around the world, considering different countries, and including cross-border scenarios.

The telecom industry has created a scalable and cost-effective technology platform that provides connectivity to over 7 billion people. How do you think this platform is relevant to your industry?

Connectivity and the telecom network are essential components of the future intelligent transport system and Industry 4.0. As a scalable architecture technology, connectivity and networks provide a cost-effective platform that can support the rapid development of new use cases and innovative applications, which reflect the intensifying demands of users. In China (the second-largest growing telecoms market), for example, more than 7 million new mobile subscriptions were established in Q3, 2015. For the same period, more than 13 million subscribers were created in India*. So, the ICT and transportation industries need to adopt a long-term global perspective; we need to know how to organize the data, and how to analyze it to avoid self-accelerating and uncontrolled data mining.

To gain this global, long-term perspective, specialists with different kinds of expertise working in a variety of industries need to be able to come together and collaborate on possible use cases. Given this fundamental requirement, Scania and Ericsson are ideal strategic partners; we can conduct the necessary research, using concrete use cases, as well as considering the demands of the entire problem space. This is in direct contrast to the traditional way of working, where each industry player developed their part of the solution in isolation.

What is the impact of big data and analytics? Can you share your views on the information model for the automotive industry, for example?

*Ericsson Mobility Report
<http://www.ericsson.com/mobility-report>



Figure 3:
Vision of a sustainable transportation system

In the automotive industry, we distinguish information models from each other on the basis of function, such as driving support, and intelligent driving functions. Each model brings with it tougher demands for both security and availability. Applications that use vehicle-to-vehicle (V2V) or vehicle-to-infrastructure (V2X) communication, for example, enable traffic to be organized in a safer manner than it is today, but they are demanding in terms of security, availability, and latency. Real-time, secure information received from traffic control signals, from sensors on the road that can detect obstacles, or from potentially hazardous situations — such as road accidents — and other systems can be used as input to the warning strategy for predictive automatic influence for car, bus and truck drivers.

Today's trucks come complete with an onboard camera system that works in conjunction with the automatic emergency braking (AEB) system. Images from the camera system can be combined with radar information, taken from sensors permanently fixed to the front of the truck. The technology needed to share the data collected by these systems with traffic control towers already exists. However, before it can be put to wide-scale use, a number of questions need to be answered, such as: how to regulate responsibility and security, and how to handle additional data analysis. Some sort of a breakthrough is needed — either by creating a standalone solution, or by creating a solution together with a communication provider and other stakeholders, in an organized and regulated manner. Drivers — both private and professional — are expecting massive improvements in terms of comfort and fuel options, as well as better functionality when it comes to increased automation in vehicles.

How should we push for a collaboration between the ICT and automotive industries in terms of innovation, both from a technological and a business model / best practices standpoint?

The question here also contains the answer. We need a neutral and independent test arena in which to develop use cases and build cooperation as

partners. The Connected Mobility Arena (CMA) project in Kista, Stockholm is just such a test arena. And so, within the context of the ITRL, the next step is to define the operating environment needed. Other cooperation areas will include the autonomous operation of mining equipment, which will require the integration of additional partners.

How and where should we collaborate on standardization, interoperability, and regulatory issues to create a system of systems?

Scania has many years of experience in building customized buses and trucks using its modular kit system. Each construction kit includes a set of smart, well-defined interfaces between the different component parts, and various performance steps. This component box/interface/API approach could be the basis of a solution that would fit well with Ericsson's approach to customized connectivity, based on network slicing; both approaches being firmly rooted in standards and best practices.

Our teamwork looks set to create a whole new ecosystem in terms of safety, resource management, and comfort. I am proud to be part of it and glad to have Ericsson on board. 🙌

AUTHOR

**Dr. Harald Ludanek**

Executive Vice President, Research and Development, Scania CV, Södertälje

◆ Attending the Clausthal University of Technology as a postgraduate engineer, German-born Harald Ludanek chose rotor dynamics and mechanical vibrations as the topic for his PhD thesis.

Today, he maintains a keen interest in technology both at work and at home. He has a few science-based hobbies, as well as a love of gardening, guitar playing, and handcrafts. But it is doubtless that it is his undying passion for engine mechanics that really drives him, and he applies this passion daily in his job as Head of R&D for Scania in Södertälje, Sweden.

He also has a fervent interest in cultivating collaboration between Scania and other key players — both within and outside the automotive industry. He is constantly

on the lookout for companies to collaborate with, for the benefit of all partners and ultimately all vehicle drivers. He believes that creating efficiencies will help to hit emissions targets, and minimize environmental impact.

IN THE PAST, Scania's development of robust, practical, reliable technology has been boosted by collaborations with car companies like Porsche. Now, Ericsson is providing the connectivity that will one day enable the truck driver to have an office and a comfortable living space all in one: Ludanek's vision for the ultimate in cabin comfort.

How then has Ludanek mastered the tricks of the truck trade?

Early on, with a doctorate in engineering, he joined Volkswagen's Research Centre in 1992, moving on in 2000 to head up the global coordination of the company's 25 worldwide development centers.

In 2002, he became Head of Technical Development and member of the executive board at Škoda auto a.S. in the Czech Republic.

He then moved on in 2007 to head up Complete Vehicle Development and Prototyping at Volkswagen

AG until September 2012, when he was appointed Executive Vice President and Head of Research and Development at Scania.

SINCE 2011, he has chaired the supervisory board of the engineering consultancy IAV GmbH, Berlin, Germany and been a member of the supervisory board of the IMF TÜV Nord in Sweden.

Having come full circle since his student days, today he lectures in automotive management and technology at Clausthal University of Technology, where he is also a member of the supervisory board.

5G transport networks

FLEXIBILITY IN

THE KEY TO MEETING THE DEMAND
FOR CONNECTIVITY

The more people have been able to achieve while on the move, the more dependent society has become on mobile broadband networks. As applications like self-driving vehicles and remotely operated machinery evolve, become more innovative, and more widespread, the level of performance that 5G networks need to deliver will inevitably rise. Keeping pace with ever-increasing demand calls for greater flexibility in all parts of the network, which in turn requires tight integration between 5G radio, transport networks, and cloud infrastructures.

PETER ÖHLÉN
BJÖRN SKUBIC
AHMAD ROSTAMI
KIM LARAQUI
FABIO CAVALIERE
BALÁZS VARGA
NEIVA FONSECA
LINDQVIST

ADVANCES IN TECHNOLOGY and a shift in human behavior are influencing how 5G networks are shaping up. With 3G, things got faster, data volumes surpassed voice, new services were developed, and people started using mobile broadband. With 4G, mobile broadband soared. Today's networks provide advanced support for data. Building on this success, 5G aims to provide unlimited access to information and the ability to share data

anywhere, anytime by anyone and anything. So, as we move deeper into the Networked Society, the connections that link things and people will become almost exclusively wireless.

■ Services like mobile broadband and media distribution will continue to evolve in line with our growing global dependence on connectivity. Networks will experience huge increases in traffic and will need to service an ever-expanding number

of connected devices — both massive MTC (IoT) and mission-critical MTC. The latter sets stringent requirements for performance characteristics like reliability and latency.

The digital and mobile transformations currently sweeping through industries worldwide are giving rise to innovative cross-sector applications that are demanding in terms of network resources. And so, 5G networks will not only need to meet a wide range of requirements derived from user demand and device development; they will also need to support advanced services — including those yet to be developed.

Limitless innovation in application development, device evolution, and network technology are shifting from a model that is operator steered to one that is user driven. Flexibility and operational scalability are key enablers for rapid innovation, short time to market for deployment of services, and speedy adaptation to the changing requirements of modern industry.

How will future networks evolve?

To ensure that networks will be able to cope with the varied landscape of future services, a variety of forums like NGMN, ITU-R, and 5G PPP are working on the definition of performance targets for 5G systems [1].

In comparison with 2015 levels, the performance projections that will have most impact on transport networks are:

- » 1000x mobile data volume per geographical area, reaching target levels of the order of Tbps per sq km
- » 1000x the number of connected devices, reaching a density of over a million terminals per sq km
- » 5x improvement in end-to-end latency, reaching to as low as 5ms — as is required by the tactile internet.

However, the maximum levels of performance will not all apply at the same time for every application or service. Instead, 5G systems will be built to meet a range of performance targets, so that different services with widely varying demands can be deployed on a single infrastructure.

Getting networks to provide such different types of connectivity, however, requires flexibility in system architecture.

Aside from meeting the stringent requirements for capacity, synchronization, timing, delay, and jitter, transport networks will also need to meet highly flexible flow and connectivity demands between sites — and in some cases even for individual user terminals [2].

Emerging 5G radio capabilities and the convergence of radio access and wireless backhaul have triggered an uptake of fixed wireless technologies as a complement to fixed broadband [3]. With hybrid access 5G networks will be able to provide the increased capacity needed to handle peak traffic for residential users. As such, 5G radio will increasingly complement and overlap with traditional fixed-broadband accesses.

Terms and abbreviations

5G PPP—5G Infrastructure Public Private Partnership | **API**—application programming interface | **BB**—baseband | **CPRI**—Common Public Radio Interface | **CWDM**—coarse wavelength division multiplexing | **DWDM**—dense wavelength division multiplexing | **EPC**—Evolved Packet Core | **FTTH**—fiber-to-the-home | **MIMO**—multiple-input, multiple-output | **MPLS**—multi-protocol label switching | **MTC**—machine-type communication | **NFV**—Network Functions Virtualization | **NGMN**—Next Generation Mobile Networks | **NG-PON2**—next-generation passive optical network | **P**—router—provider router | **PE**—router—provider edge router | **PGW**—PDN gateway | **ROADM**—reconfigurable optical add/drop multiplexer | **SDN**—software-defined networking | **SLA**—Service Level Agreement | **UE**—user equipment

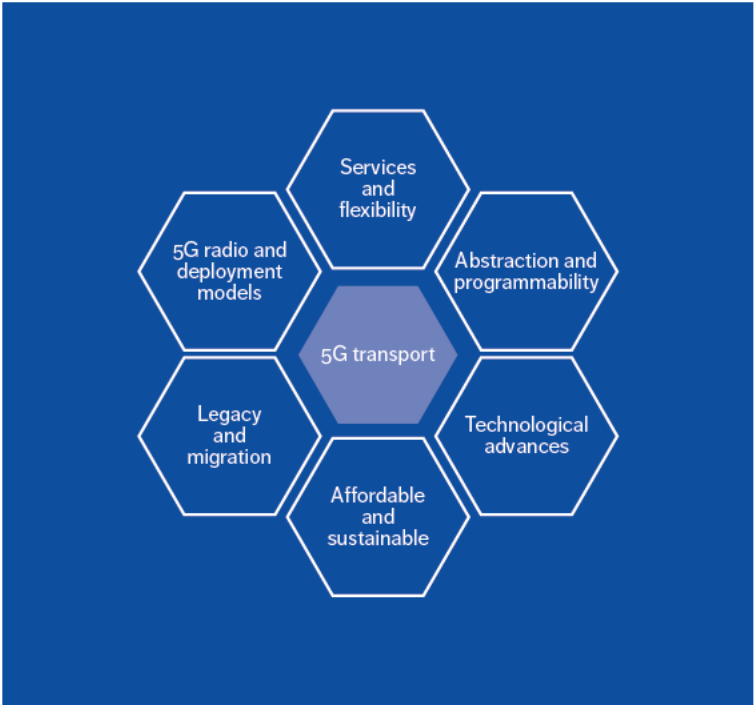


Figure 1
Landscape for 5G transport

The 5G transport network

As 5G radio-access technologies develop, transport networks will need to adapt to a new and challenging landscape, as illustrated in *Figure 1*.

Services

The expectations for 5G networks are high — providing support for a massive range of services. Industry transformation, digitalization, the global dependence on mobile broadband, MTC, the IoT, and the rise of innovative industrial applications all require new services, which has a considerable impact on the transport network. For example, a new radio-access model that supports highly scalable video distribution or massive MTC data uploading might require additional transport facilities — such as a scalable way to provide multicasting.

5G radio

How the 5G radio is deployed determines the level of flexibility needed in the transport network. Capacity, multi-site and multi-access connectivity, reliability, interference, inter-site coordination, and bandwidth requirements in the radio environment place tough demands on transport networks.

In 5G, traditional macro networks might be densified, and complemented through the addition of small cells. Higher capacity in the radio will be provided through advances in radio technology, like multi-user mimo and beamforming, as well as the availability of new and wider spectrum bands [4]. Consequently, the capacity of the 5G radio environment will reach very high levels, requiring transport networks to adapt. Not only will transport serve a large number of radio sites, but each site will support massive traffic volumes, which might be highly bursty due to the peak rate available in 5G.

For example, a UE that is connected to a number of sites simultaneously, may also be connected to several different access technologies. The device may be connected to a macro over LTE, and to a small cell using a new 5G radio-access technology. Multi-site and multi-RAT connectivity provides greater flexibility in terms of how UEs connect to the network and how E2E services are set up across

radio and transport. For example, allowing for efficient load balancing of UEs among base stations not only improves user experience, it also improves connection performance.

The impact of interference may favor deployment models where coordination can be handled more effectively. In small-cell deployments, UEs are often within reach of a number of base stations, which increases the level of interference, and at times requires radio coordination capabilities for mitigation. However, the method used for handling interference depends on how transport connectivity is deployed. In a centralized baseband deployment, tight coordination features, such as joint processing, can be implemented. In traditional Ethernet and IP-based backhaul, tight coordination requires low-latency lateral connections between participating base stations.

Centralized baseband processing tends to result in lower operational costs, which makes this approach interesting. However, it typically comes at the cost of high CPRI bandwidths in the transport network. The high bandwidth, together with stringent delay and jitter requirements, makes dedicated optical connectivity a preferred solution for fronthaul.

In 5G networks, the bandwidth requirements for fronthaul could be very high. The demand will be created by, for example, antennas for MU-MIMO and beamforming — which could use in the order of 100 antenna elements at each location. In combination with dense deployments and wider frequency bands (in the 100MHz range) traditional CPRI capacity requirements can quickly reach levels of several Tbps. A new split of RAN functionality is under investigation to satisfy requirements for cost-effective deployments and radio performance, while keeping capacity requirements on transport within a manageable range.

But some primary networking principles remain valid, such as timing and synchronization. Defining new packet-based fronthaul and midhaul interfaces requires the underlying network to include protocols and functions for time-sensitive transport services. Related standardization efforts are currently underway [5].

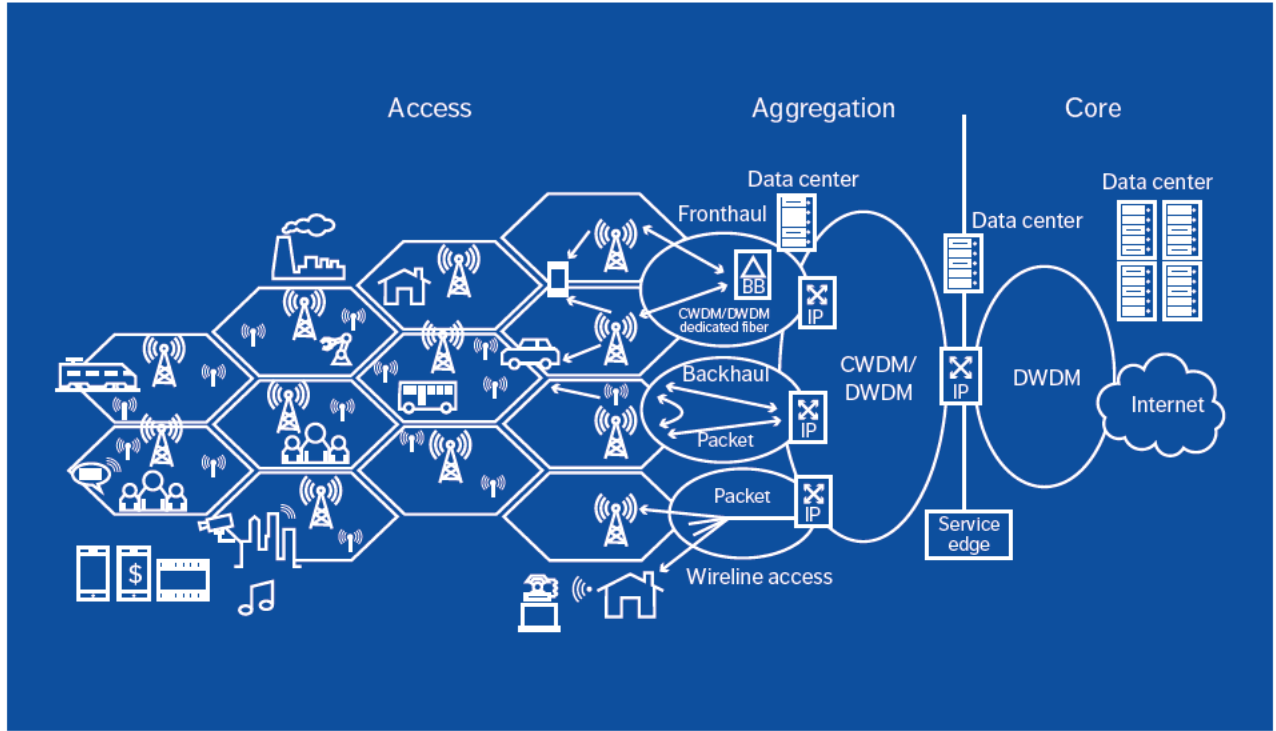


Figure 2
Main technology options to connect
RAN and transport infrastructure

Abstraction and programmability

Abstracting network resources and functionality, as well as managing services on-the-fly through programmatic APIs are the pillars of SDN, and the source of its promise to reduce network complexity, and increase flexibility.

With a new split in the RAN, some functions can be deployed on general-purpose hardware, while others, those closer to the air interface with strict real-time characteristics, should continue to be deployed on specialized hardware. Most of the

functions of the EPC will be deployed as software — following the concept of Network Functions Virtualization (NFV). Deploying network functions in this way makes it possible to build end-to-end network slices that are customized for specific services and applications. Each layer of the network slice, including the transport layer, will be designed to meet a specific set of performance characteristics.

The significance of network slices is best illustrated by comparing applications with different requirements. A network of sensors, for example,

requires the capability to capture data from a vast amount of devices. In this instance, the need for capacity and mobility is not significant. Media distribution, on the other hand, is challenged by large capacity requirements (which can be eased through distributed caching), whereas the network characteristics for remote-control applications based on real-time video are high bandwidth and low latency.

From a 5G-transport perspective, there is a need to provide efficient methods for network sharing, so that applications like these — each with their individual requirements, including mechanisms to satisfy traffic isolation and SLA fulfillment — can be supported for several clients. In addition, distributed network functions need to be connected over links that fulfill set performance levels for bandwidth, delay, and availability.

Transport networks will need to exhibit a high degree of flexibility to support new services. To this end, key features are abstraction and programmability in all aspects of networking — not just connectivity but also storage and processing.

Legacy, migration, and new technologies

The main technologies that contribute to performance enhancement and the network segment — access, aggregation, or core — they apply to are outlined in [Figure 2](#). 5G transport will be a mix of legacy and new technologies. Long-term, network evolution plans tend to include fiber-to-the-endpoint. In practice, however, providing small-cell connectivity requires that local conditions be taken into consideration, which results in the need for several technologies — such as copper, wireless links, self-backhauling, and free-space opto — to be included in the connectivity solution. Re-use of existing fixed access infrastructure [6] and systems will be important, and new technologies and systems may in turn provide more efficient use of available infrastructure. For example, additional capacity can be provided by extending the use of CWDM and DWDM closer to the access segment of the network. At the same time, interworking with IP is essential to provide end-to-end control, and to ensure that the fiber infrastructure is used efficiently.

Existing infrastructure, together with operator preferences, determines the necessary evolution steps, and how the migration process from legacy to desired architecture should proceed.

The design of 5G transport networks will need to continue to be affordable and sustainable, keeping the cost per bit transported contained. Handling legacy in a smart way, and integrating sustainable advances in technology into packet and optical networks will help to keep a lid on costs.

Programmable control and management

Flexibility through programmability is a significant characteristic that will enable 5G transport networks to support short time to market for new services and efficient scaling.

Programmability gears up networks, so they can take on innovations rapidly, and adapt to continuously changing network requirements. A couple of capabilities need to be determined to enable programmability for transport networks:

- » the required degree of flexibility or ability to reconfigure
- » the layer or layers that need to be programmable.

Determining these capabilities is a trade-off between need and gain; in other words, how does the benefit of programmability compare with the cost of the technology needed to provide it? A significant factor for transport providers in weighing up need against gain is how to address packet-optical integration. This is because extending programmability to the optical layer not only provides greater flexibility and ease of provisioning to allocate transport bandwidth; it also simplifies the process of offloading the packet layer through optical/router bypass, as well as providing improved cross-layer resilience mechanisms [7].

The telecom industry has long set itself two principal targets for transport networks: efficient resource utilization, and dynamic service provisioning and scaling. While these goals still

PROGRAMMABILITY
IN 5G TRANSPORT
NETWORKS WILL
IMPROVE FLEXIBILITY

stand, they need to be revised continually to match the changing needs of client layers. These needs include the short reaction times demanded by modern applications, and the fact that different clients will need to interface with the network at different layers. Add connection capabilities like bandwidth and latency into the mix, and the need for network programmability becomes more evident.

So just how does increased network programmability help the telecom industry meet the targets it has set for itself, given the need for different performance characteristics for different applications?

Efficient resource utilization

Transport programmability enables network operators to exploit traffic dynamicity to optimize the utilization of resources across different segments of the network.

A programmable transport network facilitates the division of transport resources into multiple (isolated) slices. These slices can be allocated to different clients — enterprises or service providers — enabling efficient sharing of resources.

Dynamic service provisioning and scaling

Being able to provision resources on the fly is particularly crucial for dynamic service chaining, which involves interconnecting distributed, virtualized network functions and ultimately facilitating dynamic service creation. In particular, establishing connection services across several networking domains has long been a challenge — here enhanced programmability can make such procedures more efficient. In most cases, flow control in the transport domain should be carried out on aggregated traffic to avoid detailed steering for individual users when it is not needed.

A programmable transport network enables the capacity allocated to a service to be scaled up or down, when and where it is needed across the network — in other words, providing elastic services.

Centralized or distributed control

Control plays an essential role in programmability. Network control can be centralized or distributed,

and networks are operated differently depending on the approach used.

Centralized control — the concept used in SDN — enables shorter service development cycles and speedier rollout of new control functionality (implementation occurs once in the central stack). For networks built with a distributed control plane, changes must be made in multiple — already deployed — control stacks (especially in multi-provider networks).

The topic of SDN is being discussed in the telecom industry as a promising toolset to facilitate network programmability. In SDN architecture, the main intelligence of network control is decoupled from data plane elements and placed into a logically centralized remote controller: the SDN controller (SDNC). As such, the SDNC provides a programmatic API, which exposes abstracted networking infrastructure capabilities to higher layer control applications and services, enabling them to dynamically program network resources.

The role of the API in SDN goes beyond traditional network control. It allows applications to be deployed on top of the control infrastructure, which enables resources to be automatically optimized across heterogeneous network domains, and new end-to-end services to be instantiated easily. The control/management system needs to provide methods for controlling resources and for exposing infrastructure capabilities — using the right abstraction with the level of detail suitable for higher layer applications.

To highlight this point, in our research we chose to exemplify the case of resource and service orchestration across multiple network domains with heterogeneous types of resources. The resulting hierarchical SDN-based control architecture, which orchestrates across three domains — transport, radio access networks (RANs), and cloud — is shown in [Figure 3](#). A management function [8], which can be partly overlapping, is included but not discussed in detail in this article.

SDN flavors

The impact of upgrading the control plane of a legacy transport network to SDN depends on a number

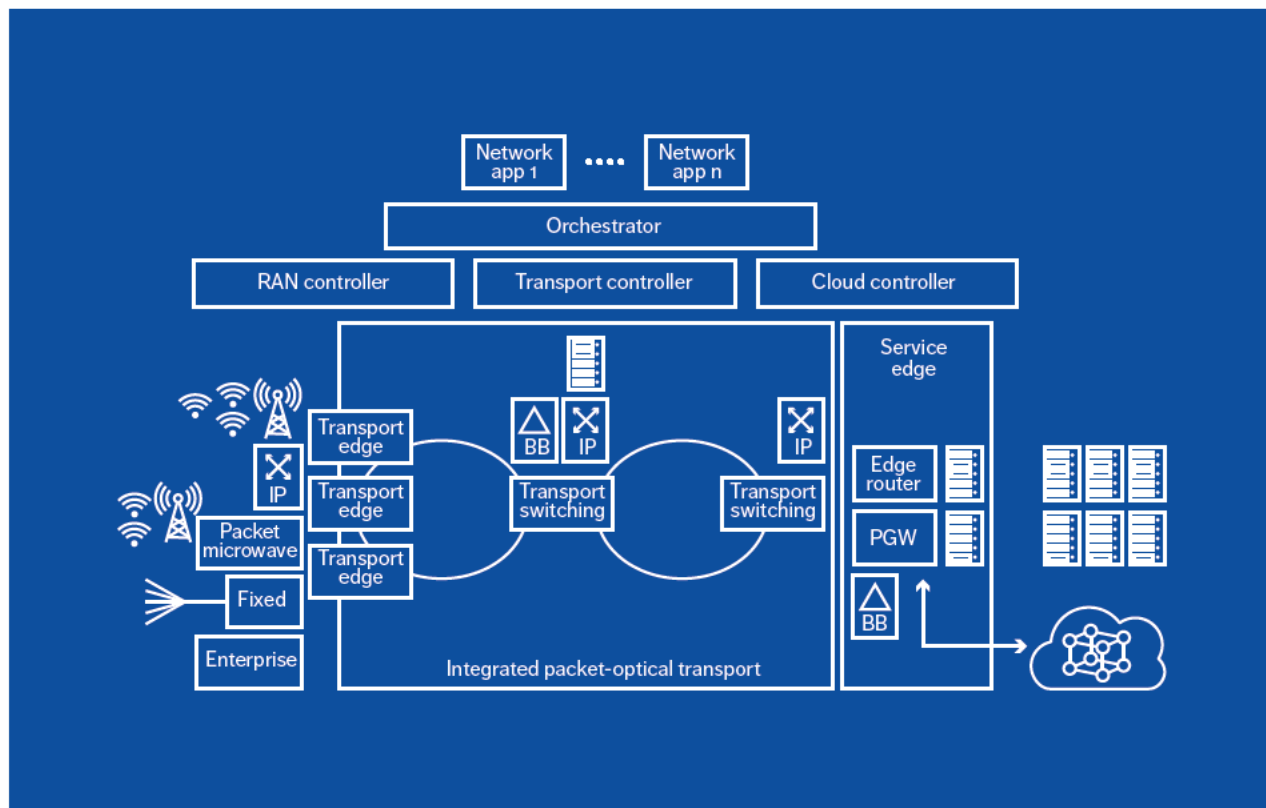


Figure 3
Hierarchical SDN control
architecture for multi-
domain orchestration

Figure 4a
Centralizing control
functionality in the optical
domain

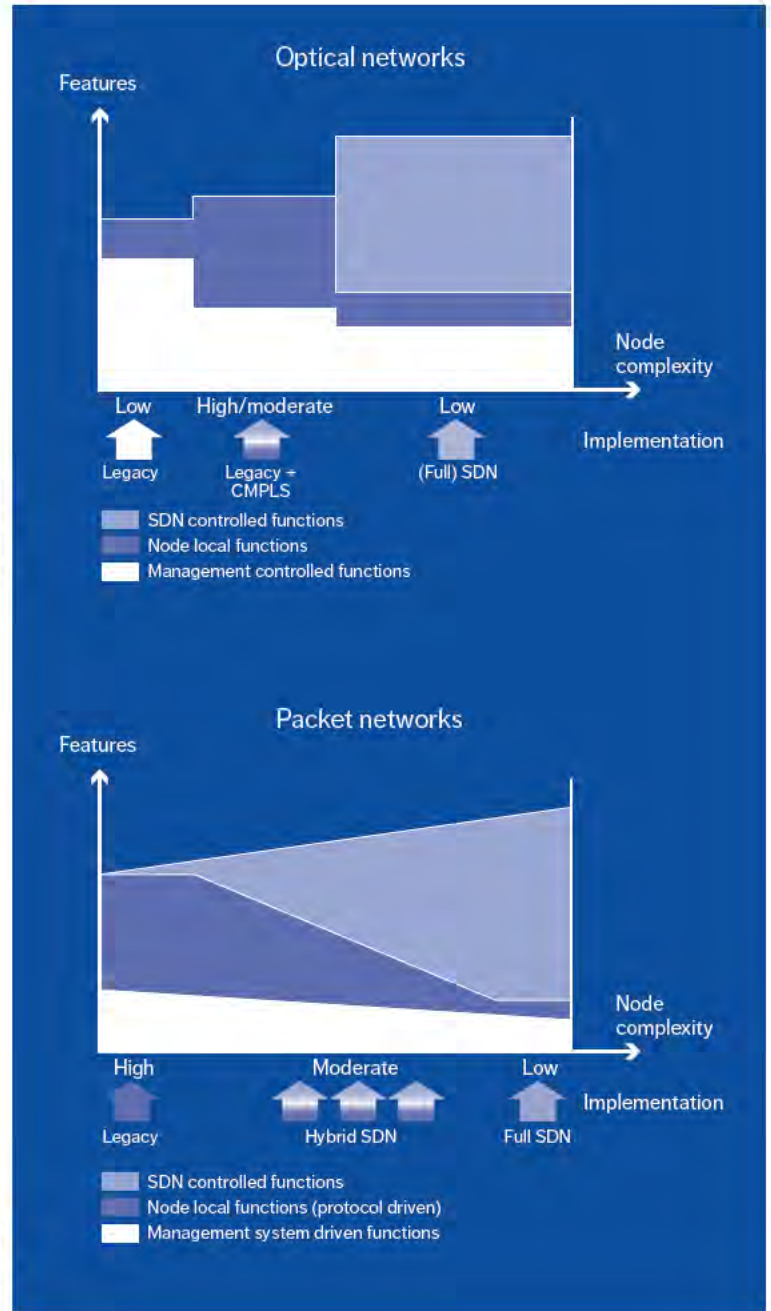


Figure 4b
Centralizing control
functionality in the packet
domain

of aspects, but primarily on the degree to which forwarding and control functions are integrated in legacy transport networks.

In legacy optical transport networks, most control functions are already separated from the data plane nodes. However, in packet-switched transport nodes, the two planes are tightly coupled. As such, introducing a fully centralized SDN control plane (full SDN) is more straightforward for optical transport than packet networks.

To integrate legacy optical transport networks, as illustrated in [Figure 4a](#), the SDNc needs to be developed along with suitable interfaces, but it does not necessarily require disruptive changes to the optical nodes.

When applied to packet networks, the disruption created by SDN is significant. A more natural approach for the packet domain would be to centralize selected elements of the control functions. The resulting hybrid-SDN alternatives are illustrated in [Figure 4b](#). Ideally, control over service-related functions should be centralized with the SDNc, while transport-related functions should be implemented locally on the node.

The decision of where to place a control-plane function or feature is operator specific, depending on many factors, like the available feature set, and operational preferences.

In packet-based transport networks, the concept of separating transport- and service-related functions is well established, where a clear logical differentiation is made between service unaware transport nodes and service nodes — such as P and PE routers in MPLS networks. Only service nodes hold service states and require implementation of service-related functions. Such separation is a future-proof concept and one that should remain intact. Any improvements in this area should focus on transport service functions, as they cause most of the challenges in building and operating networks and make the introduction of new services lengthy and costly — especially in multi-vendor environments.

Flexible transport plane

Several factors contribute to network dynamicity — the ability of a network to adapt rapidly to changing

demand. The introduction of 5G radio technologies and the launch of new services are the two main factors pushing the need for networks to be more dynamic, and consequently the need for a more flexible transport plane. There are, however, many other factors contributing to network dynamicity:

- » **resource dynamicity:** on-the-fly addition and removal of connectivity, compute, and storage resources
- » **traffic dynamicity:** responsiveness to fluctuating traffic patterns that result from user movement/migration, or variations in user activity [9]
- » **service dynamicity:** responsiveness to service usage patterns with widely varying resource requirements
- » **failures and service windows:** ability to reroute traffic and minimize impact of downtime
- » **weather conditions:** managing the effect of rain or fog on performance for microwave or free-space optical networks.

Expert opinion differs on how access and transport architectures will evolve to meet future mobile requirements, and in particular how they will provide support for small cells. Legacy networks typically consist of separate branches for fixed (residential/business services) and mobile access. Continued densification of mobile networks is likely to result in the use of several different small-cell transport technologies — each one adapted for specific network conditions.

In particular, the adoption of wireless backhaul/fronthaul technologies, such as NLOS, for provisioning connectivity to new small-cell sites, will become more prevalent. At the same time, the fixed-access infrastructure and its widespread availability will continue to be valuable for providing small-cell connectivity, pushing the need for fixed and mobile network convergence.

How fixed/mobile convergence might evolve depends on existing fixed access infrastructure. Many operators have been reluctant to invest in deep fiber technologies, like FTTN, due to the costs associated with deployment, and have instead turned to alternatives like copper-based drop links using DSL or CAT5/6. This type of architecture or active optical network relies on the presence of a large number of distributed active nodes.

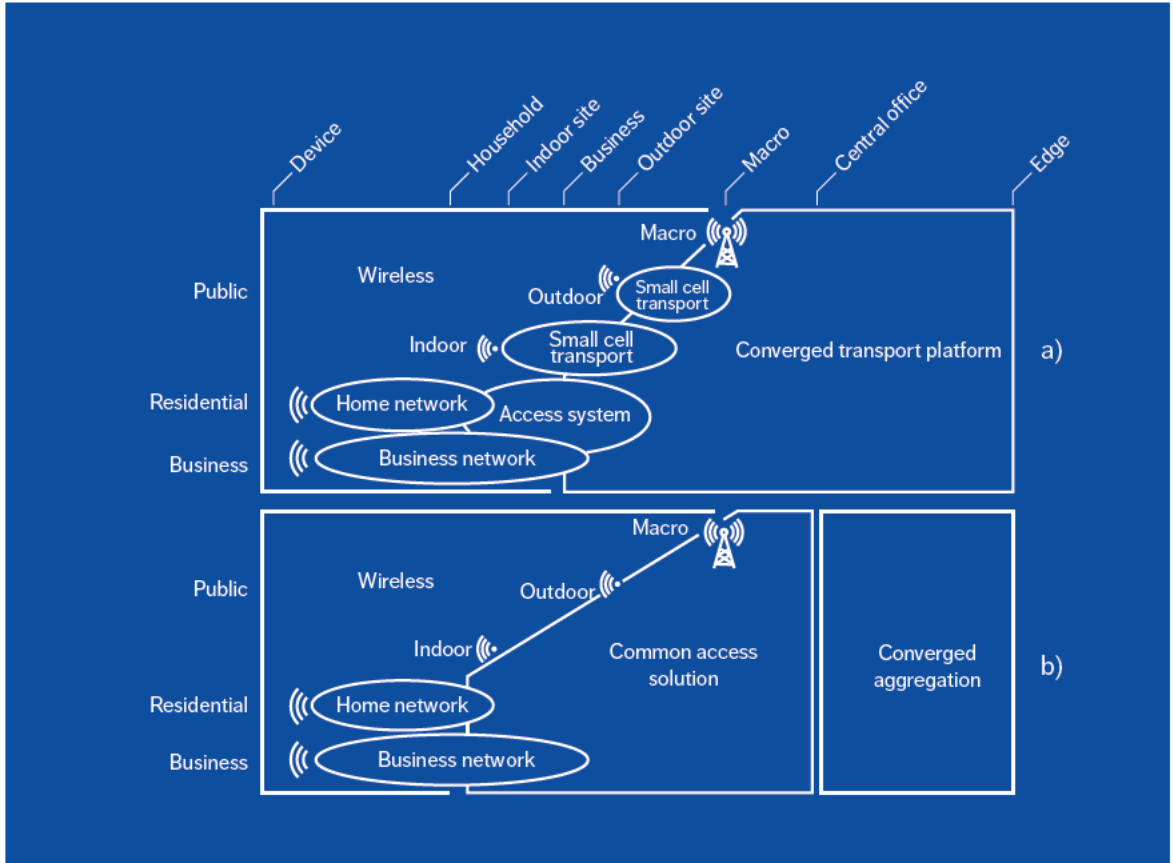


Figure 5
Different architectural avenues with scenarios based on:
(a) a converged transport platform
(b) a common access solution

Figure 5 illustrates two evolution scenarios for fixed/mobile convergence. Different options are available for providing converged access infrastructure for traditional residential and business access services, as well as IP-based backhaul and

CPRI-based fronthaul [6]. In the bottom part (b) of the illustration, the connectivity needs of the RAN are served through a common access solution. Here, the challenge is to define a system that can simultaneously meet the cost points of residential

access and the performance requirements for different RAN deployments. As illustrated in the top half (a) of Figure 5, evolving networks in this manner paves the way for a converged transport platform (possibly including small-cell fronthaul) under common control but with a somewhat diverse data plane, subject to requirements for different segments. In reality, many more scenarios and combinations exist than are illustrated here, and the choice of architectural infrastructure model, RAN deployment model, and technology are all closely interlinked. In turn, the decisions made for network architecture and technologies determine the degree of flexibility that transport can offer, whether it is at the packet or wavelength level.

In a converged scenario, one possible solution is to deploy nodes capable of providing common switching of packet and fronthaul — which today are separate domains that use different transport protocols (Ethernet and CPRI) with their own specific requirements. To multiplex Ethernet and CPRI, switching at wavelength and packet layers can be combined: the challenge is to meet latency and jitter requirements for time-sensitive applications. Deterministic delay switching using client agnostic frames is an alternative to packet for scenarios where existing DWDM/OTN metro infrastructure is used simultaneously for backhaul and fronthaul applications.

For optical interfaces, fiber rich deployments can benefit from recent developments of grey 100G and 400G optical interconnection interfaces. This is an active field of research, and many solutions, standardized and proprietary, are being explored. In some, modulation formats are natively designed to dynamically adjust the bandwidth according to real-time networking needs. Integrated photonic technologies are changing the optical communications industry outlook, promising a dramatic reduction of hardware cost, power consumption, and footprint, but also enabling flexibility at lower cost. Multi-channel transceivers and tunable lasers are among the first applications targeted by integrated photonics, as demonstrated by standardization activity on NG-PON2 and G.metro. Along the same technological trend, Ericsson

has defined a new type of device — an integrated reconfigurable optical add-drop multiplexer tailored to the specific needs of radio access. Such a device would provide flexibility in the wavelength layer and would be an order of magnitude simpler and cheaper than conventional ROADMs.

Conclusions

The first 5G network trials are already ongoing on a small scale, and commercial systems are expected in 2020. Comparing 5G with previous generations shows that it is not just a new radio-access technology — so much more is expected of it. 5G is shaping up to provide cost-effective and sustainable wireless connectivity to billions of things, people, enterprises, applications, and places around the world. To make the most of this business opportunity and deliver connectivity to billions of devices, the architecture of 5G systems — and transport in particular — needs to be built for flexibility through programmability.

Delivering the required level of flexibility, needs tighter integration between 5G radio, transport networks, and cloud infrastructure. This must be carried out with a backdrop of small-cell deployment, convergence of access and backhaul, and migration of legacy equipment and technologies — while containing costs.

When it comes to programmability, the expectations placed on SDN technology to deliver are enormous. SDN also brings service velocity with it, as well as a means to integrate transport, radio, and cloud domains. However, adopting a hybrid-SDN alternative might be best to mitigate the disruption SDN causes when applied to packet networks. This approach enables control to be centralized for service functions and distributed for transport — gaining a degree of flexibility without the disruption.

To meet capacity demands, increasing the use of DWDM closer to access will be feasible when flexible optics become more cost-efficient. In short, the major challenges for 5G transport are programmability, flexibility, and finding the right balance of packet and optical technologies to provide the capacity demanded by the Networked Society. ●

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INDUSTRIAL REMOTE OPERATION

5G rises to the challenge

Ericsson and ABB are collaborating to determine how to make the most of 5G and cellular technologies in an industrial setting. We are looking at a number of use cases, each with its own challenging set of connectivity requirements. This article presents some of the use cases being assessed, highlights the challenges posed by remote operations, and describes how 5G technology can be applied to overcome them.

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Use cases, benefits, and drivers

Power plants, mines, construction sites, and oil platforms can be hazardous environments. Industrial sites like these can be noisy and dirty, and may expose personnel to an abundance of risks associated with falling objects, harsh weather conditions, and the presence of heavy machinery and chemicals.

Business incentives like reducing the risks associated with working on remote sites have led industrial players to consider ways of minimizing the numbers of operational

personnel needed. Deploying a remote- or teleoperation for heavy machinery and other equipment is one way to cut the size of the on-site workforce. Remote operation solutions allow people to operate machinery from the safety of a control center at another site — sometimes even several hundred kilometers away.

■ With the right system design, remote operation enables an increased level of safety, and in some cases leads to more efficient use of resources. For example, operators can run a number of machines

at several different sites from the comfort of a centralized control center. Control centers can in turn be established in strategic locations; it tends to be easier to attract experts to an urban area than a remote location. Running a remote operation can also help to reduce the high cost of building the kind of infrastructure often associated with sites that are isolated. However, at times, remote operators may not be as productive as on-site manual operators owing to their reduced sense of a machine's surroundings. Operating a wheel loader in a mine on a remote basis, for example, is less efficient than handling it manually on-site, as it is harder to fill the loading shovel with as much material.

Productivity can, on the other hand, be improved by including a certain degree of automation in the solution — to help the operator with the most challenging tasks. Repetitive tasks can be almost fully automated, with operator intervention reserved for handling unexpected events, such as when an object is dropped or something gets broken. For other jobs that may be carried out more effectively by a machine than a human being — such as precise linear movements and constant contact force control — an automatic controller may be used to assist the operator. In the case of a remotely operated robotic arm, the robot and the operator can have joint control, depending on the degree of freedom and motion required.

The possible use cases for remote operations in industry are numerous, and each scenario brings its unique set of challenges.

Mining

The modern mine is crowded with vehicles and

machines performing a variety of tasks, both on the surface and underground: trucks, drills, trains, wheel loaders, and robots designed for specific tasks are all typical examples. Mines are high-risk environments, and the ability to move people and equipment from one place to another is key, given that certain areas can take a considerable amount of time to reach.

The ability to move driverless equipment into place quickly, say following a blast, is a potential time-saver when people are not permitted into the area until fumes have cleared. Benefits like this, combined with the fact that mines are typically found in remote locations, have led the mining industry to become an early adopter and developer of remote machine operation.

Construction sites

The incentives for the construction industry to implement remote operations are similar to those that apply in mining. In both industries, heavy machinery is required, such as excavators, wheel loaders, compactors, and haulers — all of which can be worked remotely to advantage. Unlike mining, machinery used in the construction industry moves from one site to the next, which requires a more flexible operating solution that can function without the need for fixed on-site infrastructure.

Ericsson's research addressing remote operations for the construction application was demonstrated at MWC in 2015 [1]. The trials leading up to the demo aimed to determine the network requirements like latency and throughput, as well as the performance needs for the audio and video equipment — with a view to ensuring that 5G will meet the specifications.

Terms and abbreviations

3GPP—3rd Generation Partnership Project | **DECOR**—dedicated core network | **E2E**—end-to-end | **FEC**—forward error correction | **IP**—Internet Protocol | **IR**—infrared | **LTE**—Long-Term Evolution | **MWC**—Mobile World Congress | **NFV**—Network Functions Virtualization | **NX**—Ericsson's 5G air interface initiative | **RAN**—radio-access network | **RTP**—Real-time Transport Protocol | **SCTP**—Stream Control Transmission Protocol | **SDN**—software-defined networking | **SLA**—Service Level Agreement | **SRTP**—Secure RTP | **TTI**—Transmission Time Interval | **UDP**—User Datagram Protocol | **UE**—User Equipment

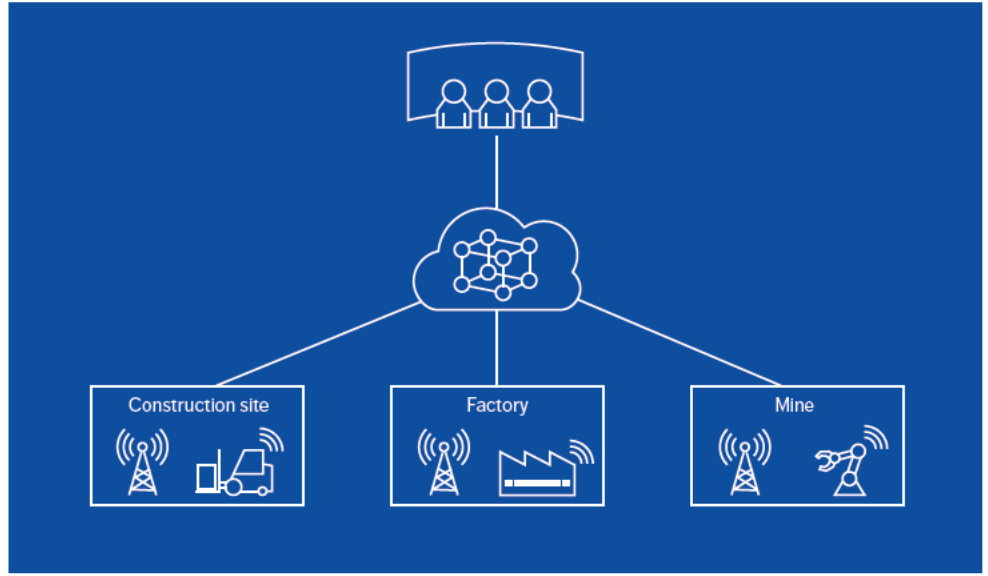


Figure 1:
Remote operation of machines

Harbors

Large cargo ships can carry over 16,000 containers. Loading and unloading is a time-consuming process often requiring a number of cranes working simultaneously for many hours at a time. Traditionally, each operator sits on-site in the control cabin of the crane, high above ground. Cranes need to be operated with speed, precision, and consistency. With smart cranes and remote operation, safety and productivity levels can be increased, while operator stress levels can be reduced. The comfort of the control room offers many benefits in terms of wellbeing, as it:

- » saves the time spent accessing a crane's control cabin
- » provides a favorable job environment with improved ergonomics
- » reduces exposure to adverse weather conditions
- » improves the security and safety of personnel

ABB has developed a solution to remotely operate cranes from a control room in the harbor, where the operator's work is facilitated by a video feed from the crane [2]. Centralization is the natural next step in the development of this solution, enabling multiple cranes situated at different sites to be operated from the same station.

Surveying and inspection

Drones, robots, and vehicles that are remotely operated are suitable for applications like land and sea inspection, where the safety issues arising from the distances covered, adverse weather conditions, and hazardous terrain can be costly to address. Remote operations work well for these types of monitoring applications, and are ideal for observing industrial and construction sites in out-of-the-way places, or large indoor venues and warehouse environments.



Figure 3:
Remote operation station
Photographer: Hans Nordlander



Figure 2:
Remote mining control center
(Garpenberg, Sweden)
Photographer: Hans Nordlander



Figure 4:
Harborside cranes for loading and unloading cargo
Photographer: Hans Nordlander

“REMOTE OPERATION ISN'T A ONE-SIZE-FITS-ALL SOLUTION. DUE TO THE RANGE OF EQUIPMENT AND THE MANY POTENTIAL SCENARIOS IN WHICH REMOTE APPLICATIONS APPLY, THE ARRAY OF USE CASES THAT COULD BENEFIT FROM REMOTE OPERATION IS EXTENSIVE.”



Video streams and other sensor data are fed back to the operator, enabling appropriate action to be taken. By combining remote inspection with remote manipulation, the level of automation can be raised. For example, a remotely operated robot in a data center can rapidly swap out a malfunctioning server, or respond to other types of hardware failures [3].

Oil and gas

The oil and gas industry operates in environments that are harsh — both for people and equipment. Inspection, servicing, and operation of equipment as well as monitoring of leaks are just some of the routine applications. Remote operation is highly applicable to this industry, but to fully reap the potential benefits, equipment must remain functional without the need for regular on-site maintenance. One of the main benefits of remote operation is a reduction in the need for people to work in hostile environments, and frequent maintenance visits would negate this benefit [4].

Remote surgery

The use of teleoperation technology is emerging in the field of medicine. It enables surgeons to perform critical specialized medical procedures remotely — allowing their vital expertise to be applied globally. While this application area is still in its infancy, it is likely to become more widespread as the technology becomes more advanced.

Challenges

For remote operation solutions to function effectively, sensory information like sounds and images needs to be transferred to the teleoperator from the equipment being controlled and its surroundings. Ensuring that audio and visual feeds are sent with minimal distortion enables the teleoperator to gain a good understanding of the remote environment, which leads to improved productivity and safety.

Remote operations would become even more efficient and intuitive if sensory data additional to the basic audio and visual information were included in the solution. Just as manual operations rely heavily

on the human ability to balance and touch things, remote operation applications — whether industrial, medical, or recreational — can benefit greatly from the incorporation of this type of sensory information. The addition of touch and balance to the operator feed can be achieved by the use of haptic interaction and force feedback. The ability for the operator to actually feel the vibrations when an object like an excavator bucket hits the ground, or to sense when a robot arm touches its target is highly valuable in terms of productivity, cost, and safety.

Additional sensors and technologies, like gyros, accelerometers, radars, lasers, lidars, and thermal and IR sensors can be used to gain more information from the remote site and provide enhanced control at the operator end.

The negative effects of bad media quality, or an imperfect representation of the remote equipment and its surrounding environment, can be alleviated to some degree through training. Before full productivity can be achieved, operators require training and experience of operating equipment remotely — even if they have previously operated the same or similar equipment on-site.

Remote operation isn't a one-size-fits-all solution. Owing to the range of equipment and the many potential scenarios in which remote applications apply, the array of use cases that could benefit from remote operation is extensive. An extra level of variation arises from the need to weave environmental parameters — such as rain, snow, dust, dirt, vibrations, and visibility — into system design. For example, remotely operating a dumper that moves cargo loads in and out of a mine is fundamentally different from performing surgery using a remote-controlled precision robot. But even less obviously contrasting examples, like operating a dumper in differing visibility conditions, can present significant challenges for the technical solution.

Communication requirements

Securing a high-quality communication link between the control station and the machines being operated is key to accurate and effective remote operation. Existing solutions tend to use cable or

WI-FI to implement the last hop of this link. Cable provides low latency and high reliability, but it is costly to install and modify — which is significant when machines are constantly being moved from one site to another, such as in the construction industry. WI-FI is a low-cost alternative that provides a certain degree of mobility — within the coverage area of the WI-FI network. Both solutions require dedicated on-site installation and a connection to the control center over the public internet or through a leased fixed-line connection.

To provide remote operation solutions with connectivity, standardized cellular systems offer a number of benefits over wired connections or WI-FI. First, using an operator-managed cellular network eliminates the need to install on-site infrastructure. Second, cellular offers widespread coverage and mobility solutions that can provide connectivity to mobile machinery and devices. Furthermore, as they use licensed frequency bands, cellular links are highly reliable, and the required level of security can be guaranteed. However, the requirements set by some use cases, which are of interest to society and certain industries, cannot easily be met by existing communication technologies.

A simple, quick and flexible on-site installation process is a basic requirement for many remote operation applications. Machines might be portable or driverless and may be required at different locations during the same working day. Job sites can be temporary and may grow, and their communication needs may change over time — which tends to be the case in construction and mining. For such environments, wireless solutions are preferable as they offer the desired level of flexibility and ease of installation, they can support equipment that is on the move, and do not require any cables.

For the most part, industrial companies expect global communications to be delivered with E2E Service Level Agreements (SLAs), which they can handle themselves to some degree. Providing E2E SLAs, however, presents a challenge given that the system may span multiple public operator networks and even infrastructure owned by the enterprise itself.

High-definition video is a fundamental element of remote operation solutions.

To deliver heavy video streams requires connection links with high minimum bitrates, especially when applications require high-resolution images, fast frame rates, stereoscopic video, immersive video, or multiple viewpoints (several camera feeds). Low media quality severely degrades the user experience, which inevitably leads to a drop in productivity. The exact bandwidth requirements are, however, highly dependent on the use case.

Like most real-time applications, remote operation requires connection links with low latency and low jitter characteristics. To operate equipment (like an excavator or a robot) efficiently on a remote basis, the time lapse between the instant an operator sends a control instruction to the moment the equipment's reaction is sensed by the operator must be as short as possible.

The toughest latency requirements occur in applications that include haptic interaction. A typical haptic control loop in a remote operation application requires latency to be below 10ms [5], and in some cases, the round trip time should not exceed a couple of milliseconds. To put this figure into perspective, current LTE networks have an average latency of 30ms, which in some cases can rise to 100ms or more if packets are delayed.

Some degree of toleration to packet loss in remote operation applications is expected. However, packet loss may result in lost or delayed control commands, which can cause machinery to stop, can be costly, and can cause damage to equipment or even injury to personnel. So, to guarantee the continuous and safe operation of machinery, the communication link and the entire solution need to be highly reliable.

System outages or hijacked equipment resulting from a cyber-attack or other security intrusion can have severe consequences. Personnel safety is

“A HIGHLY RELIABLE RADIO LINK IS NEEDED TO AVOID TRANSMISSION ERRORS AND TIME-CONSUMING RETRANSMISSIONS”

jeopardized, business continuity can be affected, and expensive equipment may be damaged. So, security is a key consideration when designing any remote operation system.

Proper audio and video feed synchronization is critical to provide the operator with a clear understanding of what is happening at the remote location. The synchronization requirements for remote operation solutions that incorporate haptic interaction and force feedback are much stronger than for a videoconference, for example. Without proper synchronization, the operator might receive confusing and contradictory messages, which has negative impact on user experience.

Mechanisms need to be in place to ensure that equipment can be stopped automatically in abnormal situations — like a machine malfunction, a collision, or the presence of unauthorized personnel. Teleoperated equipment may require additional sensors and functionality to detect potential risks and enable safe remote fault handling and recovery.

The communication requirements for remote operation can be summarized as follows:

- » ease of deployment
- » minimum bitrate
- » low latency
- » reliability
- » security
- » emergency handling and recovery

Solutions and enablers in 5G

5G innovations related to media delivery, and core, radio-access and transport networks [6] will provide the technology needed for remote operation and other industrial mission-critical cases.

RAN solutions

To deliver an acceptable level of service experience for industrial remote operation, a number of performance requirements need to be set: minimum bitrate, maximum latency, and a permitted level of packet loss. By deploying service-specific optimizations relating to scheduling, the requirements of several remote use cases may be met by modern LTE-based cellular systems. And as LTE

will continue to be enhanced with improvements such as latency reductions, it will become ever more applicable for industrial applications.

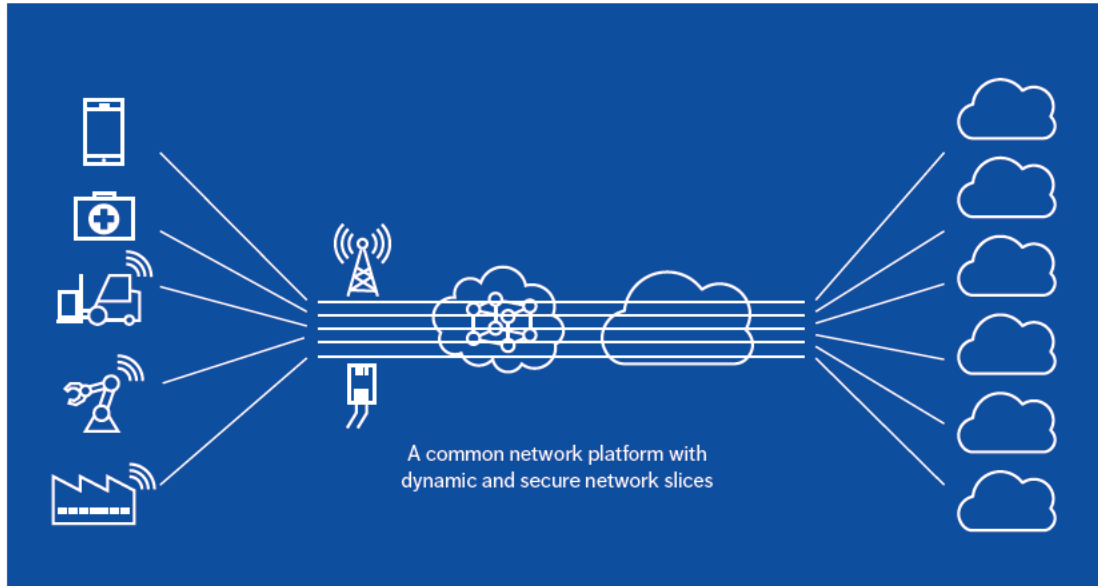
However, some demanding use cases — such as the operation of fast-moving machine parts or scenarios that require accurate real-time control — place such stringent requirements on connectivity that they cannot be met by existing cellular solutions. But 5G technologies are being developed with these requirements in mind. With market introduction due around 2020, they will be able to provide the performance capabilities necessary for demanding industrial use cases. In 5G, innovative air interfaces like NX will be developed that include sophisticated signaling methods. The evolution of LTE will be a significant part of 5G, and its technologies will coexist with NX.

If an industrial site is located within the coverage area of a mobile operator's 5G network, remote services can be provided to the site using the network's inbuilt mechanisms at the required performance level. In many cases, however, industrial sites tend to be located in areas without adequate 5G coverage. In such cases, a dedicated 5G infrastructure can be installed near the industrial site, which could be either permanent or temporary.

To support the requirements of the whole coverage area for high-load situations, special design characteristics need to be taken into consideration. The challenge arises when connections are congested or suffer from poor link rate, causing the transfer rate over the radio link to drop temporarily below the code rate of the video stream. When this occurs, queuing delays follow, which in turn degrade user experience.

Low latency and high reliability are two key design criteria for the NX-radio interface in 5G. To attain the levels of performance required for latency and reliability, a number of air interface design characteristics, like short radio frames and new coding schemes, will come into play.

To achieve low latency in the system, the time it takes to transmit a control command over the radio interface needs to be minimized. In NX, the time to transmit a single packet over the air — the Transmission Time Interval (TTI) — is expected



to be a fraction of the TTI in LTE. The TTI in LTE is defined as 1ms, whereas NX will be designed to deliver TTIs in the order of one or a few hundred microseconds [4]. Such low-order TTIs will enable short transmission times for short packages and facilitate retransmission without exceeding the latency bound.

The radio receiver needs to be able to decode received messages quickly. High-performance forward error correcting codes, such as turbo codes traditionally used for mobile broadband, are not optimal for transmission of short messages with high reliability requirements. Therefore, special forward error correcting codes such as convolutional codes are envisioned for latency-critical applications [4].

A highly reliable radio link is needed to avoid transmission errors and time-consuming retransmissions. The level of reliability needed can be achieved with high diversity order of the communication through antenna or frequency diversity, which improves the probability of signal

detection and correct reception of the transmitted radio signals.

Messages need to be transmitted over the communication link without scheduling delays. To minimize delays, service-aware scheduling algorithms can be applied to prioritize critical remote applications over other less critical communication.

Core network aspects

Traditionally, mobile core networks are optimized to deliver a specific set of operator services. This approach was successfully applied in the rapid upscaling of mobile-broadband services for global reach. By adding flexibility to the core network architecture, 5G solutions will take optimization one step further, facilitating a much wider range of services and use cases beyond mobile broadband.

One way to provide flexibility is through network slicing: the logical partition of networks into slices supporting a defined set of devices and services.

Figure 5:

Resources for different industries – logically separated through network slicing

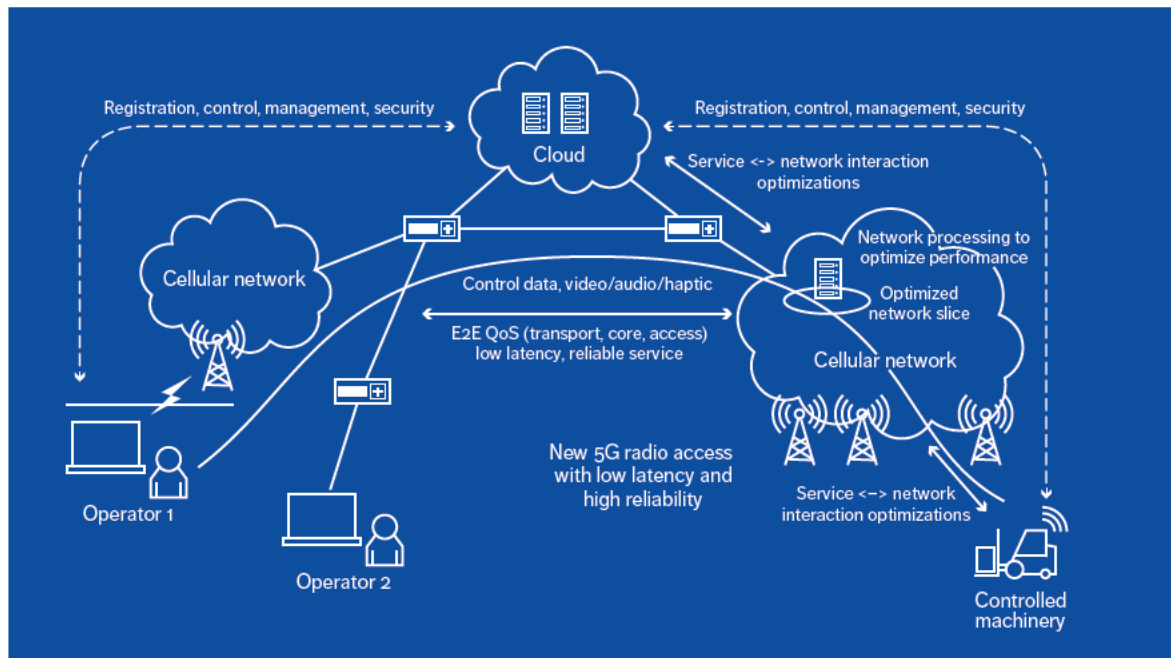


Figure 6:
Overview of 5G enablers
for industrial remote operation

Through slicing, a single physical network acts as multiple logical networks optimized for specific use cases or business needs. Resources may either be shared among several slices allocated on demand, or dedicated in advance to a given slice. The network operator decides. The functionality provided by a network slice can be tailored to a specific given use case, so the network features meet the business need and allow for cost optimization.

While physical network resources can be used to create network slices, the concept is particularly well suited to virtualized resources. Cloud technologies together with Network Functions Virtualization (NFV) provide cost-effective tools to adapt network functionality. In combination with software-defined networking (SDN), these techniques enable network operators to adjust their networks to meet the specific needs of industrial use cases.

The technologies needed to enable network slicing are emerging. In the 3GPP DECOR (dedicated core network) work, mechanisms have been defined to redirect a UE to a given network slice, based on user subscription or some other configuration information stored in the network. This work may be extended to include optimization of a network slice for a given use case.

Specifically, a number of link characteristics — like reliability, delay, and security — need to be considered to optimize network slices for industrial applications like remote operation.

A greater level of reliability can be achieved by adding system redundancy for computing, data, and network resources, and the associated control mechanisms for high availability.

Moving core network functions closer to the network edge reduces transmission delays in mobile-broadband networks that have centralized core network functionality. To cater to extreme cases, core network functionality can be colocated with RAN entities to avoid additional latency.

Deploying user services on local cloud platforms reduces latency. Extremely low latency capabilities can be provided by reusing the same execution environment for mobile-network radio and core processing, and for service functions.

The solutions that today's industry devices use for timing synchronization are independent of the mobile system. However, the air interface in 5G systems will provide accurate timing synchronization. By reusing the mobile system for timing synchronization, overall system complexity can be reduced.

A network slice may be optimized to serve a limited geographical area. If a single base station can cover the area, support for handover may not be needed. For areas covered by just a few base stations, the mobility solution may be optimized for the specific use case — and so simplification in system operations and deployment can be achieved.

The mobile network can be adjusted to use the identity schemes and security mechanisms tailored to industrial applications. For example, if an identity management scheme has been implemented on a local industrial network, the same identities could be reused in a mobile system — removing the need for an additional mobile system identity scheme.

Certain functionalities like advanced charging schemes, policy functions, and circuit switched interworking — which are fundamental to a public mobile-broadband service — are unnecessary in networks supporting industrial data applications. The resulting industrial system is more operationally efficient, which brings cost benefits.

Applications can explicitly indicate their communication requirements, which are translated into parameters for the underlying radio access and core networks. These parameters are considered in the orchestration and configuration of network functions as well as the transport network.

Industrial applications can be supported over logically partitioned network slices running on top of a generic network, or over dedicated industrial

“SPECIFICALLY, A NUMBER OF LINK CHARACTERISTICS — LIKE RELIABILITY, DELAY, AND SECURITY — NEED TO BE CONSIDERED TO OPTIMIZE NETWORK SLICES FOR INDUSTRIAL APPLICATIONS LIKE REMOTE OPERATION.”

mobile networks — independent of the public network. A dedicated custom deployment could be offered by a traditional mobile operator or by a third-party player. A hybrid solution offers additional flexibility, as standalone functions can be deployed on a dedicated network, while others can be supported by traditional operator services. The best approach can be worked out depending on the specific technical requirements and business setup of each deployment. To offer a truly flexible and global solution, the ability to set up network slices dynamically across network operator borders according to specific needs is required.

Low latency transport

To support applications like industrial remote operation over long distances (up to thousands of kilometers), transport networks need to be able to provide adequately low latency for the service at hand. Certain applications, like the excavator one, where operations take place in remote locations may require connectivity services at a given place and for a defined amount of time. The connectivity services needed to support applications like this require flexible and dynamic provisioning — possibly in several transport networks and potentially across multiple administrative and technology domains. Today, the provisioning process can be cumbersome and costly. But SDN and network orchestration promise to provide more flexible provisioning of transport services. By using these technologies, individual SDN domain controllers expose an abstraction of resources to a higher-layer controller/orchestrator, which in turn creates a global view of resources — facilitating provisioning of E2E connectivity services with given characteristics.

In theory, the maximum point-to-point distance providing a one-way-latency budget of 10ms (needed for haptic control) is given by the propagation delay of light along the surface of the earth — which for fiber corresponds to approximately 2,000km. In practice, recorded latency in transport networks is significantly greater than the theoretical value because of the lower physical layers and transport protocols. First, the actual signal path through the

transport network is longer than the direct path between two points. Measurements show that the actual path is approximately 1.5 times greater than the direct path [7]. Second, the median fiber path between routers increases the length of the signal path by an additional factor of two. Other factors that affect the practical minimum latency achievable besides propagation delay are transmission delay (which is of the order of milliseconds), processing delay (which is negligible), and queuing delay (which depends on traffic management).

To guarantee low latency, transport networks need to provide mechanisms that can apply priorities and enable optimal routing of latency-critical traffic. In practice, such mechanisms might select direct paths to minimize propagation delay or bypass certain nodes to avoid the delay incurred at intermediate hops — allowing overall latency to approach the theoretical limit.

Media delivery

Compression is a significant feature of any media-based solution that uses a mobile network to provide connectivity. The purpose of compression is to decrease bandwidth utilization, but it adds latency, and so compression algorithms need to be highly efficient. IP, UDP, and RTP are the most commonly adopted protocols for transmission of real-time application media. UDP is the best for minimizing delay, but as it is inherently unreliable, techniques such as forward error correction (FEC) or retransmission need to be used to manage packet losses. However, FEC and retransmission add to the overall delay, and so to minimize the dependence on such schemes, connectivity for remote operation should be provided over highly reliable networks.

Most remote operation applications will require a high to very high level of security. The Secure RTP (SRTP) protocol can be used instead of the RTP protocol to meet security requirements related to media delivery.

Transmission of application control signals

Application control signals in remote operation solutions include the signals traveling from the

operator to the controlled equipment, which directly or indirectly control the movements and actions of the machinery. Control signals typically originate from control equipment like a joystick or haptic device. For haptic interaction and force feedback, control signals also travel back from the controlled equipment to the operator.

Reliability is crucial when transmitting control signals, but as it often comes at the price of higher latency, some remote operation applications may benefit by using unreliable transfer mechanisms (with sufficient error handling) to transmit control signals.

The Stream Control Transmission Protocol (SCTP) is suitable for the transmission of remote operation signals, as it provides real-time characteristics and allows the level of reliability to be set. Regardless of the transport protocol used, remote-operation applications need to manage network congestion and failures as well as transmission errors swiftly and safely.

Conclusions

Examples of remote operation and control applications exist everywhere, but the benefits that can be gained in mining and construction are easier to realize than in some other industries. Increased productivity, access to specialized expertise, improved safety and wellbeing, and reduced exposure to hazardous chemicals are just some of the gains that remote operation can bring.

If configured appropriately, today's LTE networks can support some industry applications, but the needs of other, more demanding, use cases can only partly be met by existing communication solutions. 5G systems are, however, being developed to meet challenging requirements like low latency, high reliability, global coverage, and a high degree of deployment flexibility — the key drivers supporting innovative business models.

Together, Ericsson and ABB are working on remote operation and how industrial use cases can be developed into new value propositions for the Networked Society.●

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IDENTIFYING AND ADDRESSING THE VULNERABILITIES & security issues OF SDN

The promises of agility, simplified control, and real-time programmability offered by software-defined networking (SDN) are attractive incentives for operators to keep network evolution apace with advances in virtualization technologies. But do these capabilities undermine security? To answer this question, we have investigated the potential vulnerabilities of SDN. The aim is for this architecture to serve as a secure complement to cloud computing, and to ensure that networks are protected from attack by malicious intruders.

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TRADITIONAL NETWORK architecture has reached the point where its ability to adapt to dynamic environments, like those enabled by virtualization technologies, has become a hindrance. By separating the control plane from the data plane, SDN raises the level of system abstraction, which in turn opens the door for network programmability, increased speed of operations, and simplification: in short, the key to delivering on its promises, and enabling telecom networks and IT to develop in parallel.

■ At the heart of SDN architecture lies the SDN controller (SDNC). Logically positioned between network elements (NES) and SDN applications (SDN apps), the SDNC provides an interface between the two. Its centralized position enables it to provide other SDN components with a global overview of what is happening in the network; it can configure NES on the fly and determine the best path for traffic. The SDNC and the shift to centralized control set SDN architecture apart from traditional networks — in which control is distributed. Unfortunately, the centralized position of the SDNC makes it a primary surface for attack.

For the purposes of this article, we limited the scope of our study into the vulnerabilities of SDN to the single controller use case (with one controller governing the data plane), even though SDN architecture allows for several. Our discussion covers the SDN elements and their interactions in the single controller case, as well as the interactions between the SDNC and the management plane.

Why centralize?

As defined by ONF¹, a logically centralized control plane makes it possible to maintain a network-wide view of resources, which can then be exposed to the application layer. To provide such a centralized architecture, SDN uses one or more NEs that interface with the SDNC. The benefit of building networks in this way is simplified network management, and improved agility.

Centralization equips networks for programmability, which in turn increases autonomy. One possibility enabled by programmability is the automatic detection and mitigation of DDOS attacks, which results in rapid resolution of any problems that may arise. Programmability also allows network resources to be shared automatically, which — together with the capability to create virtual networks created on top of existing network infrastructure — enables automatic sharing by multiple tenants.

Benefits and vulnerabilities

SDN facilitates the integration of security appliances into networks, which can be implemented directly on top of the control plane, rather than being added as separate appliances or instantiated within multiple NEs. SDN's centralized management approach enables events within the entire network to be

collected and aggregated. The resulting broader, more coherent and more accurate image of the network's status, makes security strategies both easier to enforce and to monitor.

The ability to implement security mechanisms directly on top of the controller or on steering traffic at run time (using legacy appliances when necessary) makes it possible to dynamically add taps and sensors at various places in the network — which makes for more effective network monitoring. With an accurate picture of its status, the network can more readily detect attacks, and the number of false positives reported can be reduced. In practice, if a tap indicates to the SDNC that a device is showing signs of being hijacked by a botnet, the SDNC can steer the potentially offending traffic to an IDS for analysis and monitoring. If the traffic is deemed malicious by the IDS, the SDNC can filter it and instruct the first-hop NE accordingly.

Its ability to facilitate the collection of network-status information as well as enabling automatic detection and resolution of any breach in security, makes SDN ideal for integration into network threat intelligence centers and Service Operation Centers (SOCs). Unfortunately, the rich feature set of SDN also provides a larger attack surface compared with traditional networks — an issue documented in a number of recently published research papers².

Reference model

The overall SDN architecture comprises the following elements:

- » **NEs** — which are responsible for forwarding packets to the next appropriate NE or end host;
- » **SDNC** — which sends forwarding rules on to the NEs according to instructions it receives from SDN apps;

Terms and abbreviations

DDOS—Distributed DoS | **DOS**—Denial of Service | **GRE**—Generic Routing Encapsulation | **IDS**—intrusion detection system | **IPSEC**—Internet Protocol Security | **MM**—management module | **MPLS**—multi-protocol label switching | **NE**—network element | **ONF**—Open Networking Foundation | **RBAC**—role-based access control | **SDN**—software-defined networking | **SDNC**—SDN controller | **SLA**—Service Level Agreement | **TLS**—Transport Layer Security domain-specific modeling language

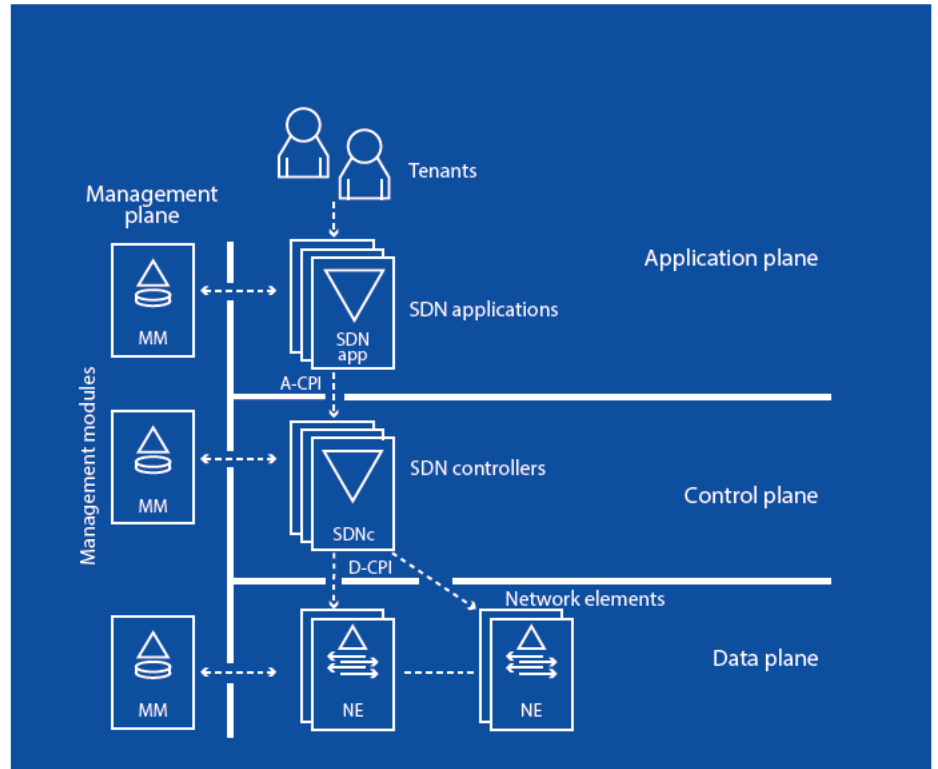


Figure 1
SDN architecture

- » **SDN apps** — which issue commands to dynamically configure the network;
- » **tenants** — the logical owners of the virtual network, who provide configuration and policy information through network apps; and
- » **management modules (MMs)** — which are responsible for device administration.

As illustrated in [Figure 1](#), the SDN architecture comprises four planes: the data plane, the control plane, the application and the management plane. The data plane carries user traffic through the different **NEs**, which are dynamically programmed to respond to the policies of the different tenants. Forwarding policies are elaborated, and sent on by the control plane to each **NE**. The management plane is dedicated to infrastructure management, physical device management as well as platform management issues such as firmware and software upgrades^{3,4}. The application plane is constituted by all applications that program the network through interactions with the **SDNC**. These applications may be independent and owned by different tenants.

Networks that are built according to SDN architecture principles need to protect a number of key security assets:

- » **availability** — the network should remain operational even under attack;
- » **performance** — the network should be able to guarantee a baseline bandwidth and latency in the event of an attack;
- » **integrity and confidentiality** — control plane and data plane integrity and isolation should be upheld between tenants.

To assure protection of these assets, a number of processes need to be in place:

Authentication and authorization

Only authenticated and authorized actors should be able to access SDN components. The granularity of authentication and authorization must be detailed enough to limit the consequences of stolen credentials or identity hijacking.

Resiliency

Networks must be able to recover as autonomously as possible from an attack, or a software or hardware failure. Alternatively, networks must be able to dynamically work around any affected functionality.

Contractual compliance

To fulfill **SLAs**, mitigation techniques must be implemented, and proof that such techniques have been activated effectively must be provided.

Multi-domain isolation

Systems must be able to isolate tenants in multiple domains, such as the resource and traffic domains.

The following forms of isolation apply:

- » **resource isolation** — prevents tenants from stealing resources, like bandwidth, from each other, and is required for **SLA** fulfillment; and
- » **traffic isolation** — required by multi-tenant deployments, so a tenant can see its own traffic only (this requirement applies to both data plane and control plane traffic).

Repudiation

All actions carried out by all system actors — both internal and external — must be logged, and the all logs need to be secured.

Transparency

Systems should provide visibility into operations and network status so they can determine the most appropriate action when issues arise. An active approach to security requires correct identification and classification of an issue so the most appropriate action to mitigate it may be chosen. Any action should be verified to ensure that it has been enforced effectively.

The potential vulnerabilities of SDN architecture are illustrated in [Figure 2](#), which for the sake of simplicity shows only a subset of the possible major attacks.

What's different about SDN security?

Many of the security issues related to SDN networks are similar to those that appear in traditional

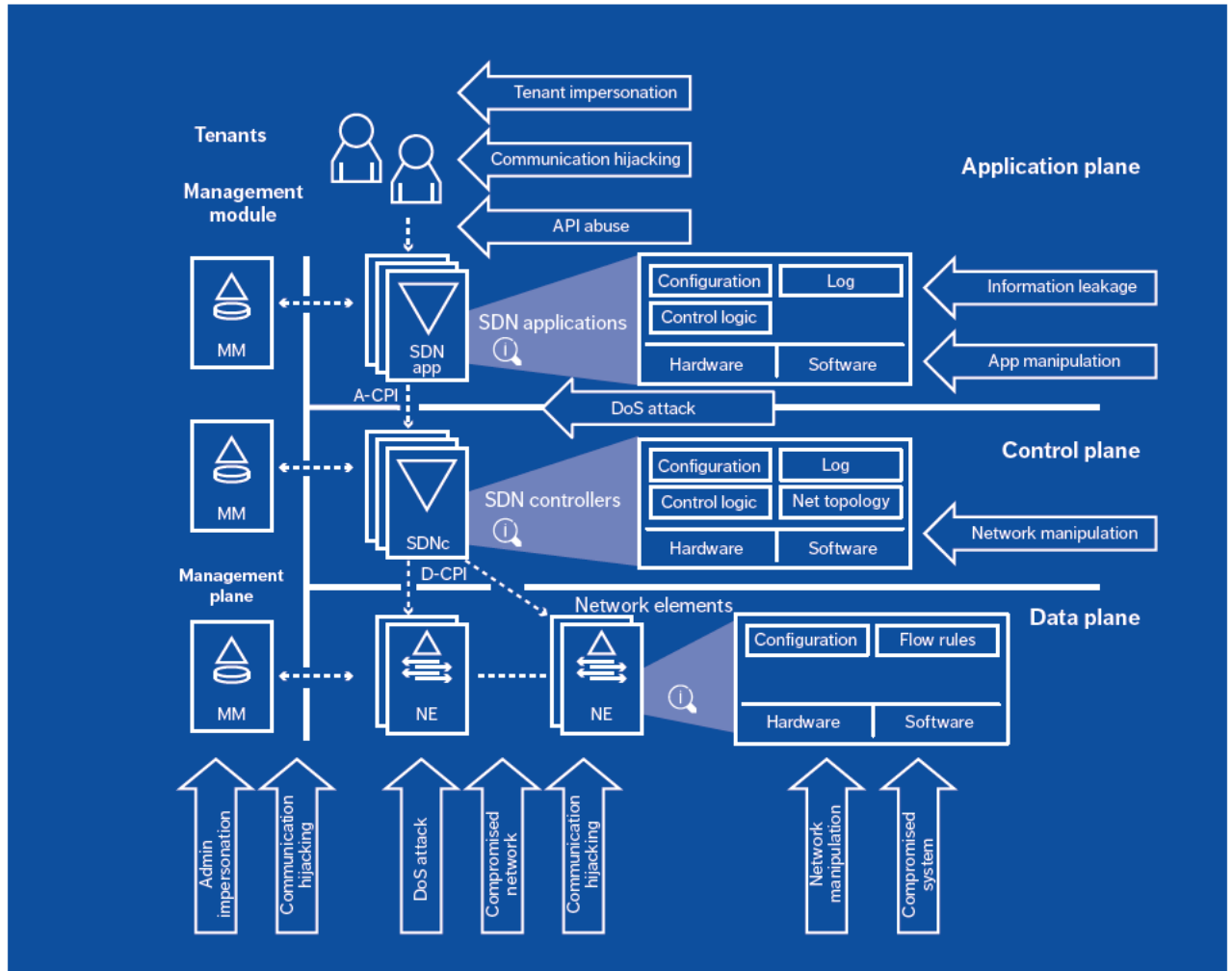


Figure 2
Potential vulnerabilities of SDN architecture

networks. What's interesting, however, is what sets SDN apart from traditional networks.

Compared with traditional networks, the separation of the control and data planes enables multi-tenancy and programmability, and introduces centralized management into the network architecture. In this new model, tenants run SDN apps that interface with the SDNC, which sends instructions to NES. From a security perspective, the ability to share and dynamically operate the same physical network is one of the key security-related differences between SDN and traditional architectures. As such, SDN security issues relate to the new control plane model, and more specifically to securing inter-component communication, and controlling the scope of applications and tenants through specific APIs and access policies.

While it may sound like there are a number of obstacles to overcome, the programmability and centralized management brought about by SDN enables a much greater level of autonomy to mitigate any security breaches — outweighing the need for additional technology.

Centralized network management

In traditional networks, NES tend to be monitored and managed individually. However, without the existence of standard protocols capable of interacting with all NES irrespective of their vendor or generation, network management has become cumbersome. The SDN approach enables coordinated monitoring and management of forwarding policies among distributed NES, resulting in a more flexible management process.

While there is a risk of the SDN control plane becoming a bottleneck, the fact that it has an overview of the entire network, makes it capable of mitigating any reported incident dynamically. For example, a DDoS attack can be detected and quickly mitigated by isolating the suspect traffic, networks or hosts. Unlike traditional DDoS appliances — which generally carry only a local view of the network — centralized elements possess a much broader view of network topology and performance, making the SDN an ideal candidate for the dynamic enforcement of a coherent security posture.

However, while it is clear that centralization provides significant benefits, it also presents a number of challenges, like the fact that the SDNC is a highly attractive attack surface. Thankfully, resiliency, authentication, and authorization address this risk, reducing the impact of attack.

Resilient control plane

The three main elements of SDN are: SDN apps, the SDNC, and NES. Given that control of the network is centralized, all communication within the control plane needs to be treated as critical, as an outage resulting from a successful attack may lead to an undesired impact on business continuity. If, for example, the SDNC is prevented from taking critical action to mitigate a DDoS attack, the entire network and all of its tenants may be affected. To avoid this, the control plane needs a greater level of resiliency built into it.

To communicate with tenant applications and NES, the SDNC exposes a set of interfaces. All these interfaces may experience heavy traffic loads, depending on the type and number of running applications. Traffic on the interfaces can be further impacted by NES, for example, forwarding packets for which they have no forwarding rules. So, in terms of dependence on the SDNC, traditional networks appear to be more robust.

An effective way to improve the resilience of the centralized control plane and prevent the spread of DDoS control-plane attacks to the rest of the network is to rate-limit NES in terms of bandwidth and resource consumption — such as CPU load, memory usage, and API calls.

Resilience can be further enhanced through proper resource dedication — where the SDNC authenticates each resource request, and subsequently checks requests against strong authorization control policies.

Strong authentication and authorization

Authentication and authorization are the processes used to identify an unknown source and then determine its access privileges. Implemented correctly, these processes can protect networks from certain types of attack, such as:

- » provision of false (statistical) feedback to the system — for example, fooling the system into believing it is under attack, resulting in unnecessary deployment of countermeasures, which consumes resources and inevitably leads to suboptimal usage;
- » modification of a valid on-path request — which results in a direct attack that alters network behavior;
- » forwarding traffic that is not meant to be forwarded, or not forwarding traffic that should be — subverting network isolation; and
- » gaining control access to any component — rendering the entire network untrustworthy.

The critical nature of the SDNC dictates that additional security measures need to be taken to protect it. At the very least, traffic must be integrity protected to prevent tampering of on-path traffic, but even this level of protection does not secure control data.

Encryption is one way of preventing control data from being leaked. But, even together with integrity protection, encryption is not sufficient to protect against man-in-the-middle-type attacks. And so, all communication within the control plane must be mutually authenticated. Security protocols like TLS and IPSEC provide a means for mutual authentication as well as for replay attack protection, confidentiality, and integrity protection.

Mutual authentication does, however, present some difficulties, such as how to bootstrap security into the system. One way to solve this is by using security certificates. How then these certificates are issued, installed, stored, and revoked then becomes the significant security difficulty. Encryption and integrity protection without mutual authentication are less useful from a security point of view.

The problem with mutual authentication is that it requires previous knowledge of the remote communicating endpoint — unless a commonly trusted third party exists.

On a small scale, mutual authentication can be implemented manually — requiring administrators to install proper certificates or shared secrets on all endpoints. However, for complex and physically separated systems — and especially in networks where many SDN components can be created

dynamically and administered by multiple parties — manual implementation may not be feasible.

The SDNC provides network configuration information through API calls to its services, which enables tenants to use SDN applications to control network behavior. This situation is somewhat alarming, given that physical hardware resources may be shared among rival tenants. While ordinary security measures — such as argument sanitization and validation — must be in place, the SDNC also needs a solid authentication, authorization and accountability infrastructure to protect the network from unauthorized changes. Strong authentication and authorization provides additional protection, as it prevents an attacker from impersonating an SDN component, especially the SDNC.

By enforcing strict authorization and accountability processes, damages can be limited, and reliable traces for forensics provided. Role-based access control (RBAC) is a commonly used approach for restricting the actions permitted by an application by assigning a role to it. Roles can be defined on a host, user or application basis.

In effect, RBAC is a security policy enforcing system. The fewer the number of permitted actions, the more limited the exploitable functionality. When implemented correctly, RBAC can be invaluable. Unfortunately, this approach is rather cumbersome in systems with very narrowly defined roles where frequent changes take place. At the other end of the scale, RBAC loses its edge if roles are too loosely defined.

For the purposes of system integrity assurance, every event that occurs in the system should be recorded in a log. How these logs are stored and secured against improper access also needs to be considered, and an external host is recommended.

Multi-tenancy

Where networks are built using SDN techniques, it is possible for the same physical network to be shared among several tenants, which can in turn manage their own virtual networks. Multi-tenancy allows for better utilization of network resources, lowering the total cost of ownership. For tenants, SDN shortens the time taken to react to changing situations

through, for example, automatic scaling of resources. To maintain an acceptable level of security, tenants should not be able to interfere with each other's networks, and need not even be aware that they are sharing network resources with others.

Tenant isolation (the separation of one tenant's resources and actions from another) is an important feature of SDN framework security.

Control plane isolation

Isolation is one way to prevent the actions of one tenant from impacting others. This is a critical business aspect that must be strongly enforced. Tenant isolation is orchestrated by the SDNC, and implemented in SDN NES through specific forwarding rules. While the burden of providing secure isolation lies with the SDNC, tenants also play an important role in sharing that burden.

The network provides isolation primarily on the link layer. If a tenant has weak network security procedures, information disclosure may occur, resulting in a breach of isolation at higher layers. For example, a rogue SDN app with privileges that span beyond isolation borders may impact overall network security by steering traffic to a third party (information disclosure) by over- or under-billing (theft of service) or by dropping traffic (DoS). The centralized nature of the SDN control plane further accentuates the impact of such attacks. Consequently, the task of providing isolation cannot be entirely offloaded onto the SDN network.

Data plane isolation

Tenants running a business on virtual networks built using SDN may be subject to the same kind of network-based attacks as in traditional networks. However, due to the shared networking infrastructure, the impact of such an attack may be divided among some or even all of these tenants. This is a new risk, which may have a commercial impact; nobody wants to open a business next to a known (or perceived) troublemaker or one that is prone to attack.

So, for the data plane, flows associated with each particular tenant must remain isolated at all times. Isolation may be performed logically through

overlay networks and enforced within the NES. For example, by tagging the ownership of traffic generated by each tenant, the traffic can be

carried over a shared infrastructure — once it has been encapsulated (tagged). Tunnels tagged for a given tenant are then forwarded to the virtual network for that tenant. Many alternative (and complementary) techniques are available for this type of encapsulation, including GRE, MPLS and IPSEC.

Tagging is one way to perform logical isolation, but IP addresses can also be used, removing the need for specific tagging techniques. Bearing in mind that separate network function instances are not required to service different tenants, some network functionality can be shared by tenants as long as isolation is preserved and enforced.

In addition to logical isolation, traffic may be encrypted with specific tenant keys. This guarantees that in the case of logical encapsulation violation, the data traffic remains isolated and information cannot be leaked.

Isolation issues need to be resolved while bearing resource consumption in mind. While traffic isolation can help with data leakage, shared resource usage also requires resource isolation. For example, the existence of a forwarding loop within one tenant may potentially impact all tenants, as the problem overloads the underlying network equipment. To counteract this problem, the SDNC must enforce resource isolation, and use measures like rate limiting to minimize the impact that a tenant can have on the network.

Programmability

One of the significant benefits brought about through SDN is programmability: the ability to configure a network efficiently, securely, and in a timely manner. SDN programmability exists in varying degrees of complexity and abstraction. At one end of the scale, programmability enables NES

“AS THE SDNC IS SO CRITICAL, ADDITIONAL SECURITY MEASURES ARE NEEDED TO PROTECT IT”

to be dynamically reprogrammed to forward data flows according to their capabilities and higher-level policies in the network. At the other end, SDN apps enable tenants to programmatically issue run-time requirements to the network. All requests are consolidated by the SDNC, which fulfills higher-level requests from the capabilities available at the lower levels. To make this task trickier, SDN apps may issue orthogonal (mutually exclusive/contradicting) requests. The automated solution may then need to dynamically reconfigure a chunk of the SDN network — and all of this must happen within seconds or less.

The primary benefit that programmability brings for networks built using the SDN architecture approach is flexible control. The ability to control a network and apply changes in a timely manner increases the network's level of agility. Such flexibility can make the network more secure, as it is constantly monitored and designed to mitigate malicious behavior in more or less real time. The downside of the flexibility provided by programmability is the significant impact it has on security.

Configuration coherency

Allowing tenants to issue programmatic changes to the network enables networks to adapt to changing conditions — increasing network agility. In practical terms, programmability can, for example, reduce the time it takes to set up a customer collaboration network from days or months to minutes or hours.

Programmability may also remove the need for manual configuration, which is prone to error. The result: the automatic reconfiguration of networks is feasible, providing the SDNC with a global view of the network, enabling it to perform sanity checking and regression testing so that new networks can be rapidly deployed.

Unfortunately, the flexibility provided by programmability allows tenants to make changes to the shared environment, which can cripple the operation of the entire network — either intentionally or unintentionally as a result of misinformation.

Ensuring coherency among the actions of the various SDN apps on the network also needs to be considered from a security point of view (as

described in⁵). Consider the case where security and load-balancing applications are instantiated for a given tenant. A coherency conflict arises, for example, when the security application decides to quarantine a server, while the load-balancing application simultaneously decides to route traffic to the quarantined server — because it appears to have low load. To avoid coherency issues, the SDNC must be able to assess and eliminate the possible side effects of the acceptable network changes by each tenant, and to feature effective conflict resolution heuristics.

Another type of conflict arises due to the complexity of virtual network topologies, and the difficulty of maintaining a coherent security policy across a network. Special care is required for traffic that needs to be forwarded to security appliances for monitoring purposes. As the traffic or parts of it can be routed over different paths, methods need to be put in place to ensure that all the traffic is covered. Consequently, monitoring is necessary on all paths. Similar issues arise in traditional networks, but the increased service velocity offered by SDN architecture may fuel this type of conflict.

Dynamicity

The dynamic and reactive nature of networks built using the SDN approach opens up new possibilities for fighting network attacks. Automated network reconfigurations, forwarding to honeypots, and black hole routing are just some of the techniques that can be employed. Service chaining is yet another technique that utilizes SDN properties and can be used to screen for malicious payload and trigger mitigating actions.

A network built using SDN techniques can do lower-layer analysis based on parameters such as data rate, source, and packet size, while the tenant can provide higher-layer analysis based on protocols, transport ports, and payload fingerprints. Once suspicious behavior has been detected, the network can use its programmability features to analyze the situation in more detail or trigger mitigating actions.

However, while the feedback system provides some advantages in terms of security, it also presents some issues. The interaction between the data

plane and the control plane breaks the fundamental SDN concept: the separation of these two planes. This in turn makes the data plane a stepping stone for attacking the control plane. As with other feedback loops, this interaction, unless managed appropriately, may lead to an oscillating situation that will eventually make the network unstable.

Conclusion

The beauty of SDN lies in its ability as a technology to make networks flexible, ensure efficient use of resources, and facilitate a much higher level of system autonomy. Like any nascent technology, SDN should be handled cautiously to avoid it becoming an attack vector. However, SDN opens up new possibilities for the implementation of improved security mechanisms in the network, offering broader visibility, programmability, as well as a centralized approach to network management. ☛

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A VISION OF THE **5G CORE:**

Flexibility

FOR NEW BUSINESS OPPORTUNITIES

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Next-generation 5G networks will cater for a wide range of new business opportunities, some of which have yet to be conceptualized. They will provide support for advanced mobile broadband services such as massive media distribution. Applications like remote operation of machinery, telesurgery, and smart metering all require connectivity, but with vastly different characteristics. The ability to provide customized connectivity will benefit many industries around the world, enabling them to bring new products and services to market rapidly, and adapt to fast-changing demands, all while continuing to offer and expand existing services. But how will future networks provide people and enterprises with the right platform, with just the right level of connectivity?

The answer flexibility. The ICT world has already started the journey to delivering elastic connectivity. Technologies like SDN and virtualization are enabling a drastic change to take place in network architecture, allowing traditional structures to be broken down into customizable elements that can be chained together programmatically to provide just the right level of connectivity, with each element running on the architecture of its choice. This is the concept of network slicing that will enable core networks to be built in a way that maximizes flexibility.

AS WE MOVE DEEPER into the Networked Society, with billions of connected devices, lots of new application scenarios, and many more services, the business potential for service providers is expanding rapidly. And 5G technologies will provide the key to tap into this potential, ensuring that customized communication can be delivered to any industry.

■ Being able to deliver the wide variety of network performance characteristics that future services will demand is one of the primary technical challenges faced by service providers today. The performance requirements placed on the network will demand connectivity in terms of data rate, latency, QoS, security, availability, and many other parameters — all of which will vary from one service to the next. But future services also present a business challenge: average revenues will differ significantly from one service to the next, and so flexibility in balancing cost-optimized implementations with those that are performance-optimized will be crucial to profitability.

In addition to the complex performance and business challenges, the 5G environment presents new challenges in terms of timing and agility. The time it takes to get new features into the network,

and time to put services into the hands of users need to be minimized, and so tools that enable fast feature introduction are a prerequisite.

Above all, overcoming the challenges requires a dynamic 5G core network.

But how do you build the core to be a dynamic, virtualized provider of customized connectivity? An important first step is a high-level vision for the 5G core network. The network architecture that meets the objectives then needs to be defined, and finally the whole concept needs to be tested using various possible deployments of the architecture.

Vision of the 5G core

The 5G core will need to be able to support a wide range of business solutions, and at the same time allow existing service offerings, like mobile broadband, to be enhanced and optimized. It will need to connect many different access technologies together, and deliver traffic to and from a wide range of device types.

Next-generation core networks will run in a business environment that is significantly different from that of today. Next-generation core networks will be designed to support the traditional operator model, but at the same time be flexible enough to support a shared-infrastructure model, as well as dedicated usage for specific industries.

Terms and abbreviations

AAA—authentication, authorization, and accounting | **APP**—application | **BSS**—business support systems | **CN**—core network | **CO**—central office | **CP**—control plane | **DC**—data center | **DM**—Device Management | **EPC**—Evolved Packet Core | **ID**—Identity | **M2M**—machine-to-machine | **MBB**—mobile broadband | **NFV**—Network Functions Virtualization | **NFVI**—NFV Infrastructure | **NFVO**—NFV Orchestration | **NX**—new radio-access technologies | **OASIS**—Organization for the Advancement of Structured Information Standards | **OS-MA**—operating system mobile application | **OSS**—operations support systems | **SDN**—software-defined networking | **SDNC**—SDN controller | **SF**—service function | **SLA**—Service Level Agreement | **TOSCA**—Topology and Orchestration Specification for Cloud Applications | **TTC**—time to customer | **TTM**—time to market | **UP**—user plane | **VIM**—Virtual Infrastructure Manager

But it's not just the core network that needs to be flexible; the whole communication ecosystem needs to work in a highly responsive manner. Agile systems and processes are needed to ensure that two crucial factors, TTM and TTC, are kept to a minimum. Service providers need to be able to create offerings quickly, and be able to tailor solutions to rapidly changing market demand (short TTM). Order processing needs to be fast, cutting the time from order to a fully active service to a minimum (rapid TTC). To build a future core network architecture that is highly flexible, modular, and scalable will require a much higher degree of programmability and automation than exists in today's networks.

The 5G core will exist in an environment that is cloud-based, with a high degree of Network Functions Virtualization for scalability, SDN for flexible networking, dynamic orchestration of network resources, and a modular and highly resilient base architecture. Full support for next-generation access networks, including NX and evolved LTE, as well WI-FI and other non-3GPP technologies are prerequisites.

Network slicing is one of the key capabilities that will enable flexibility, as it allows multiple logical networks to be created on top of a common shared physical infrastructure. The greater elasticity brought about by network slicing will help to address the cost, efficiency, and flexibility requirements imposed by future services.

Architecture and technology

Traditionally, core networks have been designed as a single network architecture serving multiple purposes, addressing a range of requirements, and supporting backward compatibility and interoperability. This one-size-fits-all approach has kept costs at a reasonable level, given that one set of vertically integrated nodes has provided all functionality.

Technology has, however, evolved. Virtualization, NFV, SDN, and advanced automation and orchestration make it possible to build networks in a more scalable, flexible, and dynamic way. Such capabilities allow today's network designers to contemplate the core in a radically different way,

providing greater possibilities for tailored and optimized solutions.

The concept of flexibility applies not only to the hardware and software parts of the network, but also to its management. For example, setting up a network instance that uses different network functions optimized to deliver a specific service needs to be automated. Flexible management will enable future networks to support new types of business offerings that previously would have made no technical or economic sense.

High-level architecture

Network slicing allows networks to be logically separated, with each slice providing customized connectivity, and all slices running on the same, shared infrastructure. This is a much more flexible solution than a single physical network providing a maximum level of connectivity. Virtualization and SDN are the key technologies that make network slicing possible. As shown in [Figure 1](#), network slices are logically separated and isolated systems that can be designed with different architectures, but can share functional components. One slice may be designed for evolved MBB services providing access to LTE, evolved LTE and NX devices; another may be designed for an industry application with an optimized core network control plane, different authentication schemes, and lightweight user plane handling. Together, the two slices can support a more comprehensive set of services and enable new offerings that are cost-effective to operate.

To support a specific set of services efficiently, a network slice should be assigned different types of resources, such as infrastructure — including VPNs, cloud services, and access — as well as resources for the core network in the form of VNFS.

As illustrated in [Figure 2](#), network slicing supports business expansion due to the fact that it lowers the risks associated with introducing and running new services — the isolated nature of slices protects existing services running on the same physical infrastructure from any impact. An additional benefit of network slicing is that it supports migration, as new technologies or architectures can be launched on isolated slices.

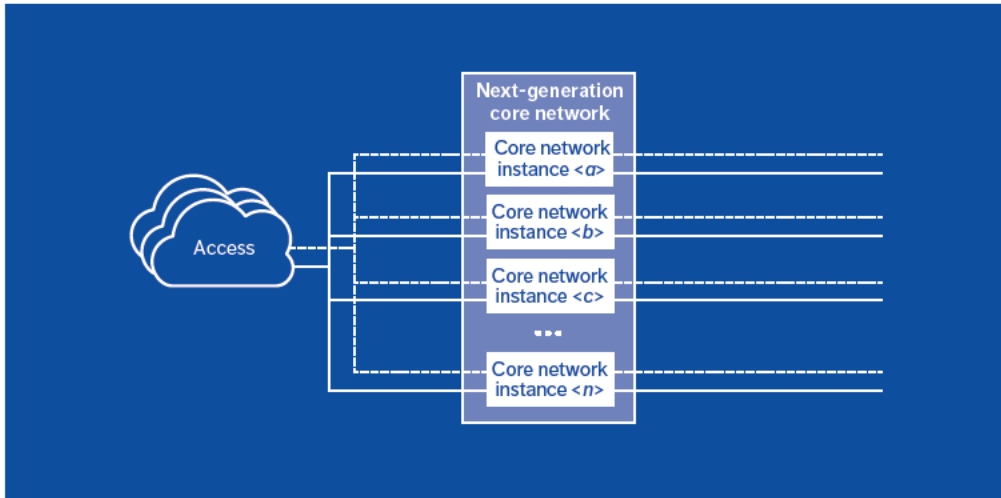


Figure 1:
The next-generation
core network, comprising
various slices

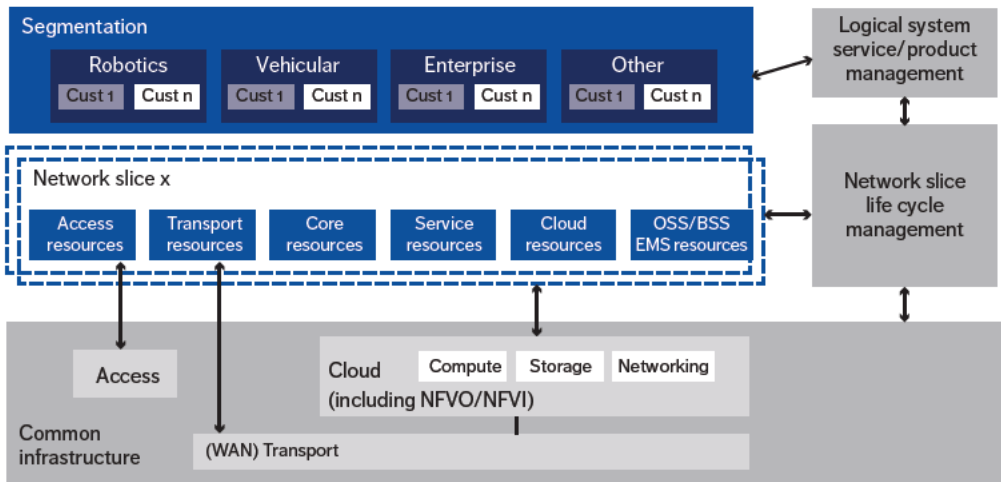


Figure 2:
Network slicing supports
business expansion

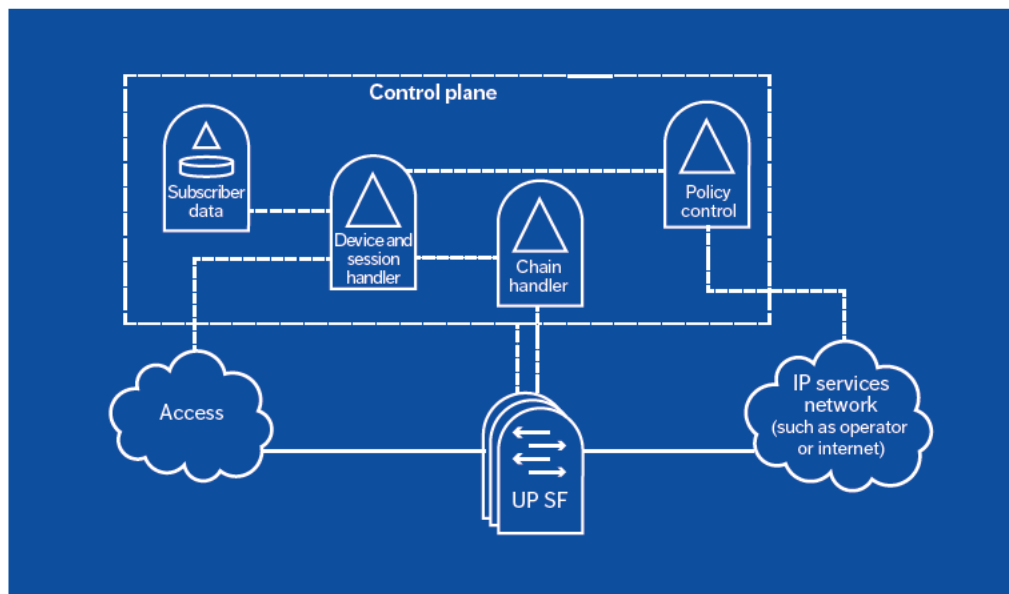


Figure 3:
5G control-plane
architecture

Evolving standards should allow network architecture to develop — in a radical or revolutionary way. By steering away from the one-size-fits-all approach, evolving standards will allow for a whole palette of architectures from which different network slices can be designed. The introduction of a selection mechanism like Dedicated Core Network (DCN) [1] — which allows for multiple parallel architectures — is one step in the right direction.

Control- and user-plane separation

Many aspects of the 5G network — not just the deployment architecture — need to be flexible to allow for business expansion. It is likely that networks will need to be deployed using different hardware technologies, with different feature sets placed at different physical locations in the network — depending on the use case. Special attention must be paid to the design of the user plane to meet requirements for high bandwidth, which may apply on an individual subscriber basis or as an aggregated target. For example, in some use cases, the majority of user-plane traffic may require only very simple processing, which can be run on low-cost hardware, whereas the remainder of the traffic might require more advanced processing. Cost-efficient scaling of the user plane to handle the increasing individual and aggregated bandwidths is a key component of a 5G core network.

Supporting the separation of the control- and user-plane functions is one of the most significant principles of the 5G core-network architecture. Separation allows control- and user-plane resources to be scaled independently, and it supports migration to cloud-based deployments. By separating user- and control-plane resources, the planes may also be established in different locations. For example, the control plane can be placed in a central site, which makes management and operation less complex. And the user plane can be distributed over a number of local sites, bringing it closer to the user. This is beneficial, as it shortens the round-trip-time between the user and the network service, and reduces the amount of bandwidth required between sites. Content caching is a good

example of how locating functions on a local site reduces the required bandwidth between sites.

As separation of the control plane and the user plane is a fundamental concept of SDN, the flexibility of 5G core networks will improve significantly by adopting SDN technologies.

The control plane, illustrated in [Figure 3](#), can be agnostic of many user-plane aspects, such as physical deployment, and L2 and L3 transport specifics. Typical control-plane functionality includes capabilities like the maintenance of location information, policy negotiation, and session authentication. As such, there is a natural separation at this level.

User-plane functionality, which can be seen as a chain of functions, can be deployed to suit a specific use case. Given that the connectivity needs of each use case varies, the most cost-efficient unique deployment can be created for each scenario. For example, the connectivity needs for an M2M service with small payload volume and low mobility are quite different from the needs of an MBB service with high payload volume and high mobility. An MBB service can be broken down into several sub-services, such as video streaming and web browsing, which can in turn be implemented by separate sub chains within the network slice. Such additional decomposition within the user-plane domain further increases the flexibility of the core network.

The strict separation of the control and user planes enables different execution platforms to be used for each. Similarly, different user planes can be deployed with different execution platforms, even within a user plane — all depending on which solution is most cost-efficient. In the above MBB example, one sub chain of services may run on general-purpose CPUs, whereas another sub chain of services that requires simple user-data processing can be executed on low-cost hardware.

Governing the network

It is clear that enabling business expansion requires greater flexibility in the way networks are built. And as we have illustrated, network slicing is a key enabler to achieving greater flexibility. However, increasing flexibility may lead to greater complexity

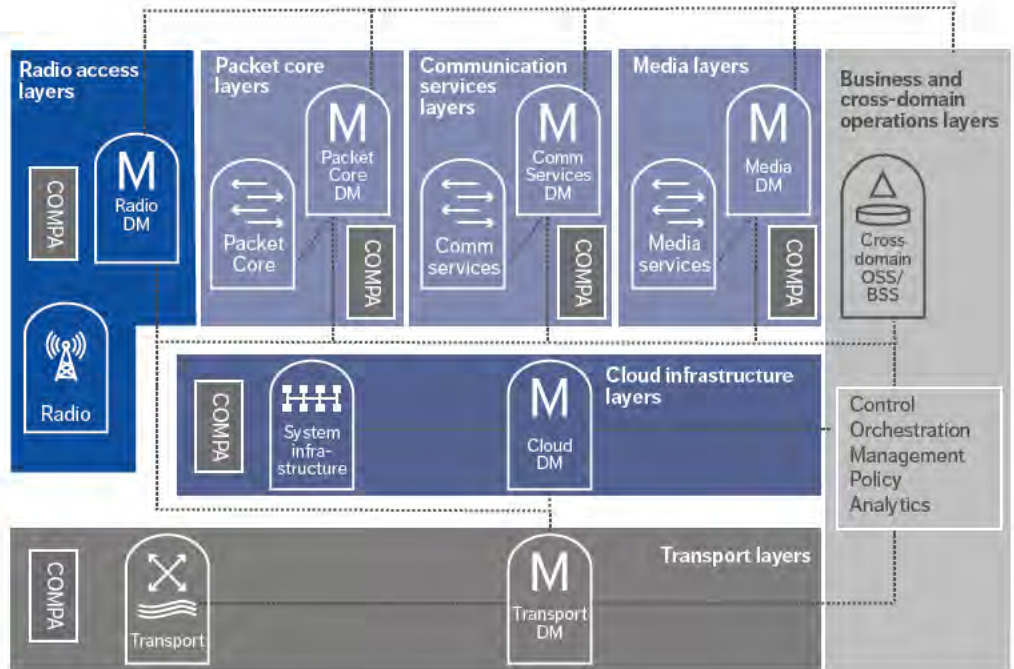


Figure 4:
Separation of concerns

at all levels of the system, which in turn tends to raise the cost of operations and lengthen lead times. Automation is an essential way of avoiding this spiral of complexity.

Network governance is addressed in three main steps of the life cycle: creation, activation, and runtime.

- » Creation of new (or customization of existing) services with minimum TTM — the ability to break down the overall solution into components is necessary, so that services and slice types can be designed, verified, and validated rapidly.
- » Activation of a service with minimum TTC — the ability to complete activation in a fully automated way will minimize lead times.
- » Runtime — exposing the right capabilities to the user, service and SLA monitoring, and adapting to changing conditions (such as scaling and failovers) enable

scalability for new services, all of which need to be fully automated.

As illustrated in *Figure 4*, the fundamental architectural principles for achieving flexibility are separation of concerns, abstraction, and programmability [2][3].

The capability offered by network slicing to deliver different categories of connectivity is handled by two-layered governance functionalities: one layer focuses on services and products (such as business-to-business offerings that can be implemented using network slicing); and one focuses on the network slices themselves — as illustrated in *Figure 5*. By creating slices based on performance characteristics like a low latency slice, or a high capacity, low throughput, and high-speed slice, innovative offers can be created by

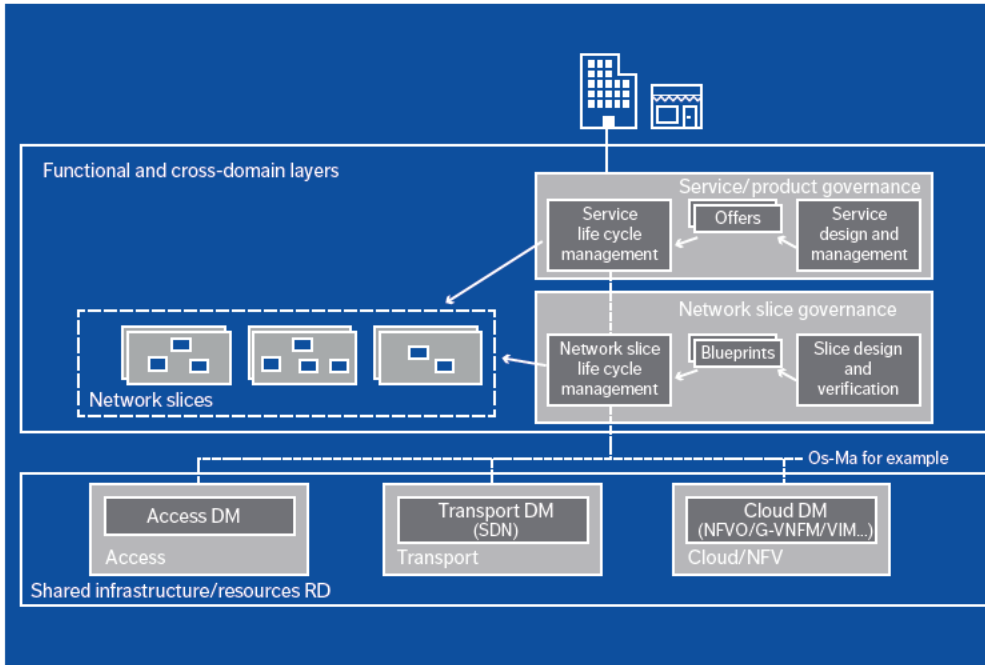


Figure 5:
Governance functions for
network slices, services,
and products

bundling slice capabilities. Each offer would include governance capabilities such as SLAs (and how to translate wanted service levels into technical control parameters for a network slice), business policies, and control of exposure of capabilities from within the slice.

The life cycle management provided by these capabilities extends from design and creation of network slice types and services, through activation for individual customers, to runtime monitoring and updates (if needed).

The governance layers handle life cycle management of network slices, aided by a blueprint. A blueprint defines the setup of the slice, including: the components that need to be instantiated, the features to enable, configurations to apply, resource assignments, and all associated workflows — including all aspects of the life cycle (such as

upgrades and changes). The blueprint contains machine-readable parts, similar to OASIS TOSCA models, which support automation.

To instantiate a new customer service, a new network slice is created, or in some cases, an existing slice is reconfigured. The slice can be independently managed, and it comprises a set of resources or components, which may be traditional, such as an EPC, or a new type of architecture — such as a CP/UP separation. Network slices typically contain management capabilities, some of which may be under the control of the service provider, and some under the control of the customer — depending on the business model. The governance layer uses a number of systems and interfaces to facilitate the creation and configuration of resources, such as northbound API interfaces exposed by an NFVO or an SDNC, or APIs exposed by network functions and

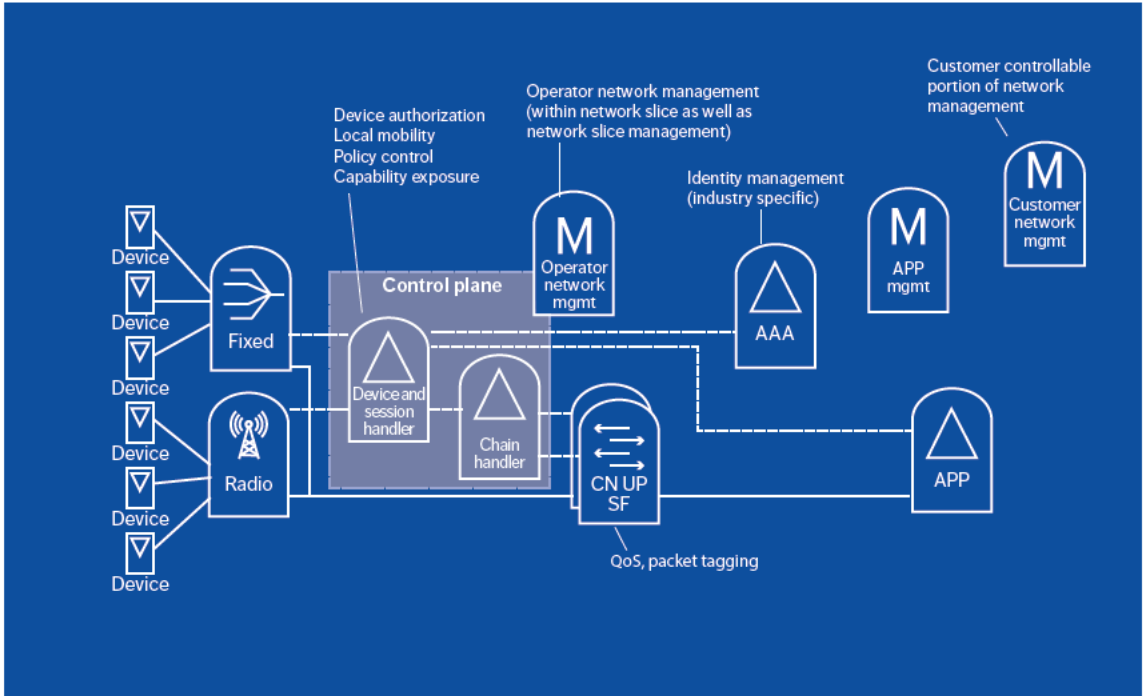


Figure 6:
Functional architecture of a possible application

components within the slice. Flexibility is key in the automation/orchestration system, which can be achieved, for example, by using plugins.

Deployment scenarios

Applying the concept of network slicing widens the choices for application support — different slices can be deployed independently for quite different purposes, both functionally as well as for performance reasons. The following use case describes the functionality, architecture, and variety in deployments of one such network slice.

In our example use case (see Figure 6), the

processes involved in an industry application require low-delay communication between the application controller and the devices in the system — which could be sensors or actuators. Deployment is local to minimize delay and the number of points of failure. To ensure high availability, all relevant resources can be duplicated using a hot standby or load-sharing scheme; deploying duplication locally limits the need for redundancy in the global part of the network. The network slice could include cloud resources to host the industry application together with the necessary network functions in the operator data center.

Core network control-plane functions may

be limited to device authorization, local mobility management, and policy control. What user-plane functions are needed depends on the nature of the application and possibly other usage. For example, a QoS function may be in place to ensure that traffic prioritization is upheld; the application traffic is assigned with one or more priorities (such as real-time dependent versus background application traffic), and management traffic is assigned another priority.

If radio access is shared among several network slices, the user plane may include functions to ensure that uplink user-plane traffic is separated for the different network slices. The different user-plane service functions (UPSFs) can be deployed as chains, and packets can be tagged to ensure they pass through the desired chain.

From a management perspective, responsibilities are shared between the operator and the customer (the industry enterprise using communication). The customer is responsible for the application, and device and identity management in the system, and the operator is responsible for the physical infrastructure — including data centers, transport, and nodes. Management of the slice may be shared between the two, allowing the customer (within the framework of an SLA) to manage capabilities for network functions that support the application — such as local mobility management.

As shown in [Figure 7](#), the application controller, the management of the application controller, and the customer part of network management could all be deployed close to the industry site, if only local communication is needed. Operator-controlled network-management functions tend to be deployed at a central location in the operator network.

The decision of where to locate core network control-plane functions (1), (2), or (3) is governed not only by performance requirements, such as core-plane delay and reliability, but by other factors such as the number of local industries that need to be supported by the network slice, their geographical spread, and the organization of the operator. Operational parameters from the perspective of the operator also come into play. As identity management is industry specific, this function could

be carried out from the same location as application controller management.

The deployment of a network slice for an industry process that operates on a regional, national, or even multinational basis is shown in [Figure 8](#). The management of the application controller for this case should be deployed centrally, with room for some local control over application controller management capabilities.

For performance and reliability reasons, the application controller for the industry process should be deployed close to each of the industry sites. Customer-controlled network management functions could be placed on the same industry site as the management of the application controller.

Conclusion

Evolved virtualization, network programmability, and 5G use cases will change everything about network design, from planning and construction through deployment. Network functions will no longer be located according to traditional vertical groupings in single network nodes, but will instead be distributed to provide connectivity where it is needed.

To support the wide range of performance requirements demanded by new business opportunities, multiple access technologies, a wide variety of services, and lots of new device types, the 5G core will be highly flexible.

Minimizing cost for service providers and industries that depend on connectivity is a key part of the design for this flexible and dynamic core — enabling to keep costs under control, while networks adapt as quickly as business models change.

Technologies like SDN will be used in innovative ways, to set up a network slice, and to implement additional user-plane modifications. Cloud technology together with advanced analytics capabilities, NFV, and SDN provide a common distributed platform on which networks can be instantiated. The technology boost provided by a flexible core, with end-to-end network slices at the center, will increase the value of networks built on a common infrastructure and platform. *

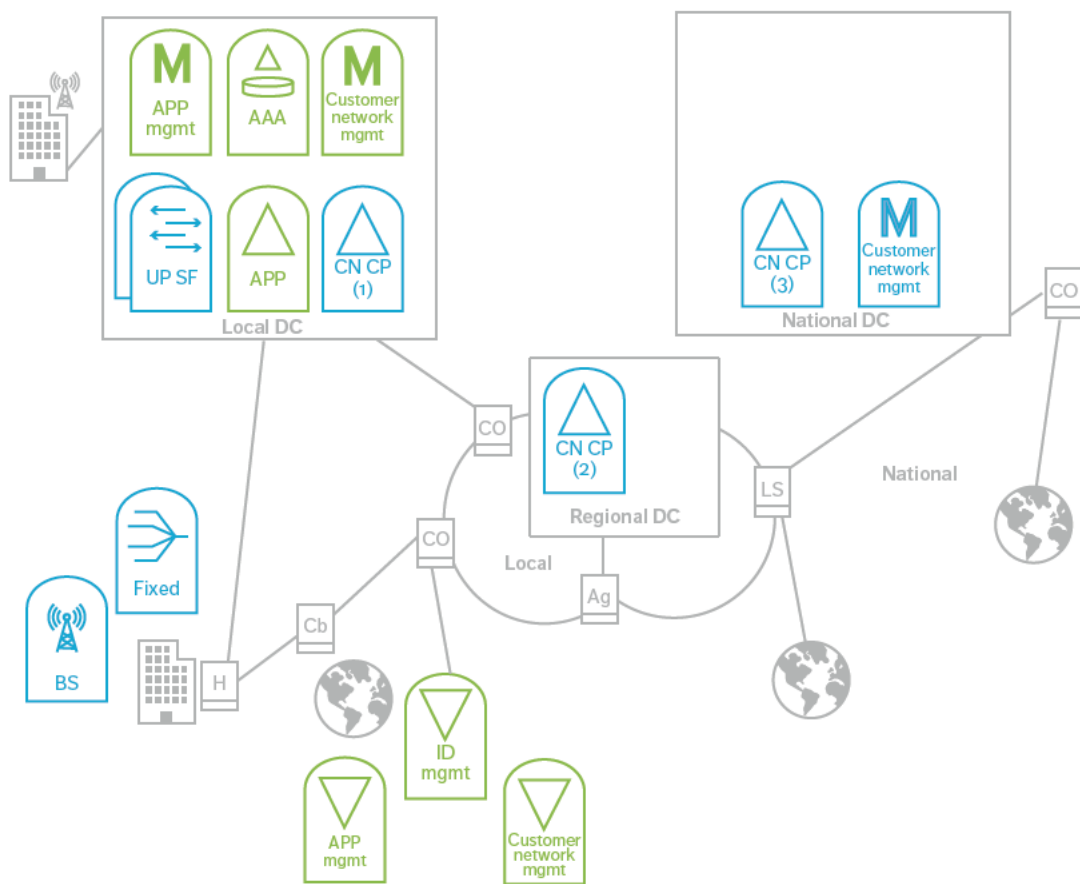


Figure 7:

Low-latency application, local industry

(1), (2), and (3) are possible operator sites for core-network and control-plane functions

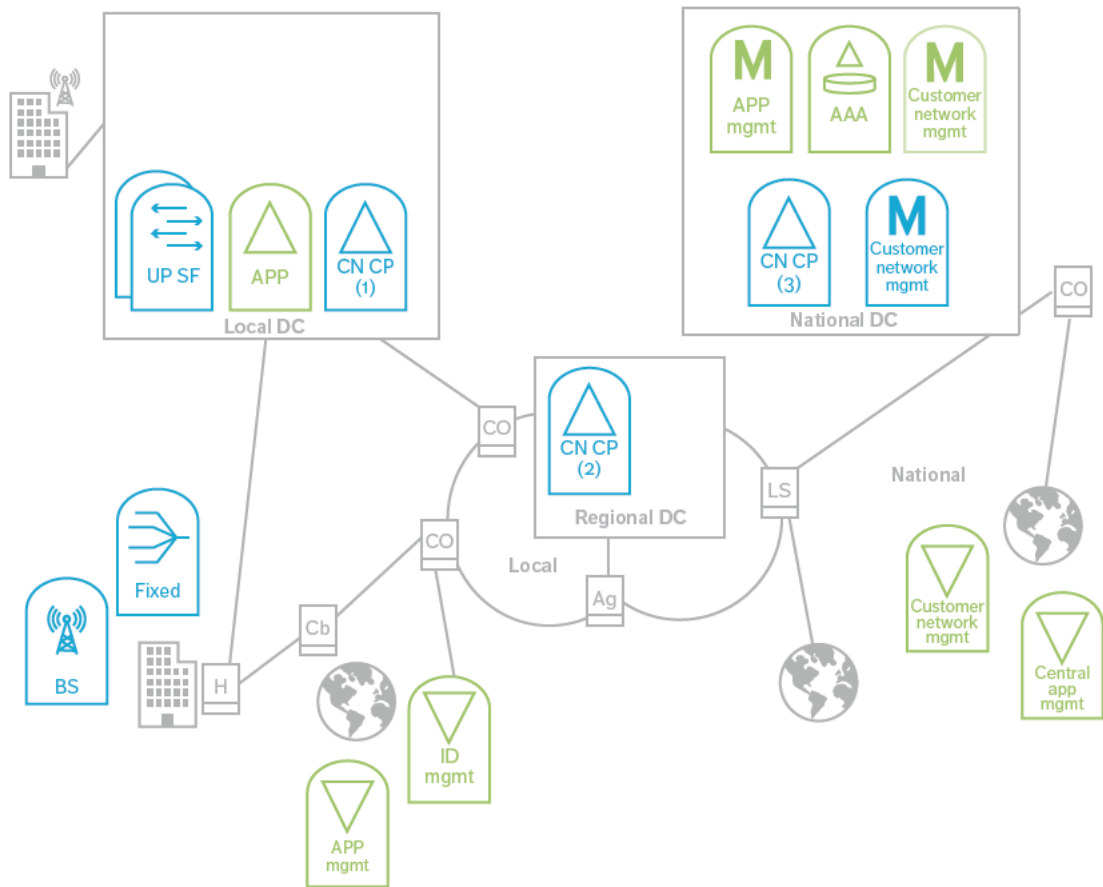


Figure 8:

Low latency application, regional, national, or multi-national industry
 (1), (2), and (3) are possible operator sites for core network control plane functions

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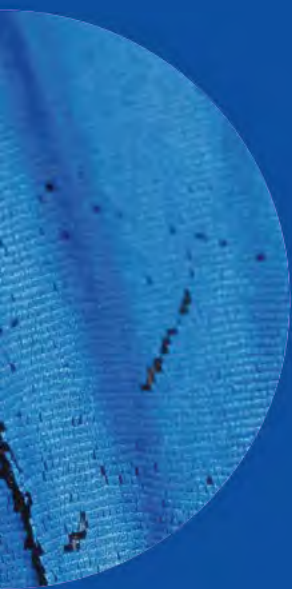
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